

FUNDAMENTALS OF THERMAL RADIATION

So far, we have considered the conduction and convection modes of heat transfer, which are related to the nature of the media involved and the presence of fluid motion, among other things. We now turn our attention to the third mechanism of heat transfer: *radiation*, which is characteristically different from the other two.

We start this chapter with a discussion of *electromagnetic waves* and the *electromagnetic spectrum*, with particular emphasis on *thermal radiation*. Then we introduce the idealized *blackbody*, *blackbody radiation*, and *blackbody radiation function*, together with the *Stefan–Boltzmann law*, *Planck’s law*, and *Wien’s displacement law*.

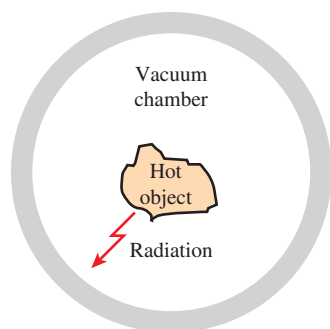
Radiation is emitted by every point on a plane surface in all directions into the hemisphere above the surface. The quantity that describes the magnitude of radiation emitted or incident in a specified direction in space is the *radiation intensity*. Various radiation fluxes such as *emissive power*, *irradiation*, and *radiosity* are expressed in terms of intensity. This is followed by a discussion of radiative properties of materials such as *emissivity*, *absorptivity*, *reflectivity*, and *transmissivity* and their dependence on wavelength, direction, and temperature. The *greenhouse effect* is presented as an example of the consequences of the wavelength dependence of radiation properties. We end this chapter with a discussion of *atmospheric* and *solar radiation*.



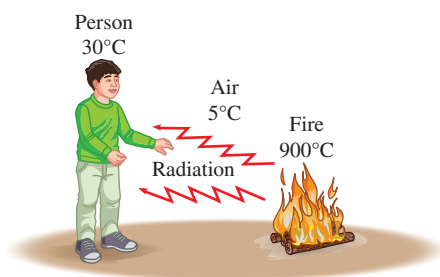
OBJECTIVES

When you finish studying this chapter, you should be able to:

- Classify electromagnetic radiation, and identify thermal radiation,
- Understand the idealized blackbody, and calculate the total and spectral blackbody emissive power,
- Calculate the fraction of radiation emitted in a specified wavelength band using the blackbody radiation functions,
- Understand the concept of radiation intensity, and define spectral directional quantities using intensity,
- Develop a clear understanding of the properties emissivity, absorptivity, reflectivity, and transmissivity on spectral, directional, and total basis,
- Apply Kirchhoff’s law to determine the absorptivity of a surface when its emissivity is known,
- Model the atmospheric radiation by the use of an effective sky temperature, and appreciate the importance of the greenhouse effect.

**FIGURE 12-1**

A hot object in a vacuum chamber loses heat by radiation only.

**FIGURE 12-2**

Unlike conduction and convection, heat transfer by radiation can occur between two bodies, even when they are separated by a medium colder than both.

12-1 ■ INTRODUCTION

Consider a hot object that is suspended in an evacuated chamber whose walls are at room temperature (Fig. 12-1). The hot object will eventually cool down and reach thermal equilibrium with its surroundings. That is, it will lose heat until its temperature reaches the temperature of the walls of the chamber. Heat transfer between the object and the chamber could not have taken place by conduction or convection, because these two mechanisms cannot occur in a vacuum. Therefore, heat transfer must have occurred through another mechanism that involves the emission of the internal energy of the object. This mechanism is *radiation*.

Radiation differs from the other two heat transfer mechanisms in that it does not require the presence of a material medium to take place. In fact, energy transfer by radiation is fastest (at the speed of light), and it suffers no attenuation in a *vacuum*. Also, radiation transfer occurs in solids as well as liquids and gases. In most practical applications, all three modes of heat transfer occur concurrently at varying degrees. But heat transfer through an evacuated space can occur only by radiation. For example, the energy of the sun reaches the earth by radiation.

You will recall that heat transfer by conduction or convection takes place in the direction of decreasing temperature; that is, from a high-temperature medium to a lower-temperature one. It is interesting that radiation heat transfer can occur between two bodies separated by a medium colder than both bodies (Fig. 12-2). For example, solar radiation reaches the surface of the earth after passing through cold air layers at high altitudes. Also, the radiation-absorbing surfaces inside a greenhouse reach high temperatures even when its plastic or glass cover remains relatively cool.

The theoretical foundation of radiation was established in 1864 by physicist James Clerk Maxwell, who postulated that accelerated charges or changing electric currents give rise to electric and magnetic fields. These rapidly moving fields are called **electromagnetic waves** or **electromagnetic radiation**, and they represent the energy emitted by matter as a result of the changes in the electronic configurations of the atoms or molecules. In 1887, Heinrich Hertz experimentally demonstrated the existence of such waves. Electromagnetic waves transport energy just like other waves, and all electromagnetic waves travel at the *speed of light* in a vacuum, which is $c_0 = 2.9979 \times 10^8$ m/s. Electromagnetic waves are characterized by their *frequency* ν or *wavelength* λ . These two properties in a medium are related by

$$\lambda = \frac{c}{\nu} \quad (12-1)$$

where c is the speed of propagation of a wave in that medium. The speed of propagation in a medium is related to the speed of light in a vacuum by $c = c_0/n$, where n is the *index of refraction* of that medium. The refractive index is essentially unity for air and most gases, about 1.5 for glass, and 1.33 for water. The commonly used unit of wavelength is the *micrometer* (μm) or *micron*, where $1 \mu\text{m} = 10^{-6}$ m. Unlike the wavelength and the speed of propagation, the frequency of an electromagnetic wave depends only on the source and is independent of the medium through which the wave travels. The *frequency* (the number of oscillations per second) of an electromagnetic

wave can range from less than a million Hz to a septillion Hz or higher, depending on the source. Note from Eq. 12–1 that the wavelength and the frequency of electromagnetic radiation are inversely proportional.

It has proven useful to view electromagnetic radiation as the propagation of a collection of discrete packets of energy called **photons** or **quanta**, as proposed by Max Planck in 1900 in conjunction with his *quantum theory*. In this view, each photon of frequency ν is considered to have an energy of

$$e = h\nu = \frac{hc}{\lambda} \quad (12-2)$$

where $h = 6.626069 \times 10^{-34}$ J·s is *Planck's constant*. Note from the second part of Eq. 12–2 that the energy of a photon is inversely proportional to its wavelength. Therefore, shorter-wavelength radiation possesses larger photon energies. It is no wonder that we try to avoid very-short-wavelength radiation such as gamma rays and X-rays since they are highly destructive.

12-2 ■ THERMAL RADIATION

Although all electromagnetic waves have the same general features, waves of different wavelength differ significantly in their behavior. The electromagnetic radiation encountered in practice covers a wide range of wavelengths, varying from less than 10^{-10} μm for cosmic rays to more than 10^{10} μm for electrical power waves. The **electromagnetic spectrum** also includes gamma rays, X-rays, ultraviolet radiation, visible light, infrared radiation, thermal radiation, microwaves, and radio waves, as shown in Fig. 12–3.

Different types of electromagnetic radiation are produced through various mechanisms. For example, *gamma rays* are produced by nuclear reactions, *X-rays* by the bombardment of metals with high-energy electrons, *microwaves* by special types of electron tubes such as klystrons and magnetrons, and *radio waves* by the excitation of some crystals or by the flow of alternating current through electric conductors.

The short-wavelength gamma rays and X-rays are primarily of concern to nuclear engineers, while the long-wavelength microwaves and radio waves are of concern to electrical engineers. The type of electromagnetic radiation that is pertinent to heat transfer is the **thermal radiation** emitted as a result of energy transitions of molecules, atoms, and electrons of a substance. Temperature is a measure of the strength of these activities at the microscopic level, and the rate of thermal radiation emission increases with increasing temperature. Thermal radiation is continuously emitted by all matter whose temperature is above absolute zero. That is, everything around us, such as walls, furniture, and our friends constantly emits (and absorbs) radiation (Fig. 12–4). Thermal radiation is also defined as the portion of the electromagnetic spectrum that extends from about 0.1 to 100 μm , since the radiation emitted by bodies due to their temperature falls almost entirely into this wavelength range. Thus, thermal radiation includes the entire visible and infrared (IR) radiation bands as well as a portion of the ultraviolet (UV) radiation band.

What we call **light** is simply the *visible* portion of the electromagnetic spectrum that lies between 0.40 and 0.76 μm . Light is characteristically no different than other electromagnetic radiation, except that it happens to trigger the

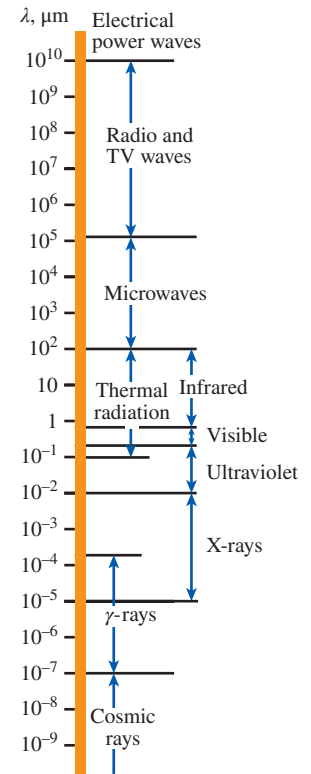


FIGURE 12-3

The electromagnetic wave spectrum.

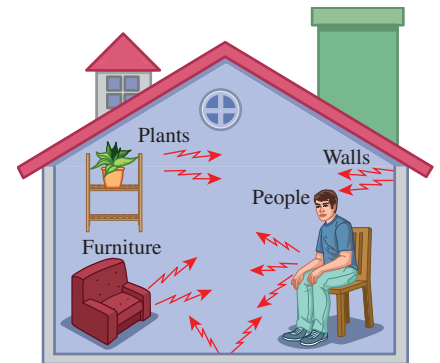


FIGURE 12-4

Everything around us constantly emits thermal radiation.

TABLE 12-1

The wavelength ranges of different colors

Color	Wavelength Band
Violet	0.40–0.44 μm
Blue	0.44–0.49 μm
Green	0.49–0.54 μm
Yellow	0.54–0.60 μm
Orange	0.60–0.67 μm
Red	0.63–0.76 μm

sensation of seeing in the human eye. Light, or the visible spectrum, consists of narrow bands of color from violet (0.40–0.44 μm) to red (0.63–0.76 μm), as shown in Table 12-1.

A body that emits some radiation in the visible range is called a light source. The sun is obviously our primary light source. The electromagnetic radiation emitted by the sun is known as **solar radiation**, and nearly all of it falls into the wavelength band 0.3–3 μm . Almost *half* of solar radiation is light (i.e., it falls into the visible range), with the remaining portion being ultraviolet and infrared.

The radiation emitted by bodies at room temperature falls into the **infrared** region of the spectrum, which extends from 0.76 to 100 μm . Bodies start emitting noticeable visible radiation at temperatures above 800 K. The tungsten filament of a lightbulb must be heated to temperatures above 2000 K before it can emit any significant amount of radiation in the visible range.

The **ultraviolet** radiation includes the low-wavelength end of the thermal radiation spectrum and lies between the wavelengths 0.01 and 0.40 μm . Ultraviolet rays are to be avoided since they can kill microorganisms and cause serious damage to humans and other living beings. About 12 percent of solar radiation is in the ultraviolet range, and it would be devastating if it were to reach the surface of the earth. Fortunately, the ozone (O_3) layer in the atmosphere acts as a protective blanket and absorbs most of this ultraviolet radiation. The ultraviolet rays that remain in sunlight are still sufficient to cause serious sunburns to sun worshippers, and prolonged exposure to direct sunlight is the leading cause of skin cancer, which can be lethal. Recent discoveries of “holes” in the ozone layer have prompted the international community to ban the use of ozone-destroying chemicals such as the refrigerant Freon-12 in order to save the earth. Ultraviolet radiation is also produced artificially in fluorescent lamps for use in medicine as a bacteria killer and in tanning parlors as an artificial tanner.

Microwave ovens utilize electromagnetic radiation in the **microwave** region of the spectrum generated by microwave tubes called *magnetrons*. Microwaves in the range of 10^2 – 10^5 μm are very suitable for use in cooking since they are *reflected* by metals, *transmitted* by glass and plastics, and *absorbed* by food (especially water) molecules. Thus, the electric energy converted to radiation in a microwave oven eventually becomes part of the internal energy of the food. The fast and efficient cooking of microwave ovens has made them one of the essential appliances in modern kitchens (Fig. 12-5).

Radars and cordless telephones also use electromagnetic radiation in the microwave region. The wavelength of the electromagnetic waves used in radio and TV broadcasting usually ranges between 1 and 1000 m in the **radio wave** region of the spectrum.

In heat transfer studies, we are interested in the energy emitted by bodies because of their temperature only. Therefore, we limit our consideration to *thermal radiation*, which we simply call *radiation*. The relations developed below are restricted to thermal radiation only and may not be applicable to other forms of electromagnetic radiation.

The electrons, atoms, and molecules of all solids, liquids, and gases above absolute zero temperature are constantly in motion, and thus radiation is constantly emitted, as well as being absorbed or transmitted throughout the entire volume of matter. That is, radiation is a **volumetric phenomenon**. However,

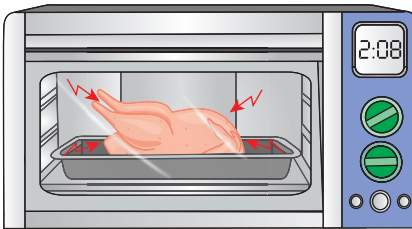


FIGURE 12-5

Food is heated or cooked in a microwave oven by absorbing the electromagnetic radiation energy generated by the magnetron of the oven.

for opaque (nontransparent) solids such as metals, wood, and rocks, radiation is considered to be a **surface phenomenon**, since the radiation emitted by the interior regions can never reach the surface, and the radiation incident on such bodies is usually absorbed within a few microns from the surface (Fig. 12–6). Note that the radiation characteristics of surfaces can be changed completely by applying thin layers of coatings on them.

12–3 ■ BLACKBODY RADIATION

A body at a thermodynamic (or absolute) temperature above zero emits radiation in all directions over a wide range of wavelengths. The amount of radiation energy emitted from a surface at a given wavelength depends on the material of the body and the condition of its surface as well as the surface temperature. Therefore, different bodies may emit different amounts of radiation per unit surface area, even when they are at the same temperature. Thus, it is natural to be curious about the *maximum* amount of radiation that can be emitted by a surface at a given temperature. Satisfying this curiosity requires the definition of an idealized body, called a *blackbody*, to serve as a standard against which the radiative properties of real surfaces may be compared.

A **blackbody** is defined as a *perfect emitter and absorber of radiation*. At a specified temperature and wavelength, no surface can emit more energy than a blackbody. A blackbody absorbs *all* incident radiation, regardless of wavelength and direction. Also, a blackbody emits radiation energy uniformly in all directions per unit area normal to direction of emission (Fig. 12–7). That is, a blackbody is a *diffuse* emitter. The term *diffuse* means “independent of direction.”

The radiation energy emitted by a blackbody per unit time and per unit surface area was determined experimentally by Joseph Stefan in 1879 and expressed as

$$E_b(T) = \sigma T^4 \quad (\text{W/m}^2) \quad (12-3)$$

where $\sigma = 5.670 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is the *Stefan–Boltzmann constant* and T is the absolute temperature of the surface in K. This relation was theoretically verified in 1884 by Ludwig Boltzmann. Equation 12–3 is known as the **Stefan–Boltzmann law** and E_b is called the **blackbody emissive power**. Note that the emission of thermal radiation is proportional to the *fourth power* of the absolute temperature.

Although a blackbody would appear *black* to the eye, a distinction should be made between the idealized blackbody and an ordinary black surface. Any surface that absorbs light (the visible portion of radiation) would appear black to the eye, and a surface that reflects it completely would appear white. Considering that visible radiation occupies a very narrow band of the spectrum from 0.4 to 0.76 μm , we cannot make any judgments about the blackness of a surface on the basis of visual observations. For example, snow and white paint reflect light and thus appear white. But they are essentially black for infrared radiation since they strongly absorb long-wavelength radiation. Surfaces coated with lampblack paint approach idealized blackbody behavior.

Another type of body that closely resembles a blackbody is a *large cavity with a small opening*, as shown in Fig. 12–8. Radiation coming in through the opening of area A undergoes multiple reflections, and thus it has several

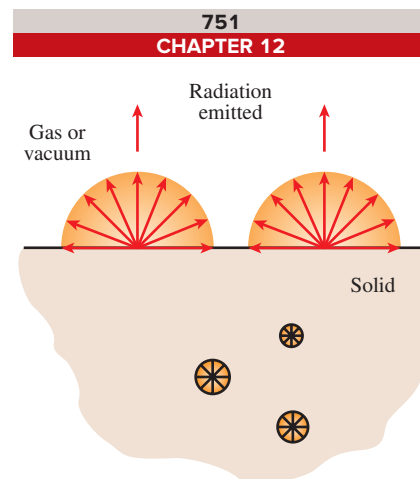


FIGURE 12–6

Radiation in opaque solids is considered a surface phenomenon since the radiation emitted only by the molecules at the surface can escape the solid.

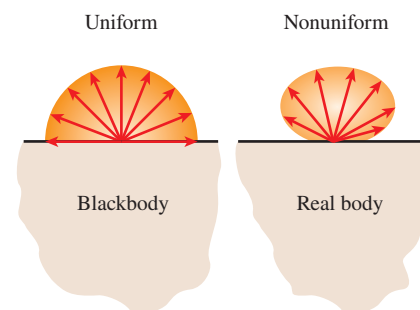


FIGURE 12–7

A blackbody is said to be a *diffuse* emitter since it emits radiation energy uniformly in all directions.

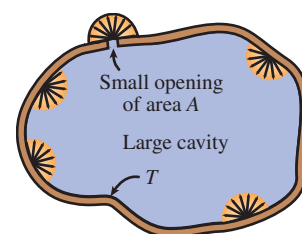


FIGURE 12–8

A large isothermal cavity at temperature T with a small opening of area A closely resembles a blackbody of surface area A at the same temperature.

chances to be absorbed by the interior surfaces of the cavity before any part of it can possibly escape. Also, if the surface of the cavity is isothermal at temperature T , the radiation emitted by the interior surfaces streams through the opening after undergoing multiple reflections, and thus it has a diffuse nature. Therefore, the cavity acts as a perfect absorber and perfect emitter, and the opening will resemble a blackbody of surface area A at temperature T , regardless of the actual radiative properties of the cavity.

The Stefan–Boltzmann law in Eq. 12–3 gives the *total* blackbody emissive power E_b , which is the sum of the radiation emitted over all wavelengths. Sometimes we need to know the **spectral blackbody emissive power**, which is *the amount of radiation energy emitted by a blackbody at a thermodynamic temperature T per unit time, per unit surface area, and per unit wavelength about the wavelength λ* . For example, we are more interested in the amount of radiation an incandescent lightbulb emits in the visible wavelength spectrum than we are in the total amount emitted.

The relation for the spectral blackbody emissive power $E_{b\lambda}$ was developed by Max Planck in 1901 in conjunction with his famous quantum theory. This relation is known as **Planck's law** and is expressed as

$$E_{b\lambda}(\lambda, T) = \frac{C_1}{\lambda^5 [\exp(C_2/\lambda T) - 1]} \quad (\text{W/m}^2 \cdot \mu\text{m}) \quad (12-4)$$

where

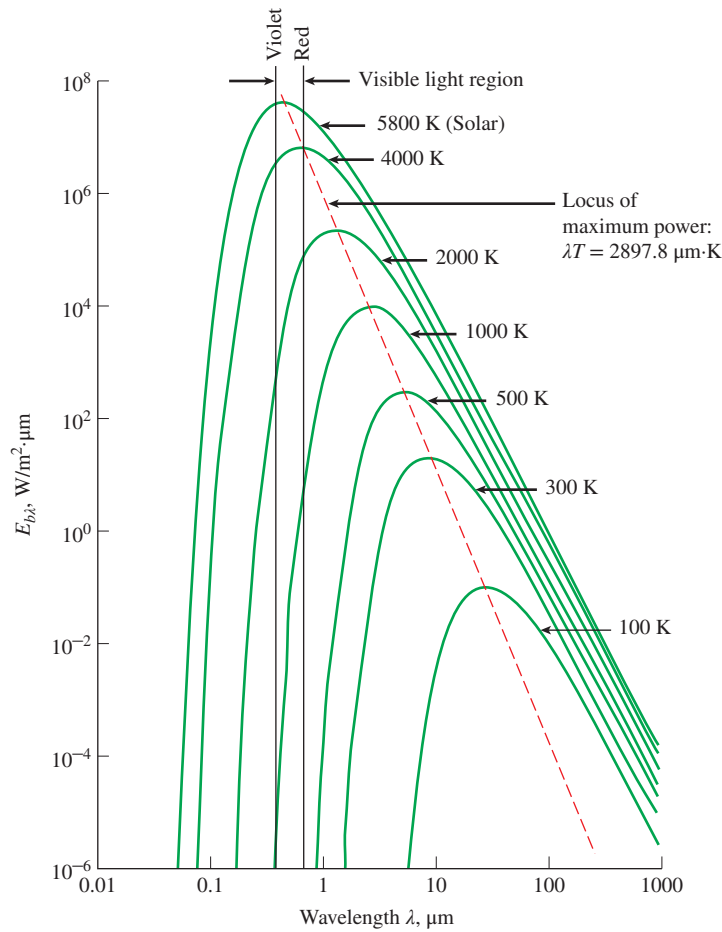
$$C_1 = 2\pi hc_0^2 = 3.74177 \times 10^8 \text{ W} \cdot \mu\text{m}^4/\text{m}^2$$

$$C_2 = hc_0/k = 1.43878 \times 10^4 \mu\text{m} \cdot \text{K}$$

Also, T is the absolute temperature of the surface, λ is the wavelength of the radiation emitted, and $k = 1.38065 \times 10^{-23} \text{ J/K}$ is *Boltzmann's constant*. This relation is valid for a surface in a *vacuum* or a *gas*. For other media, it needs to be modified by replacing C_1 with C_1/n^2 , where n is the index of refraction of the medium. Note that the term *spectral* indicates dependence on wavelength.

The variation of the spectral blackbody emissive power with wavelength is plotted in Fig. 12–9 for selected temperatures. Several observations can be made from this figure:

1. The emitted radiation is a continuous function of *wavelength*. At any specified temperature, it increases with wavelength, reaches a peak, and then decreases with increasing wavelength.
2. At any wavelength, the amount of emitted radiation *increases* with increasing temperature.
3. As temperature increases, the curves shift to the left to the shorter-wavelength region. Consequently, a larger fraction of the radiation is emitted at *shorter wavelengths* at higher temperatures.
4. The radiation emitted by the *sun*, which is considered to be a blackbody at 5780 K (or roughly at 5800 K), reaches its peak in the visible region of the spectrum. Therefore, the sun is in tune with our eyes. On the other hand, surfaces at $T \leq 800 \text{ K}$ emit almost entirely in the infrared region and thus are not visible to the eye unless they reflect light coming from other sources.

**FIGURE 12-9**

The variation of the blackbody emissive power with wavelength for several temperatures.

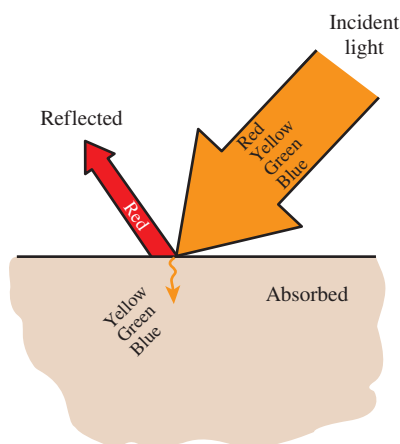
As the temperature increases, the peak of the curve in Fig. 12-9 shifts toward shorter wavelengths. The wavelength at which the peak occurs for a specified temperature is given by **Wien's displacement law** as

$$(\lambda T)_{\text{max power}} = 2897.8 \mu\text{m}\cdot\text{K} \quad (12-5)$$

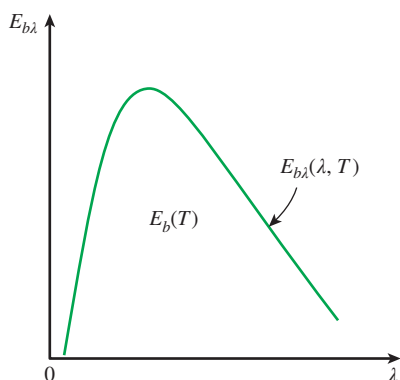
This relation was originally developed by Willy Wien in 1894 using classical thermodynamics, but it can also be obtained by differentiating Eq. 12-4 with respect to λ while holding T constant and setting the result equal to zero. A plot of Wien's displacement law, which is the locus of the peaks of the radiation emission curves, is also given in Fig. 12-9.

The peak of the solar radiation, for example, occurs at $\lambda = 2897.8/5780 = 0.50 \mu\text{m}$, which is near the middle of the visible range. The peak of the radiation emitted by a surface at room temperature ($T = 298 \text{ K}$) occurs at $9.72 \mu\text{m}$, which is well into the infrared region of the spectrum.

An electrical resistance heater starts radiating heat soon after it is plugged in, and we can feel the emitted radiation energy by holding our hands against the heater. But this radiation is entirely in the infrared region and thus cannot

**FIGURE 12-10**

A surface that reflects red while absorbing the remaining parts of the incident light appears red to the eye.

**FIGURE 12-11**

On an $E_{b\lambda}$ - λ chart, the area under a curve for a given temperature represents the total radiation energy emitted by a blackbody at that temperature.

be sensed by our eyes. The heater will appear dull red when its temperature reaches about 1000 K, since it starts emitting a detectable amount (about $1 \text{ W/m}^2 \cdot \mu\text{m}$) of visible red radiation at that temperature. As the temperature rises even more, the heater appears bright red and is said to be *red hot*. When the temperature reaches about 1500 K, the heater emits enough radiation in the entire visible range of the spectrum to appear almost *white* to the eye, and it is called *white hot*.

Although it cannot be sensed directly by the human eye, infrared radiation can be detected by infrared cameras, which transmit the information to microprocessors to display visual images of objects at night. *Rattlesnakes* can sense the infrared radiation or the “body heat” coming off warm-blooded animals, and thus they can see at night without using any instruments. Similarly, honeybees are sensitive to ultraviolet radiation. A surface that reflects all of the light appears *white*, while a surface that absorbs all of the light incident on it appears black. (Then how do we see a black surface?)

It should be clear from this discussion that the color of an object is not due to emission, which is primarily in the infrared region, unless the surface temperature of the object exceeds about 1000 K. Instead, the color of a surface depends on the absorption and reflection characteristics of the surface and is due to selective absorption and reflection of the incident visible radiation coming from a light source such as the sun or an incandescent lightbulb. A piece of clothing containing a pigment that reflects red while absorbing the remaining parts of the incident light appears “red” to the eye (Fig. 12-10). Leaves appear “green” because their cells contain the pigment chlorophyll, which strongly reflects green while absorbing other colors.

It is left as an exercise to show that integration of the *spectral* blackbody emissive power $E_{b\lambda}$ over the entire wavelength spectrum gives the *total* blackbody emissive power E_b :

$$E_b(T) = \int_0^\infty E_{b\lambda}(\lambda, T) d\lambda = \sigma T^4 \quad (\text{W/m}^2) \quad (12-6)$$

Thus, we obtained the Stefan–Boltzmann law (Eq. 12-3) by integrating Planck’s law (Eq. 12-4) over all wavelengths. Note that on an $E_{b\lambda}$ - λ chart, $E_{b\lambda}$ corresponds to any value on the curve, whereas E_b corresponds to the area under the entire curve for a specified temperature (Fig. 12-11). Also, the term *total* means “integrated over all wavelengths.”

EXAMPLE 12-1 Radiation Emission from a Black Ball

Consider a 20-cm-diameter spherical ball at 800 K suspended in air as shown in Fig. 12-12. Assuming the ball closely approximates a blackbody, determine (a) the total blackbody emissive power, (b) the total amount of radiation emitted by the ball in 5 min, (c) the spectral blackbody emissive power at a wavelength of $3 \mu\text{m}$, and (d) the wavelength for the maximum spectral blackbody emissive power at the ball surface temperature of 800 K.

SOLUTION An isothermal sphere is suspended in air. The total blackbody emissive power, the total radiation emitted in 5 min, the spectral blackbody emissive power at $3 \mu\text{m}$, and the wavelength for the maximum spectral blackbody emissive power at the ball surface temperature of 800 K are to be determined.

Assumptions 1 The ball behaves as a blackbody. 2 Uniform surface temperature.

Analysis (a) The total blackbody emissive power is determined from the Stefan–Boltzmann law to be

$$E_b = \sigma T^4 = (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(800 \text{ K})^4 = \mathbf{23.2 \text{ kW/m}^2}$$

That is, the ball emits 23.2 kJ of energy in the form of electromagnetic radiation per second per m² of the surface area of the ball.

(b) The total amount of radiation energy emitted from the entire ball in 5 min is determined by multiplying the blackbody emissive power obtained above by the total surface area of the ball and the given time interval:

$$A_s = \pi D^2 = \pi(0.2 \text{ m})^2 = 0.1257 \text{ m}^2$$

$$\Delta t = (5 \text{ min}) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) = 300 \text{ s}$$

$$\begin{aligned} Q_{\text{rad}} &= E_b A_s \Delta t = (23.2 \text{ kW/m}^2)(0.1257 \text{ m}^2)(300 \text{ s}) \left(\frac{1 \text{ kJ}}{1 \text{ kW} \cdot \text{s}} \right) \\ &= \mathbf{875 \text{ kJ}} \end{aligned}$$

That is, the ball loses 875 kJ of its internal energy in the form of electromagnetic waves to the surroundings in 5 min, which is enough energy to heat 20 kg of water from 0°C to 100°C. Note that the surface temperature of the ball cannot remain constant at 800 K unless there is an equal amount of energy flow to the surface from the surroundings or from the interior regions of the ball through some mechanisms such as chemical or nuclear reactions.

(c) The spectral blackbody emissive power at a wavelength of 3 μm is determined from Planck's distribution law to be

$$\begin{aligned} E_{b\lambda} &= \frac{C_1}{\lambda^5 \left[\exp \left(\frac{C_2}{\lambda T} \right) - 1 \right]} = \frac{3.74177 \times 10^8 \text{ W} \cdot \mu\text{m}^4/\text{m}^2}{(3 \mu\text{m})^5 \left[\exp \left(\frac{1.43878 \times 10^4 \mu\text{m} \cdot \text{K}}{(3 \mu\text{m})(800 \text{ K})} \right) - 1 \right]} \\ &= \mathbf{3846 \text{ W/m}^2 \cdot \mu\text{m}} \end{aligned}$$

(d) The wavelength for the maximum spectral blackbody emissive power at the ball surface temperature of 800 K is determined from Wien's displacement law to be

$$(\lambda T)_{\text{max power}} = 2897.8 \mu\text{m} \cdot \text{K}$$

So,

$$\lambda_{\text{max power}} = \frac{2897.8 \mu\text{m} \cdot \text{K}}{800 \text{ K}} = \mathbf{3.62 \mu\text{m}}$$

Discussion The maximum spectral blackbody emissive power at the $\lambda_{\text{max power}} = 3.62 \mu\text{m}$ is 4229 W/m²·μm. This is about 10 percent higher than the value calculated in part (c) for a wavelength of 3 μm.

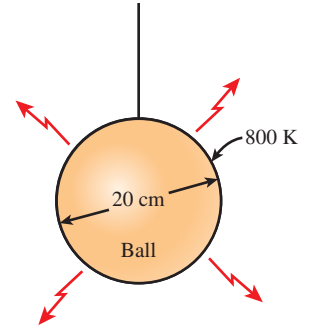


FIGURE 12-12

The spherical ball considered in Example 12-1.

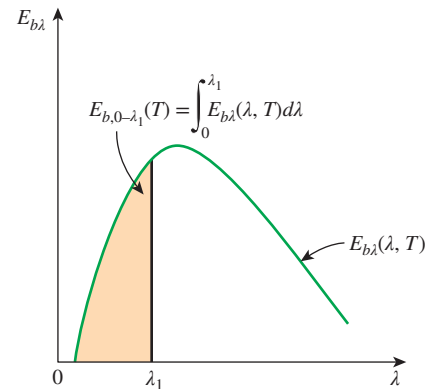


FIGURE 12-13

On an $E_{b\lambda}$ – λ chart, the area under the curve to the left of the $\lambda = \lambda_1$ line represents the radiation energy emitted by a blackbody in the wavelength range 0– λ_1 for the given temperature.

The Stefan–Boltzmann law $E_b(T) = \sigma T^4$ gives the *total* radiation emitted by a blackbody at all wavelengths from $\lambda = 0$ to $\lambda = \infty$. But we are often interested in the amount of radiation emitted over *some wavelength band*. For example, an incandescent lightbulb is judged on the basis of the radiation it emits in the visible range rather than the radiation it emits at all wavelengths.

The radiation energy emitted by a blackbody per unit area over a wavelength band from $\lambda = 0$ to λ is determined from (Fig. 12-13)

$$E_{b,0-\lambda}(T) = \int_0^{\lambda} E_{b\lambda}(\lambda, T) d\lambda \quad (\text{W/m}^2) \quad (12-7)$$

It looks like we can determine $E_{b,0-\lambda}$ by substituting the $E_{b\lambda}$ relation from Eq. 12-4 and performing this integration. But it turns out that this integration does not have a simple closed-form solution, and performing a numerical integration each time we need a value of $E_{b,0-\lambda}$ is not practical. Therefore, we define a dimensionless quantity f_{λ} called the **blackbody radiation function** as

$$f_{\lambda}(T) = \frac{\int_0^{\lambda} E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4} \quad (12-8)$$

The function f_{λ} represents *the fraction of radiation emitted from a blackbody at temperature T in the wavelength band from $\lambda = 0$ to λ* . In reference to Fig. 12-13, f_{λ} can be interpreted as the ratio of the shaded region (radiation energy emitted by the blackbody in the wavelength range $0 - \lambda_1$) over the total area under the curve (total radiation energy emitted by the blackbody in the wavelength range $0 - \infty$). The values of f_{λ} are listed in Table 12-2 as a function of λT , where λ is in μm and T is in K.

TABLE 12-2Blackbody radiation functions f_{λ}

$\lambda T, \mu\text{m}\cdot\text{K}$	f_{λ}	$\lambda T, \mu\text{m}\cdot\text{K}$	f_{λ}
200	0.000000	6200	0.754140
400	0.000000	6400	0.769234
600	0.000000	6600	0.783199
800	0.000016	6800	0.796129
1000	0.000321	7000	0.808109
1200	0.002134	7200	0.819217
1400	0.007790	7400	0.829527
1600	0.019718	7600	0.839102
1800	0.039341	7800	0.848005
2000	0.066728	8000	0.856288
2200	0.100888	8500	0.874608
2400	0.140256	9000	0.890029
2600	0.183120	9500	0.903085
2800	0.227897	10,000	0.914199
3000	0.273232	10,500	0.923710
3200	0.318102	11,000	0.931890
3400	0.361735	11,500	0.939959
3600	0.403607	12,000	0.945098
3800	0.443382	13,000	0.955139
4000	0.480877	14,000	0.962898
4200	0.516014	15,000	0.969981
4400	0.548796	16,000	0.973814
4600	0.579280	18,000	0.980860
4800	0.607559	20,000	0.985602
5000	0.633747	25,000	0.992215
5200	0.658970	30,000	0.995340
5400	0.680360	40,000	0.997967
5600	0.701046	50,000	0.998953
5800	0.720158	75,000	0.999713
6000	0.737818	100,000	0.999905

The fraction of radiation energy emitted by a blackbody at temperature T over a finite wavelength band from $\lambda = \lambda_1$ to $\lambda = \lambda_2$ is determined from (Fig. 12–14)

$$f_{\lambda_1-\lambda_2}(T) = \frac{\int_0^{\lambda_2} E_{b\lambda}(\lambda, T) d\lambda - \int_0^{\lambda_1} E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4} = f_{\lambda_2}(T) - f_{\lambda_1}(T) \quad (12-9)$$

where $f_{\lambda_1}(T)$ and $f_{\lambda_2}(T)$ are blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$, respectively.

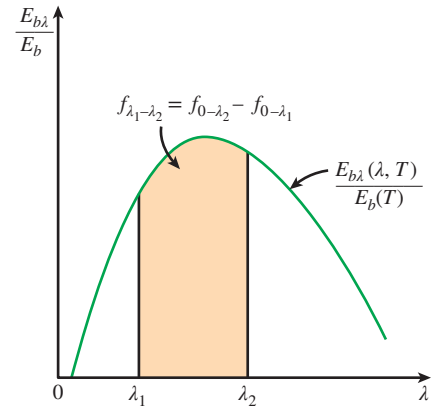


FIGURE 12-14

Graphical representation of the fraction of radiation emitted in the wavelength band from λ_1 to λ_2 .

EXAMPLE 12-2 Light Emitted by the Sun and by a Lightbulb

Charge-coupled device (CCD) image sensors, which are common in modern digital cameras, respond differently to light sources with different spectral distributions. Daylight and incandescent light sources may be approximated as blackbodies at the effective surface temperatures of 5800 K and 2800 K, respectively. Determine the fraction of radiation emitted within the visible spectrum wavelengths, from 0.40 μm (violet) to 0.76 μm (red), for each of the lighting sources.

SOLUTION For specified blackbody temperatures, the fraction of visible radiation emitted by the sun and the filament of an incandescent lightbulb are to be determined.

Assumptions The sun and the incandescent light filament behave as blackbodies.

Analysis The visible range of the electromagnetic spectrum extends from $\lambda_1 = 0.40 \mu\text{m}$ to $\lambda_2 = 0.76 \mu\text{m}$. For the sun at $T = 5800 \text{ K}$, the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$ are determined from Table 12–2 to be (Fig. 12–15)

$$\lambda_1 T = (0.40 \mu\text{m})(5800 \text{ K}) = 2320 \mu\text{m} \cdot \text{K} \rightarrow f_{\lambda_1, \text{daylight}} = 0.124509$$

$$\lambda_2 T = (0.76 \mu\text{m})(5800 \text{ K}) = 4408 \mu\text{m} \cdot \text{K} \rightarrow f_{\lambda_2, \text{daylight}} = 0.550015$$

Then the fraction of visible radiation emitted by the sun becomes

$$f_{\lambda_1-\lambda_2, \text{daylight}} = 0.550015 - 0.124509 = \mathbf{0.426} \text{ or } \mathbf{42.6 \text{ percent}}$$

For an incandescent lightbulb at $T = 2800 \text{ K}$, the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$ are determined from Table 12–2 to be

$$\lambda_1 T = (0.40 \mu\text{m})(2800 \text{ K}) = 1120 \mu\text{m} \cdot \text{K} \rightarrow f_{\lambda_1, \text{incandescent}} = 0.001409$$

$$\lambda_2 T = (0.76 \mu\text{m})(2800 \text{ K}) = 2128 \mu\text{m} \cdot \text{K} \rightarrow f_{\lambda_2, \text{incandescent}} = 0.088590$$

Then the fraction of radiation the lightbulb emits in the visible range becomes

$$f_{\lambda_1-\lambda_2, \text{incandescent}} = 0.088590 - 0.001409 = \mathbf{0.087} \text{ or } \mathbf{8.7 \text{ percent}}$$

Discussion Note that almost half of the radiation emitted by the sun is in the visible range, and thus the sun is a very efficient light source. But less than 10 percent of the radiation emitted by the incandescent lightbulb is in the form of visible light, and thus incandescent lightbulbs are inefficient as light sources. Consequently, they are being replaced by the highly efficient fluorescent and LED light sources.

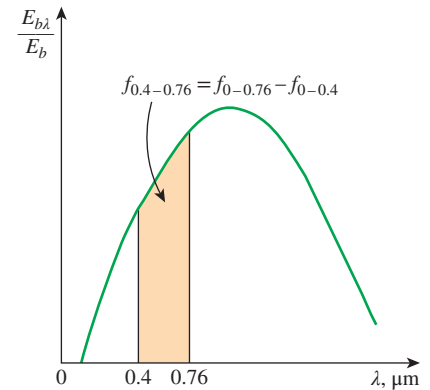
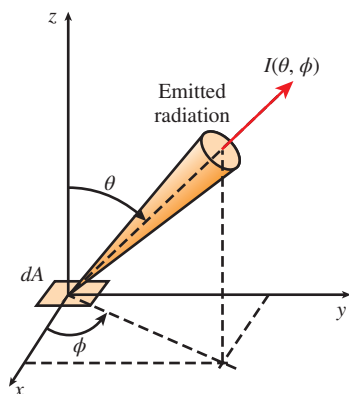
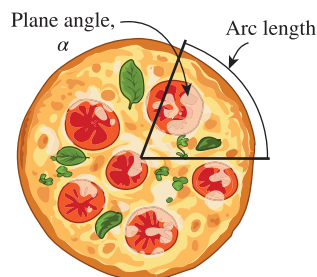
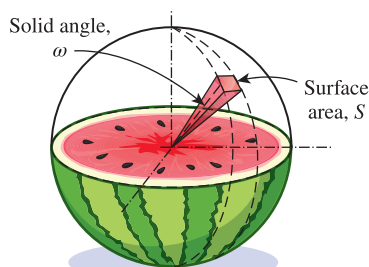


FIGURE 12-15

Graphical representation of the fraction of radiation emitted in the visible range in Example 12–2.

**FIGURE 12-16**

Radiation intensity is used to describe the variation of radiation energy with direction.

A slice of pizza of plane angle α A slice of watermelon of solid angle ω **FIGURE 12-17**

Describing the size of a slice of pizza by a plane angle and the size of a watermelon slice by a solid angle.

12-4 ■ RADIATION INTENSITY

Radiation is emitted by all parts of a plane surface in all directions into the hemisphere above the surface, and the directional distribution of emitted (or incident) radiation is usually not uniform. Therefore, we need a quantity that describes the magnitude of radiation emitted (or incident) in a specified direction in space. This quantity is *radiation intensity*, denoted by I . Before we can describe a directional quantity, we need to specify direction in space. The direction of radiation passing through a point is best described in spherical coordinates in terms of the zenith angle θ and the azimuth angle ϕ , as shown in Fig. 12-16. Radiation intensity is used to describe how the emitted radiation varies with the zenith and azimuth angles.

If all surfaces emitted radiation uniformly in all directions, the *emissive power* would be sufficient to quantify radiation, and we would not need to deal with intensity. The radiation emitted by a blackbody per unit normal area is the same in all directions, and thus there is no directional dependence. But this is not the case for real surfaces. Before we define intensity, we need to quantify the size of an opening in space.

Solid Angle

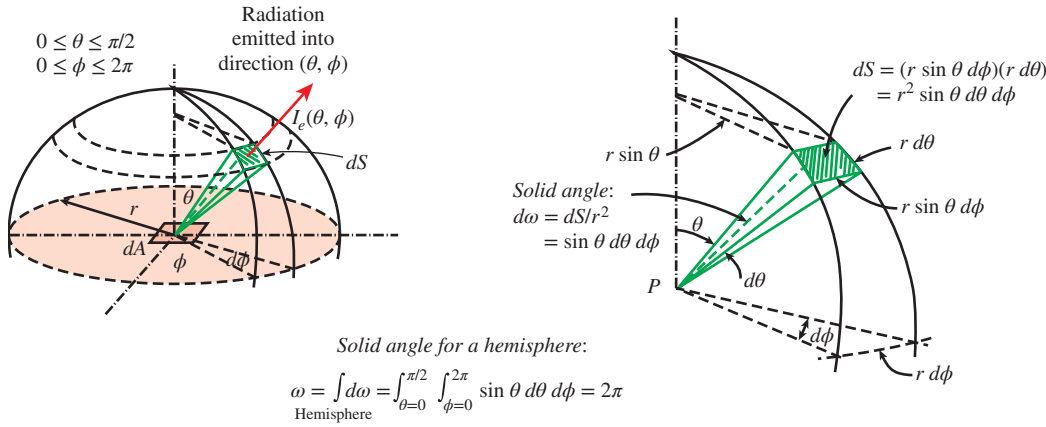
Let us try to quantify the size of a slice of pizza. One way of doing that is to specify the arc length of the outer edge of the slice and to form the slice by connecting the end points of the arc to the center. A more general approach is to specify the angle of the slice at the center, as shown in Fig. 12-17. An angle of 90° (or $\pi/2$ radians), for example, always represents a quarter pizza, no matter what the radius is. For a circle of unit radius, the length of an arc is equivalent in magnitude to the *plane angle* it subtends (both are 2π for a complete circle of radius $r = 1$).

Now consider a watermelon, and let us attempt to quantify the size of a slice. Again we can do it by specifying the outer surface area of the slice (the green part) or by working with angles for generality. Connecting all points at the edges of the slice to the center in this case will form a three-dimensional body (like a cone whose tip is at the center), and thus the angle at the center in this case is properly called the **solid angle**. The solid angle is denoted by ω , and its unit is the *steradian* (sr). In analogy to plane angle, we can say that *the area of a surface on a sphere of unit radius is equivalent in magnitude to the solid angle it subtends* (both are 4π for a sphere of radius $r = 1$).

This can be shown easily by considering a differential surface area on a sphere $dS = r^2 \sin \theta d\theta d\phi$, as shown in Fig. 12-18, and integrating it from $\theta = 0$ to $\theta = \pi$, and from $\phi = 0$ to $\phi = 2\pi$. We get

$$S = \int_{\text{sphere}} dS = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin \theta d\theta d\phi = 2\pi r^2 \int_{\theta=0}^{\pi} \sin \theta d\theta = 4\pi r^2 \quad (12-10)$$

which is the formula for the area of a sphere. For $r = 1$ it reduces to $S = 4\pi$, and thus the solid angle associated with a sphere is $\omega = 4\pi$ sr. For a hemisphere, which is more relevant to radiation emitted or received by a surface, it is $\omega = 2\pi$ sr.

**FIGURE 12-18**

The emission of radiation from a differential surface element into the surrounding hemispherical space through a differential solid angle.

The differential solid angle $d\omega$ subtended by a differential area dS on a sphere of radius r can be expressed as

$$d\omega = \frac{dS}{r^2} = \sin \theta \, d\theta \, d\phi \quad (12-11)$$

Note that the area dS is normal to the direction of viewing since dS is viewed from the center of the sphere. In general, the differential solid angle $d\omega$ subtended by a differential surface area dA when viewed from a point at a distance r from dA is expressed as

$$d\omega = \frac{dA_n}{r^2} = \frac{dA \cos \alpha}{r^2} \quad (12-12)$$

where α is the angle between the normal of the surface and the direction of viewing, and thus $dA_n = dA \cos \alpha$ is the normal (or projected) area to the direction of viewing.

Small surfaces viewed from relatively large distances can approximately be treated as differential areas in solid angle calculations. For example, the solid angle subtended by a 5 cm^2 plane surface when viewed from a point at a distance of 80 cm along the normal of the surface is

$$\omega \cong \frac{A_n}{r^2} = \frac{5 \text{ cm}^2}{(80 \text{ cm})^2} = 7.81 \times 10^{-4} \text{ sr}$$

If the surface is tilted so that the normal of the surface makes an angle of $\alpha = 60^\circ$ with the line connecting the point of viewing to the center of the surface, the projected area would be $dA_n = dA \cos \alpha = (5 \text{ cm}^2) \cos 60^\circ = 2.5 \text{ cm}^2$, and the solid angle in this case would be half of the value just determined.

Intensity of Emitted Radiation

Consider the emission of radiation by a differential area element dA of a surface, as shown in Fig. 12-18. Radiation is emitted in all directions into the hemispherical space, and the radiation streaming through the surface area dS is proportional to the solid angle $d\omega$ subtended by dS . It is also proportional to the radiating area dA as seen by an observer on dS , which varies from a maximum of dA when dS is at the top directly above dA ($\theta = 0^\circ$) to a minimum of zero when dS is at the bottom ($\theta = 90^\circ$). Therefore, the effective area of dA for emission in the direction of θ is the projection of dA on a plane normal to θ ,

which is $dA \cos \theta$. Radiation intensity in a given direction is based on a unit area *normal* to that direction to provide a common basis for the comparison of radiation emitted in different directions.

The **radiation intensity** for emitted radiation $I_e(\theta, \phi)$ is defined as *the rate at which radiation energy $d\dot{Q}_e$ is emitted in the (θ, ϕ) direction per unit area normal to this direction and per unit solid angle about this direction*. That is,

$$I_e(\theta, \phi) = \frac{d\dot{Q}_e}{dA \cos \theta \cdot d\omega} = \frac{d\dot{Q}_e}{dA \cos \theta \sin \theta d\theta d\phi} \quad (\text{W/m}^2 \cdot \text{sr}) \quad (12-13)$$

The *radiation flux* for emitted radiation is the **emissive power** E (the rate at which radiation energy is emitted per unit area of the emitting surface), which can be expressed in differential form as

$$dE = \frac{d\dot{Q}_e}{dA} = I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi \quad (12-14)$$

Noting that the hemisphere above the surface intercepts all the radiation rays emitted by the surface, the emissive power from the surface into the hemisphere surrounding it can be determined by integration as

$$E = \int_{\text{hemisphere}} dE = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi \quad (\text{W/m}^2) \quad (12-15)$$

The intensity of radiation emitted by a surface, in general, varies with direction (especially with the zenith angle θ). But many surfaces in practice can be approximated as being diffuse. For a *diffusely emitting* surface, the intensity of the emitted radiation is independent of direction and thus $I_e = \text{constant}$.

Noting that $\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} \cos \theta \sin \theta d\theta d\phi = \pi$, the emissive power relation in Eq. 12-15 reduces in this case to

$$\text{Diffusely emitting surface:} \quad E = \pi I_e \quad (\text{W/m}^2) \quad (12-16)$$

Note that the factor in Eq. 12-16 is π . You might have expected it to be 2π since intensity is radiation energy per unit solid angle, and the solid angle associated with a hemisphere is 2π . The reason for the factor being π is that the emissive power is based on the *actual* surface area, whereas the intensity is based on the *projected* area (and thus the factor $\cos \theta$ that accompanies it), as shown in Fig. 12-19.

For a *blackbody*, which is a diffuse emitter, Eq. 12-16 can be expressed as

$$\text{Blackbody:} \quad E_b = \pi I_b \quad (12-17)$$

where $E_b = \sigma T^4$ is the blackbody emissive power. Therefore, the intensity of the radiation emitted by a blackbody at absolute temperature T is

$$\text{Blackbody:} \quad I_b(T) = \frac{E_b(T)}{\pi} = \frac{\sigma T^4}{\pi} \quad (\text{W/m}^2 \cdot \text{sr}) \quad (12-18)$$

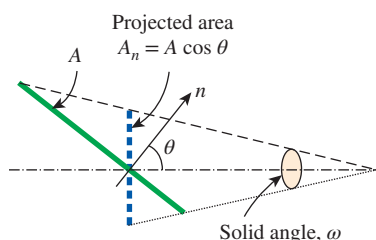


FIGURE 12-19

Radiation intensity is based on projected area, and thus the calculation of radiation emission from a surface involves the projection of the surface.

Incident Radiation

All surfaces emit radiation, but they also receive radiation emitted or reflected by other surfaces. The intensity of incident radiation $I_i(\theta, \phi)$ is defined as *the rate at which radiation energy dG is incident from the (θ, ϕ) direction per unit area of the receiving surface normal to this direction and per unit solid angle about this direction* (Fig. 12–20). Here θ is the angle between the direction of incident radiation and the normal of the surface.

The radiation flux incident on a surface from *all directions* is called **irradiation** G and is expressed as

$$G = \int_{\text{hemisphere}} dG = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_i(\theta, \phi) \cos \theta \sin \theta d\theta d\phi \quad (\text{W/m}^2) \quad (12-19)$$

Therefore, irradiation represents the rate at which radiation energy is incident on a surface per unit area of the surface. When the incident radiation is diffuse and thus $I_i = \text{constant}$, Eq. 12–19 reduces to

$$\text{Diffusely incident radiation:} \quad G = \pi I_i \quad (\text{W/m}^2) \quad (12-20)$$

Again note that irradiation is based on the *actual* surface area (and thus the factor $\cos \theta$), whereas the intensity of incident radiation is based on the *projected* area.

Radiosity

Surfaces emit radiation as well as reflecting it, and thus the radiation leaving a surface consists of emitted and reflected components, as shown in Fig. 12–21. The calculation of radiation heat transfer between surfaces involves the *total* radiation energy streaming away from a surface, with no regard for its origin. Thus, we need to define a quantity that represents *the rate at which radiation energy leaves a unit area of a surface in all directions*. This quantity is called the **radiosity** J and is expressed as

$$J = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{e+r}(\theta, \phi) \cos \theta \sin \theta d\theta d\phi \quad (\text{W/m}^2) \quad (12-21)$$

where I_{e+r} is the sum of the emitted and reflected intensities. For a surface that is both a diffuse emitter and a diffuse reflector, $I_{e+r} = \text{constant}$, and the radiosity relation reduces to

$$\text{Diffuse emitter and reflector:} \quad J = \pi I_{e+r} \quad (\text{W/m}^2) \quad (12-22)$$

For a blackbody, radiosity J is equivalent to the emissive power E_b since a blackbody absorbs all radiation incident on it, and there is no reflected component in radiosity.

Spectral Quantities

So far we have considered *total* radiation quantities (quantities integrated over all wavelengths), and we have made no reference to wavelength dependence. This lumped approach is adequate for many radiation problems encountered

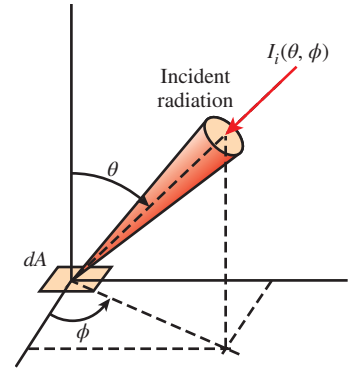


FIGURE 12-20

Radiation incident on a surface in the direction (θ, ϕ) .

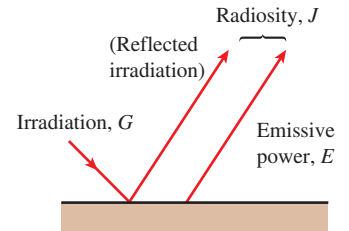


FIGURE 12-21

The three kinds of radiation flux (in W/m^2): emissive power, irradiation, and radiosity.

in practice. But sometimes it is necessary to consider the variation of radiation with wavelength as well as direction and to express quantities at a certain wavelength λ or per unit wavelength interval about λ . Such quantities are referred to as *spectral* quantities to draw attention to wavelength dependence. The modifier “spectral” is used to indicate “at a given wavelength.”

The *spectral radiation intensity* $I_{\lambda}(\lambda, \theta, \phi)$, for example, is simply the total radiation intensity $I(\theta, \phi)$ per unit wavelength interval about λ . The **spectral intensity** for emitted radiation $I_{\lambda,e}(\lambda, \theta, \phi)$ can be defined as *the rate at which radiation energy $d\dot{Q}_e$ is emitted at the wavelength λ in the (θ, ϕ) direction per unit area normal to this direction, per unit solid angle about this direction, and it can be expressed as*

$$I_{\lambda,e}(\lambda, \theta, \phi) = \frac{d\dot{Q}_e}{dA \cos \theta \cdot d\omega \cdot d\lambda} \quad (\text{W/m}^2 \cdot \text{sr} \cdot \mu\text{m}) \quad (12-23)$$

Then the *spectral emissive power* becomes

$$E_{\lambda} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda,e}(\lambda, \theta, \phi) \cos \theta \sin \theta \, d\theta \, d\phi \quad (\text{W/m}^2) \quad (12-24)$$

Similar relations can be obtained for spectral irradiation G_{λ} and spectral radiosity J_{λ} by replacing $I_{\lambda,e}$ in this equation with $I_{\lambda,i}$ and $I_{\lambda,e+r}$, respectively.

When the variation of spectral radiation intensity I_{λ} with wavelength λ is known, the total radiation intensity I for emitted, incident, and emitted + reflected radiation can be determined by integration over the entire wavelength spectrum as (Fig. 12-22)

$$I_e = \int_0^{\infty} I_{\lambda,e} \, d\lambda, \quad I_i = \int_0^{\infty} I_{\lambda,i} \, d\lambda, \quad \text{and} \quad I_{e+r} = \int_0^{\infty} I_{\lambda,e+r} \, d\lambda \quad (12-25)$$

These intensities can then be used in Eqs. 12-15, 12-19, and 12-21 to determine the emissive power E , irradiation G , and radiosity J , respectively.

Similarly, when the variations of spectral radiation fluxes E_{λ} , G_{λ} , and J_{λ} with wavelength λ are known, the total radiation fluxes can be determined by integration over the entire wavelength spectrum as

$$E = \int_0^{\infty} E_{\lambda} \, d\lambda, \quad G = \int_0^{\infty} G_{\lambda} \, d\lambda, \quad \text{and} \quad J = \int_0^{\infty} J_{\lambda} \, d\lambda \quad (12-26)$$

When the surfaces and the incident radiation are *diffuse*, the spectral radiation fluxes are related to spectral intensities as

$$E_{\lambda} = \pi I_{\lambda,e}, \quad G_{\lambda} = \pi I_{\lambda,i}, \quad \text{and} \quad J_{\lambda} = \pi I_{\lambda,e+r} \quad (12-27)$$

Note that the relations for spectral and total radiation quantities are of the same form.

The spectral intensity of radiation emitted by a blackbody at a thermodynamic temperature T at a wavelength λ was determined by Max Planck and is expressed as

$$I_{b\lambda}(\lambda, T) = \frac{2hc_0^2}{\lambda^5 [\exp(hc_0/\lambda kT) - 1]} \quad (\text{W/m}^2 \cdot \text{sr} \cdot \mu\text{m}) \quad (12-28)$$

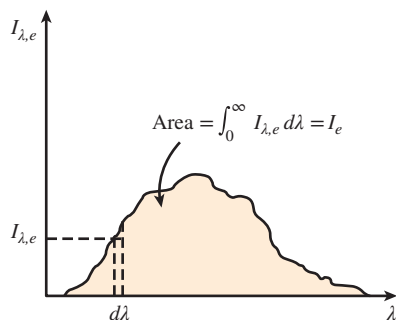


FIGURE 12-22

Integration of a “spectral” quantity for all wavelengths gives the “total” quantity.

where $h = 6.626069 \times 10^{-34}$ J·s is the Planck constant, $k = 1.38065 \times 10^{-23}$ J/K is the Boltzmann constant, and $c_0 = 2.9979 \times 10^8$ m/s is the speed of light in a vacuum. Then the spectral blackbody emissive power is, from Eq. 12-27,

$$E_{b\lambda}(\lambda, T) = \pi I_{b\lambda}(\lambda, T) \quad (12-29)$$

A simplified relation for $E_{b\lambda}$ is given by Eq. 12-4.

EXAMPLE 12-3 Radiation Incident on a Small Surface

A small surface of area $A_1 = 3 \text{ cm}^2$ emits radiation as a blackbody at $T_1 = 600 \text{ K}$. Part of the radiation emitted by A_1 strikes another small surface of area $A_2 = 5 \text{ cm}^2$ oriented as shown in Fig. 12-23. Determine the solid angle subtended by A_2 when viewed from A_1 , and the rate at which radiation emitted by A_1 strikes A_2 .

SOLUTION A surface is subjected to radiation emitted by another surface. The solid angle subtended and the rate at which emitted radiation is received are to be determined.

Assumptions 1 Surface A_1 emits diffusely as a blackbody. 2 Both A_1 and A_2 can be approximated as differential surfaces since both are very small compared to the square of the distance between them.

Analysis Approximating both A_1 and A_2 as differential surfaces, the solid angle subtended by A_2 when viewed from A_1 can be determined from Eq. 12-12 to be

$$\omega_{2-1} \cong \frac{A_{n,2}}{r^2} = \frac{A_2 \cos \theta_2}{r^2} = \frac{(5 \text{ cm}^2) \cos 40^\circ}{(75 \text{ cm})^2} = \mathbf{6.81 \times 10^{-4} \text{ sr}}$$

since the normal of A_2 makes 40° with the direction of viewing. Note that the solid angle subtended by A_2 would be maximum if A_2 were positioned normal to the direction of viewing. Also, the point of viewing on A_1 is taken to be a point in the middle, but it can be any point since A_1 is assumed to be very small.

The radiation emitted by A_1 that strikes A_2 is equivalent to the radiation emitted by A_1 through the solid angle ω_{2-1} . The intensity of the radiation emitted by A_1 is

$$I_1 = \frac{E_b(T_1)}{\pi} = \frac{\sigma T_1^4}{\pi} = \frac{(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(600 \text{ K})^4}{\pi} = 2339 \text{ W/m}^2 \cdot \text{sr}$$

This value of intensity is the same in all directions since a blackbody is a diffuse emitter. Intensity represents the rate of radiation emission per unit area normal to the direction of emission per unit solid angle. Therefore, the rate of radiation energy emitted by A_1 in the direction of θ_1 through the solid angle ω_{2-1} is determined by multiplying I_1 by the area of A_1 normal to θ_1 and the solid angle ω_{2-1} . That is,

$$\begin{aligned} \dot{Q}_{1-2} &= I_1 (A_1 \cos \theta_1) \omega_{2-1} \\ &= (2339 \text{ W/m}^2 \cdot \text{sr})(3 \times 10^{-4} \cos 55^\circ \text{ m}^2)(6.81 \times 10^{-4} \text{ sr}) \\ &= \mathbf{2.74 \times 10^{-4} \text{ W}} \end{aligned}$$

Therefore, the radiation emitted from surface A_1 will strike surface A_2 at a rate of $2.74 \times 10^{-4} \text{ W}$.

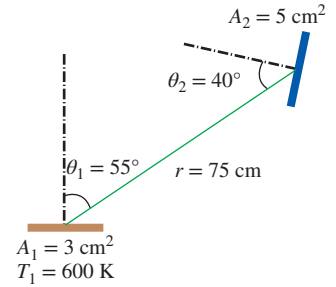


FIGURE 12-23
Schematic for Example 12-3.

Discussion The total rate of radiation emission from surface A_1 is $\dot{Q}_e = A_1 \sigma T_1^4 = 2.204 \text{ W}$. Therefore, the fraction of emitted radiation that strikes A_2 is $2.74 \times 10^{-4} / 2.204 = 0.00012$ (or 0.012 percent). Noting that the solid angle associated with a hemisphere is 2π , the fraction of the solid angle subtended by A_2 is $6.81 \times 10^{-4} / (2\pi) = 0.000108$ (or 0.0108 percent), which is 0.9 times the fraction of emitted radiation. Therefore, the fraction of the solid angle a surface occupies does not represent the fraction of radiation energy the surface will receive even when the intensity of emitted radiation is constant. This is because radiation energy emitted by a surface in a given direction is proportional to the *projected area* of the surface in that direction and reduces from a maximum at $\theta = 0^\circ$ (the direction normal to surface) to zero at $\theta = 90^\circ$ (the direction parallel to surface).

12-5 ■ RADIATIVE PROPERTIES

Most materials encountered in practice, such as metals, wood, and bricks, are *opaque* to thermal radiation, and radiation is considered to be a *surface phenomenon* for such materials. That is, thermal radiation is emitted or absorbed within the first few microns of the surface, and thus we speak of radiative properties of *surfaces* for opaque materials.

Some other materials, such as glass and water, allow visible radiation to penetrate to considerable depths before any significant absorption takes place. Radiation through such *semitransparent* materials obviously cannot be considered to be a surface phenomenon since the entire volume of the material interacts with radiation. On the other hand, both glass and water are practically opaque to infrared radiation. Therefore, materials can exhibit different behavior at different wavelengths, and the dependence on wavelength is an important consideration in the study of radiative properties such as emissivity, absorptivity, reflectivity, and transmissivity of materials.

In Sec. 12-4, we defined a *blackbody* as a perfect emitter and absorber of radiation and said that no body can emit more radiation than a blackbody at the same temperature. Therefore, a blackbody can serve as a convenient *reference* in describing the emission and absorption characteristics of real surfaces.

Emissivity

The **emissivity** of a surface represents *the ratio of the radiation emitted by the surface at a given temperature to the radiation emitted by a blackbody at the same temperature*. The emissivity of a surface is denoted by ε , and it varies between zero and one, $0 \leq \varepsilon \leq 1$. Emissivity is a measure of how closely a real surface approximates a blackbody, for which $\varepsilon = 1$.

The emissivity of a real surface is not a constant. Rather, it varies with the *temperature* of the surface as well as the *wavelength* and the *direction* of the emitted radiation. Therefore, different emissivities can be defined for a surface, depending on the effects considered. The most elemental emissivity of a surface at a given temperature is the **spectral directional emissivity**, which is defined as the ratio of the intensity of radiation emitted by the surface at a specified wavelength in a specified direction to the intensity of radiation emitted by a blackbody at the same temperature at the same wavelength. That is,

$$\varepsilon_{\lambda, \theta}(\lambda, \theta, \phi, T) = \frac{I_{\lambda, e}(\lambda, \theta, \phi, T)}{I_{b\lambda}(\lambda, T)} \quad (12-30)$$

where the subscripts λ and θ are used to designate *spectral* and *directional* quantities, respectively. Note that blackbody radiation intensity is independent of direction, and thus it has no functional dependence on θ and ϕ .

The **total directional emissivity** is defined in a like manner by using total intensities (intensities integrated over all wavelengths) as

$$\varepsilon_\theta(\theta, \phi, T) = \frac{I_e(\theta, \phi, T)}{I_b(T)} \quad (12-31)$$

In practice, it is usually more convenient to work with radiation properties averaged over all directions, called *hemispherical properties*. Noting that the integral of the rate of radiation energy emitted at a specified wavelength per unit surface area over the entire hemisphere is *spectral emissive power*, the **spectral hemispherical emissivity** can be expressed as

$$\varepsilon_\lambda(\lambda, T) = \frac{E_\lambda(\lambda, T)}{E_{b\lambda}(\lambda, T)} \quad (12-32)$$

Note that the emissivity of a surface at a given wavelength can be different at different temperatures since the spectral distribution of emitted radiation (and thus the amount of radiation emitted at a given wavelength) changes with temperature.

Finally, the **total hemispherical emissivity** is defined in terms of the radiation energy emitted over all wavelengths in all directions as

$$\varepsilon(T) = \frac{E(T)}{E_b(T)} \quad (12-33)$$

Therefore, the total hemispherical emissivity (or simply the “average emissivity”) of a surface at a given temperature represents the ratio of the total radiation energy emitted by the surface to the radiation emitted by a blackbody of the same surface area at the same temperature.

Noting from Eqs. 12-26 and 12-32 that $E = \int_0^\infty E_\lambda d\lambda$ and $E_\lambda(\lambda, T) = \varepsilon_\lambda(\lambda, T) E_{b\lambda}(\lambda, T)$, the total hemispherical emissivity can also be expressed as

$$\varepsilon(T) = \frac{E(T)}{E_b(T)} = \frac{\int_0^\infty \varepsilon_\lambda(\lambda, T) E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4} \quad (12-34)$$

since $E_b(T) = \sigma T^4$. To perform this integration, we need to know the variation of spectral emissivity with wavelength at the specified temperature. The integrand is usually a complicated function, and the integration has to be performed numerically. However, the integration can be performed quite easily by dividing the spectrum into a sufficient number of *wavelength bands* and assuming the emissivity to remain constant over each band; that is, by expressing the function $\varepsilon_\lambda(\lambda, T)$ as a step function. This simplification offers great convenience for little sacrifice of accuracy, since it allows us to transform the integration into a summation in terms of blackbody emission functions.

As an example, consider the emissivity function plotted in Fig. 12-24. It seems like this function can be approximated reasonably well by a step function of the form

$$\varepsilon_\lambda = \begin{cases} \varepsilon_1 = \text{constant}, & 0 \leq \lambda < \lambda_1 \\ \varepsilon_2 = \text{constant}, & \lambda_1 \leq \lambda < \lambda_2 \\ \varepsilon_3 = \text{constant}, & \lambda_2 \leq \lambda < \infty \end{cases} \quad (12-35)$$

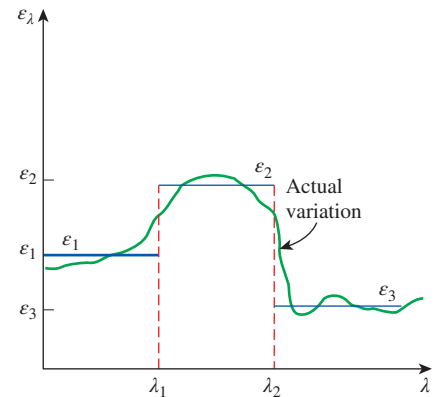


FIGURE 12-24

Approximating the actual variation of emissivity with wavelength by a step function.

Real surface:

$$\epsilon_\theta \neq \text{constant}$$

$$\epsilon_\lambda \neq \text{constant}$$

Diffuse surface:

$$\epsilon_\theta = \text{constant}$$

Gray surface:

$$\epsilon_\lambda = \text{constant}$$

Diffuse, gray surface:

$$\epsilon = \epsilon_\lambda = \epsilon_\theta = \text{constant}$$

FIGURE 12–25

The effect of diffuse and gray approximations on the emissivity of a surface.

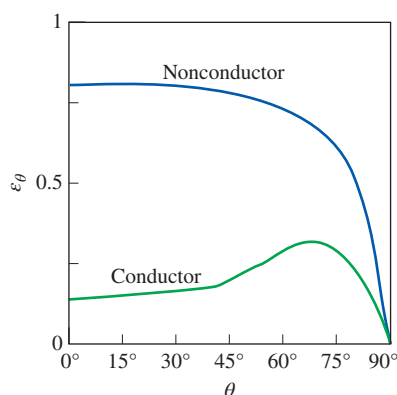


FIGURE 12–26

Typical variations of emissivity with direction for electrical conductors and nonconductors.

Then the average emissivity can be determined from Eq. 12–34 by breaking the integral into three parts and utilizing the definition of the blackbody radiation function as

$$\begin{aligned} \epsilon(T) &= \frac{\epsilon_1 \int_0^{\lambda_1} E_{b\lambda} d\lambda}{E_b} + \frac{\epsilon_2 \int_{\lambda_1}^{\lambda_2} E_{b\lambda} d\lambda}{E_b} + \frac{\epsilon_3 \int_{\lambda_2}^{\infty} E_{b\lambda} d\lambda}{E_b} \\ &= \epsilon_1 f_{0-\lambda_1}(T) + \epsilon_2 f_{\lambda_1-\lambda_2}(T) + \epsilon_3 f_{\lambda_2-\infty}(T) \end{aligned} \quad (12-36)$$

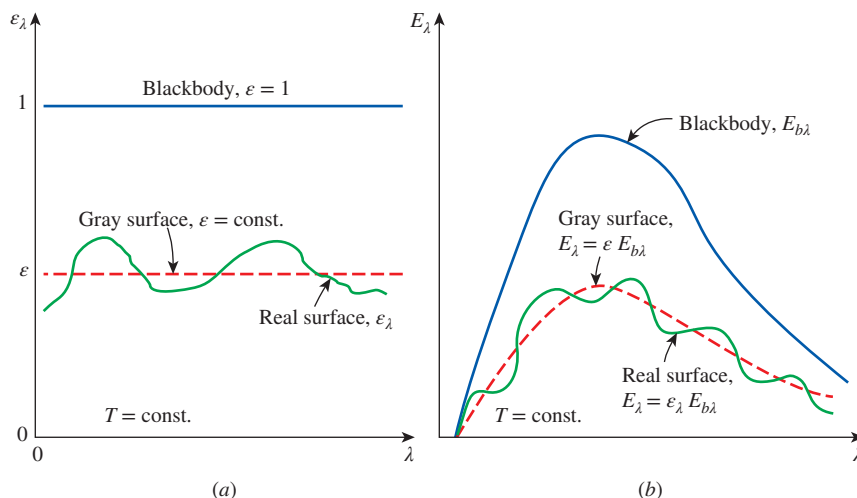
Radiation is a complex phenomenon as it is, and the consideration of the wavelength and direction dependence of properties, assuming sufficient data exist, makes it even more complicated. Therefore, the *diffuse* and *gray* approximations are often used in radiation calculations. A surface is said to be *diffuse* if its properties are *independent of direction*, and *gray* if its properties are *independent of wavelength*. Therefore, the emissivity of a gray, diffuse surface is simply the total hemispherical emissivity of that surface because of independence of direction and wavelength (Fig. 12–25).

A few comments about the validity of the diffuse approximation are in order. Although real surfaces do not emit radiation in a perfectly diffuse manner as a blackbody does, they often come close. The variation of emissivity with direction for both electrical conductors and nonconductors is given in Fig. 12–26. Here θ is the angle measured from the normal of the surface, and thus $\theta = 0$ for radiation emitted in a direction normal to the surface. Note that ϵ_θ remains nearly constant for about $\theta < 40^\circ$ for conductors such as metals and for $\theta < 70^\circ$ for nonconductors such as plastics. Therefore, the directional emissivity of a surface in the normal direction is representative of the hemispherical emissivity of the surface. In radiation analysis, it is common practice to assume the surfaces to be diffuse emitters with an emissivity equal to the value in the normal ($\theta = 0$) direction.

The effect of the gray approximation on emissivity and emissive power of a real surface is illustrated in Fig. 12–27. Note that the radiation emission from a real surface, in general, differs from the Planck distribution, and the emission curve may have several peaks and valleys. A gray surface should emit as much

FIGURE 12–27

Comparison of the emissivity (*a*) and emissive power (*b*) of a real surface with those of a gray surface and a blackbody at the same temperature.



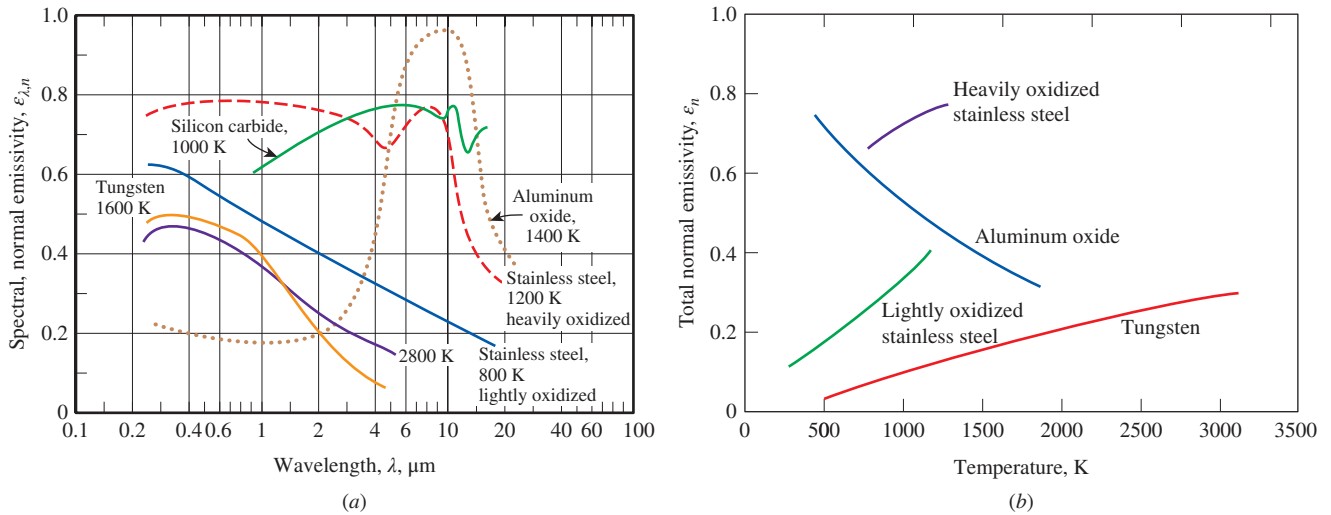


FIGURE 12-28

The variation of normal emissivity with (a) wavelength and (b) temperature for various materials.

radiation as the real surface it represents at the same temperature. Therefore, the areas under the emission curves of the real and gray surfaces must be equal.

The emissivities of common materials are listed in Tables A-18 and A-19, and the variation of emissivity with wavelength and temperature is illustrated in Fig. 12-28. Typical ranges of emissivity of various materials are given in Fig. 12-29. Note that metals generally have low emissivities, as low as 0.02 for polished surfaces, and nonmetals such as ceramics and organic materials have high ones. The emissivity of metals increases with temperature. Also, oxidation causes significant increases in the emissivity of metals. Heavily oxidized metals can have emissivities comparable to those of nonmetals.

Care should be exercised in the use and interpretation of radiation property data reported in the literature, since the properties strongly depend on the surface conditions such as oxidation, roughness, type of finish, and cleanliness. Consequently, there is considerable discrepancy and uncertainty in the reported values. This uncertainty is largely due to the difficulty in characterizing and describing the surface conditions precisely.

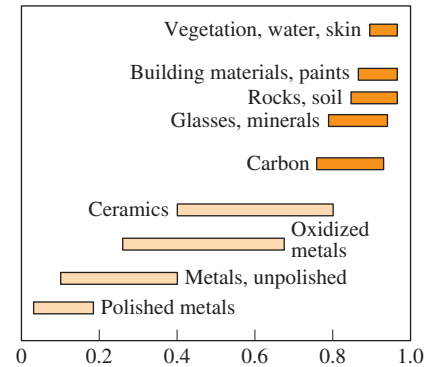


FIGURE 12-29

Typical ranges of emissivity for various materials.

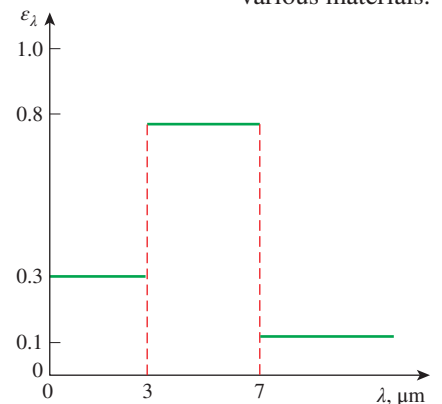


FIGURE 12-30

The spectral emissivity of the surface considered in Example 12-4.

EXAMPLE 12-4 Emissivity of a Surface and Emissive Power

The spectral emissivity function of an opaque surface at 800 K is approximated as (Fig. 12-30)

$$\epsilon_\lambda = \begin{cases} \epsilon_1 = 0.3, & 0 \leq \lambda < 3 \mu\text{m} \\ \epsilon_2 = 0.8, & 3 \mu\text{m} \leq \lambda < 7 \mu\text{m} \\ \epsilon_3 = 0.1, & 7 \mu\text{m} \leq \lambda < \infty \end{cases}$$

Determine the average emissivity of the surface and its emissive power.

SOLUTION The variation of emissivity of a surface at a specified temperature with wavelength is given. The average emissivity of the surface and its emissive power are to be determined.

Analysis The variation of the emissivity with wavelength is given as a step function. Therefore, the average emissivity of the surface can be determined from Eq. 12–34 by breaking the integral into three parts,

$$\begin{aligned}\varepsilon(T) &= \frac{\varepsilon_1 \int_0^{\lambda_1} E_{b\lambda} d\lambda}{\sigma T^4} + \frac{\varepsilon_2 \int_{\lambda_1}^{\lambda_2} E_{b\lambda} d\lambda}{\sigma T^4} + \frac{\varepsilon_3 \int_{\lambda_2}^{\infty} E_{b\lambda} d\lambda}{\sigma T^4} \\ &= \varepsilon_1 f_{0-\lambda_1}(T) + \varepsilon_2 f_{\lambda_1-\lambda_2}(T) + \varepsilon_3 f_{\lambda_2-\infty}(T) \\ &= \varepsilon_1 f_{\lambda_1} + \varepsilon_2 (f_{\lambda_2} - f_{\lambda_1}) + \varepsilon_3 (1 - f_{\lambda_2})\end{aligned}$$

where f_{λ_1} and f_{λ_2} are blackbody radiation functions and are determined from Table 12–2 to be

$$\lambda_1 T = (3 \mu\text{m})(800 \text{ K}) = 2400 \mu\text{m}\cdot\text{K} \rightarrow f_{\lambda_1} = 0.140256$$

$$\lambda_2 T = (7 \mu\text{m})(800 \text{ K}) = 5600 \mu\text{m}\cdot\text{K} \rightarrow f_{\lambda_2} = 0.701046$$

Note that $f_{0-\lambda_1} = f_{\lambda_1} - f_0 = f_{\lambda_1}$ since $f_0 = 0$, and $f_{\lambda_2-\infty} = f_{\infty} - f_{\lambda_2} = 1 - f_{\lambda_2}$ since $f_{\infty} = 1$. Substituting,

$$\begin{aligned}\varepsilon &= 0.3 \times 0.140256 + 0.8(0.701046 - 0.140256) + 0.1(1 - 0.701046) \\ &= \mathbf{0.521}\end{aligned}$$

That is, the surface will emit as much radiation energy at 800 K as a gray surface having a constant emissivity of $\varepsilon = 0.521$. The emissive power of the surface is

$$E = \varepsilon \sigma T^4 = 0.521(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(800 \text{ K})^4 = \mathbf{12,100 \text{ W/m}^2}$$

Discussion Note that the surface emits 12.1 kJ of radiation energy per second per m² area of the surface.

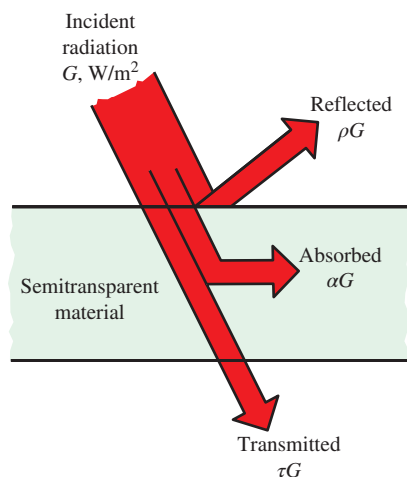


FIGURE 12–31

The absorption, reflection, and transmission of incident radiation by a semitransparent material.

Absorptivity, Reflectivity, and Transmissivity

Everything around us constantly emits radiation, and the emissivity represents the emission characteristics of those bodies. This means that every body, including our own, is constantly bombarded by radiation coming from all directions over a range of wavelengths. Recall that radiation flux *incident on a surface* is called **irradiation** and is denoted by G .

When radiation strikes a surface, part of it is absorbed, part of it is reflected, and the remaining part, if any, is transmitted, as illustrated in Fig. 12–31. *The fraction of irradiation absorbed by the surface* is called the **absorptivity** α , *the fraction reflected by the surface* is called the **reflectivity** ρ , and *the fraction transmitted* is called the **transmissivity** τ . That is,

$$\text{Absorptivity:} \quad \alpha = \frac{\text{Absorbed radiation}}{\text{Incident radiation}} = \frac{G_{\text{abs}}}{G}, \quad 0 \leq \alpha \leq 1 \quad (12-37)$$

$$\text{Reflectivity:} \quad \rho = \frac{\text{Reflected radiation}}{\text{Incident radiation}} = \frac{G_{\text{ref}}}{G}, \quad 0 \leq \rho \leq 1 \quad (12-38)$$

$$\text{Transmissivity:} \quad \tau = \frac{\text{Transmitted radiation}}{\text{Incident radiation}} = \frac{G_{\text{tr}}}{G}, \quad 0 \leq \tau \leq 1 \quad (12-39)$$

where G is the radiation flux incident on the surface, and G_{abs} , G_{ref} , and G_{tr} are the absorbed, reflected, and transmitted portions of it, respectively. The first

law of thermodynamics requires that the sum of the absorbed, reflected, and transmitted radiation be equal to the incident radiation. That is,

$$G_{\text{abs}} + G_{\text{ref}} + G_{\text{tr}} = G$$

Dividing each term of this relation by G yields

$$\alpha + \rho + \tau = 1 \quad (12-40)$$

For idealized blackbodies which are perfect absorbers, $\rho = 0$ and $\tau = 0$, and Eq. 12-40 reduces to $\alpha = 1$. For opaque surfaces such as most solids and liquids, $\tau = 0$, and thus

$$\alpha + \rho = 1 \quad (12-41)$$

For most gases, the reflectance is absent, $\rho = 0$, and Eq. 12-40 reduces in this case to

$$\alpha + \tau = 1 \quad (12-42)$$

The preceding two equations are important property relations since the knowledge of one property (either α or ρ in Eq. 12-41 and α or τ in Eq. 12-42) implies the knowledge of the other property.

These definitions are for *total hemispherical* properties, since G represents the radiation flux incident on the surface from all directions over the hemispherical space and over all wavelengths. Thus, α , ρ , and τ are the *average* properties of a medium for all directions and all wavelengths. However, like emissivity, these properties can also be defined for a specific wavelength and/or direction. For example, the **spectral directional absorptivity** and **spectral directional reflectivity** of a surface are defined, respectively, as the absorbed and reflected fractions of the intensity of radiation incident at a specified wavelength in a specified direction as

$$\alpha_{\lambda, \theta}(\lambda, \theta, \phi) = \frac{I_{\lambda, \text{abs}}(\lambda, \theta, \phi)}{I_{\lambda, i}(\lambda, \theta, \phi)} \quad \text{and} \quad \rho_{\lambda, \theta}(\lambda, \theta, \phi) = \frac{I_{\lambda, \text{ref}}(\lambda, \theta, \phi)}{I_{\lambda, i}(\lambda, \theta, \phi)} \quad (12-43)$$

Likewise, the **spectral hemispherical absorptivity** and **spectral hemispherical reflectivity** of a surface are defined as

$$\alpha_{\lambda}(\lambda) = \frac{G_{\lambda, \text{abs}}(\lambda)}{G_{\lambda}(\lambda)} \quad \text{and} \quad \rho_{\lambda}(\lambda) = \frac{G_{\lambda, \text{ref}}(\lambda)}{G_{\lambda}(\lambda)} \quad (12-44)$$

where G_{λ} is the spectral irradiation (in $\text{W}/\text{m}^2 \cdot \mu\text{m}$) incident on the surface, and $G_{\lambda, \text{abs}}$ and $G_{\lambda, \text{ref}}$ are the reflected and absorbed portions of it, respectively.

Similar quantities can be defined for the transmissivity of semitransparent materials. For example, the **spectral hemispherical transmissivity** of a medium can be expressed as

$$\tau_{\lambda}(\lambda) = \frac{G_{\lambda, \text{tr}}(\lambda)}{G_{\lambda}(\lambda)} \quad (12-45)$$

The average absorptivity, reflectivity, and transmissivity of a surface can also be defined in terms of their spectral counterparts as

$$\alpha = \frac{\int_0^{\infty} \alpha_{\lambda} G_{\lambda} d\lambda}{\int_0^{\infty} G_{\lambda} d\lambda}, \quad \rho = \frac{\int_0^{\infty} \rho_{\lambda} G_{\lambda} d\lambda}{\int_0^{\infty} G_{\lambda} d\lambda}, \quad \tau = \frac{\int_0^{\infty} \tau_{\lambda} G_{\lambda} d\lambda}{\int_0^{\infty} G_{\lambda} d\lambda} \quad (12-46)$$

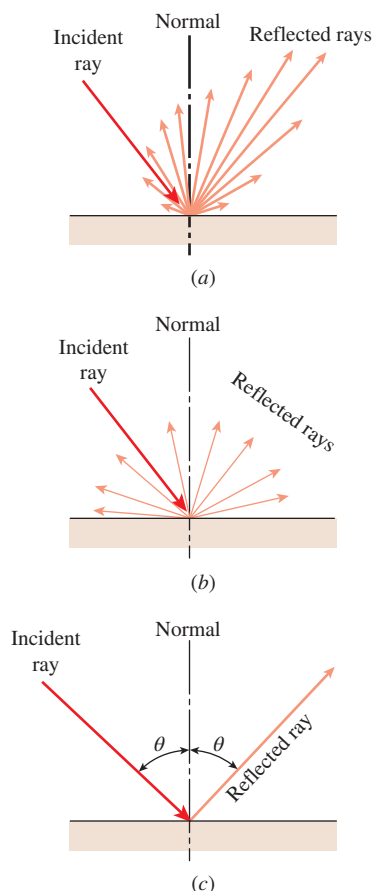


FIGURE 12-32

Different types of reflection from a surface: (a) actual or irregular, (b) diffuse, and (c) specular or mirrorlike.

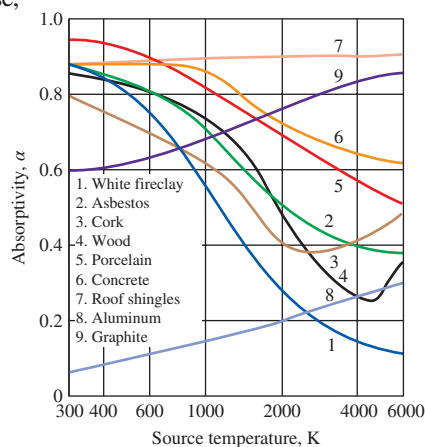


FIGURE 12-33

Variation of absorptivity with the temperature of the source of irradiation for various common materials at room temperature.

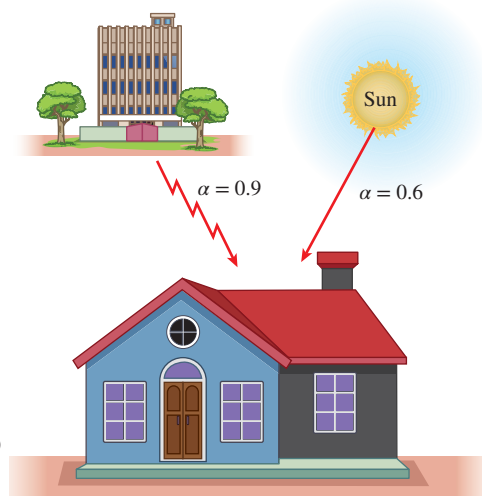


FIGURE 12-34

The absorptivity of a material may be quite different for radiation originating from sources at different temperatures.

The reflectivity differs somewhat from the other properties in that it is *bidirectional* in nature. That is, the value of the reflectivity of a surface depends not only on the direction of the incident radiation but also the direction of reflection. Therefore, the reflected rays of a radiation beam incident on a real surface in a specified direction form an irregular shape, as shown in Fig. 12–32. Such detailed reflectivity data do not exist for most surfaces, and even if they did, they would be of little value in radiation calculations since this would usually add more complication to the analysis.

In practice, for simplicity, surfaces are assumed to reflect in a perfectly *specular* or *diffuse* manner. In **specular** (or *mirrorlike*) **reflection**, the angle of reflection equals the angle of incidence of the radiation beam. In **diffuse reflection**, radiation is reflected equally in all directions, as shown in Fig. 12–32. Reflection from smooth and polished surfaces approximates specular reflection, whereas reflection from rough surfaces approximates diffuse reflection. In radiation analysis, smoothness is defined relative to wavelength. A surface is said to be *smooth* if the height of the surface roughness is much smaller than the wavelength of the incident radiation.

Unlike emissivity, the absorptivity of a material is practically independent of surface temperature. However, the absorptivity depends strongly on the temperature of the source from which the incident radiation is originating. This is also evident from Fig. 12–33, which shows the absorptivities of various materials at room temperature as functions of the temperature of the radiation source. For example, the absorptivity of the concrete roof of a house is about 0.6 for solar radiation (source temperature: 5780 K) and 0.9 for radiation originating from the surrounding trees and buildings (source temperature: 300 K), as illustrated in Fig. 12–34.

Notice that the absorptivity of aluminum increases with the source temperature, a characteristic for metals, and the absorptivity of electric nonconductors, in general, decreases with temperature. This decrease is most pronounced for

surfaces that appear white to the eye. For example, the absorptivity of a white painted surface is low for solar radiation, but it is rather high for infrared radiation.

Kirchhoff's Law

Consider a small body of surface area A_s , emissivity ε , and absorptivity α at temperature T contained in a large isothermal enclosure at the same temperature, as shown in Fig. 12–35. Recall that a large isothermal enclosure forms a blackbody cavity regardless of the radiative properties of the enclosure surface, and the body in the enclosure is too small to interfere with the blackbody nature of the cavity. Therefore, the radiation incident on any part of the surface of the small body is equal to the radiation emitted by a blackbody at temperature T . That is, $G = E_b(T) = \sigma T^4$, and the radiation absorbed by the small body per unit of its surface area is

$$G_{\text{abs}} = \alpha G = \alpha \sigma T^4$$

The radiation emitted by the small body is

$$E_{\text{emit}} = \varepsilon \sigma T^4$$

Considering that the small body is in thermal equilibrium with the enclosure, the net rate of heat transfer to the body must be zero. Therefore, the radiation emitted by the body must be equal to the radiation absorbed by it:

$$A_s \varepsilon \sigma T^4 = A_s \alpha \sigma T^4$$

Thus, we conclude that for any surface in the enclosure,

$$\varepsilon(T) = \alpha(T) \quad (12-47)$$

That is, *the total hemispherical emissivity of a surface at temperature T is equal to its total hemispherical absorptivity for radiation coming from a blackbody at the same temperature*. This relation, which greatly simplifies the radiation analysis, was first developed by Gustav Kirchhoff in 1860 and is now called **Kirchhoff's law**. Note that this relation is derived under the condition that the surface temperature is equal to the temperature of the source of irradiation, and the reader is cautioned against using it when considerable difference (more than a few hundred degrees) exists between the surface temperature and the temperature of the source of irradiation.

For any surface in the enclosure, the preceding derivation can also be repeated for radiation at a specified wavelength to obtain the *spectral* form of Kirchhoff's law:

$$\varepsilon_\lambda(T) = \alpha_\lambda(T) \quad (12-48)$$

This relation is valid when the irradiation or the emitted radiation is independent of direction. The form of Kirchhoff's law that involves no restrictions is the *spectral directional* form expressed as $\varepsilon_{\lambda, \theta}(T) = \alpha_{\lambda, \theta}(T)$. That is, the emissivity of a surface at a specified wavelength, direction, and temperature is always equal to its absorptivity at the same wavelength, direction, and temperature.

It is very tempting to use Kirchhoff's law in radiation analysis since the relation $\varepsilon = \alpha$ together with $\rho = 1 - \alpha$ enables us to determine all three properties of an opaque surface from a knowledge of only *one* property.

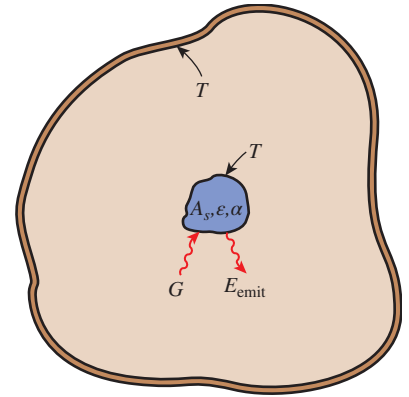
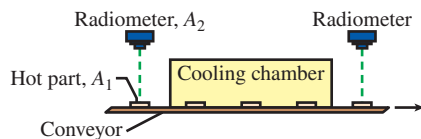


FIGURE 12–35

The small body contained in a large isothermal enclosure used in the development of Kirchhoff's law.

**FIGURE 12-36**

Schematic for Example 12-5.

Although Eq. 12-47 gives acceptable results in most cases, in practice, care should be exercised when there is considerable difference between the surface temperature and the temperature of the source of incident radiation.

PtD**EXAMPLE 12-5****Prevention of Thermal Burn Hazard by Using Radiation Sensors to Monitor Temperatures of Parts on a Conveyor**

Hot opaque parts ($A_1 = 15 \text{ cm}^2$, $m = 0.5 \text{ kg}$, $c_v = 450 \text{ J/kg}\cdot\text{K}$, $\varepsilon = 0.85$), coming out of a production line and on their way to packaging, are cooled in an air-chilled chamber (Fig. 12-36). One of the reasons for cooling the parts prior to packing is to alleviate the risk of thermal burns. For the parts to be safe for handling by the operators, each part should be cooled to below 45°C . To monitor the temperatures of the parts, two identical radiometers are placed at a distance of 1 m from the parts on the conveyor at the entrance and exit of the cooling chamber. Throughout the entire process, irradiation on the parts from the surroundings is estimated to be 300 W/m^2 . Inside the cooling chamber, 215 W of heat is removed from each part in the duration of 1 minute . If the radiometer at the cooling chamber entrance detects an irradiation of 0.21 W/m^2 , determine the irradiation that the radiometer at the cooling chamber exit would measure. Would the parts exiting the cooling chamber be safe for handling?

SOLUTION In this example, the concepts of Prevention through Design (PtD) are applied in conjunction with the concepts of radiation intensity and radiative properties.

Assumptions 1 Both part surface and radiometer can be approximated as differential surfaces since both are very small compared to the square of the distance between them. 2 Temperatures of parts are uniform. 3 Kirchhoff's law is applicable. 4 Parts are opaque and their properties are constant.

Properties The specific heat and the emissivity of the parts are given as $c_v = 450 \text{ J/kg}\cdot\text{K}$ and $\varepsilon = 0.85$. The Stefan-Boltzmann constant is $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$.

Analysis From Kirchhoff's law, the emissivity of the parts is

$$\varepsilon = \alpha = 0.85$$

The parts are opaque ($\tau = 0$) and the reflectivity of the parts is

$$\rho = 1 - \alpha = 1 - 0.85 = 0.15$$

The solid angle subtended by both radiometers when viewed from the parts at the entrance and exit of the cooling chamber is

$$\omega_{2-1} \cong \frac{A_{n,2}}{L^2} = \frac{A_2 \cos \theta_2}{L^2}$$

The intensities of the radiation emitted and reflected (radiosity) by the parts A_1 at the entrance and exit of the cooling chamber are

$$I_{1,\text{in}} = \frac{J_{\text{in}}}{\pi} = \frac{E_{\text{in}} + G_{\text{ref}}}{\pi} = \frac{\varepsilon \sigma T_{1,\text{in}}^4 + \rho G_1}{\pi} \text{ and } I_{1,\text{out}} = \frac{J_{\text{out}}}{\pi} = \frac{E_{\text{out}} + G_{\text{ref}}}{\pi} = \frac{\varepsilon \sigma T_{1,\text{out}}^4 + \rho G_1}{\pi}$$

The irradiances measured by the radiometers A_2 at the entrance and exit of the cooling chamber are

$$G_{2,\text{in}} = \frac{\dot{Q}_{1-2,\text{in}}}{A_2} = \frac{I_{1,\text{in}}(A_1 \cos \theta_1) \omega_{2-1}}{A_2} = \frac{(\varepsilon \sigma T_{1,\text{in}}^4 + \rho G_1) A_1 (\cos \theta_1) (\cos \theta_2)}{\pi L^2}$$

and

$$G_{2,\text{out}} = \frac{\dot{Q}_{1-2,\text{out}}}{A_2} = \frac{I_{1,\text{out}}(A_1 \cos \theta_1) \omega_{2-1}}{A_2} = \frac{(\varepsilon \sigma T_{1,\text{out}}^4 + \rho G_1) A_1 (\cos \theta_1) (\cos \theta_2)}{\pi L^2}$$

With both radiometers placed normal to the direction of viewing from each part that passes them ($\theta_1 = \theta_2 = 0^\circ$), we have

$$G_{2,\text{in}} = \frac{(\varepsilon \sigma T_{1,\text{in}}^4 + \rho G_1) A_1}{\pi L^2} \quad \text{and} \quad G_{2,\text{out}} = \frac{(\varepsilon \sigma T_{1,\text{out}}^4 + \rho G_1) A_1}{\pi L^2}$$

From the given irradiation $G_{2,\text{in}}$ detected by the radiometer at the cooling chamber entrance, the temperature of the parts at that location is

$$\begin{aligned} T_{1,\text{in}} &= \left[\left(\frac{G_{2,\text{in}} \pi L^2}{A_1} - \rho G_1 \right) \left(\frac{1}{\varepsilon \sigma} \right) \right]^{1/4} \\ &= \left\{ \left[\frac{(0.21 \text{ W/m}^2) \pi (1.5 \text{ m})^2}{15 \times 10^{-4} \text{ m}^2} - (0.15)(300 \text{ W/m}^2) \frac{1}{(0.85)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)} \right] \right\}^{1/4} \\ &= 374 \text{ K} = 101^\circ \text{C} \end{aligned}$$

The temperature of the parts at the exit of the cooling chamber can be determined from the change in the internal energy and the heat removed from the parts during the cooling process:

$$Q = \dot{Q} \Delta t = \Delta U = mc_v (T_{1,\text{in}} - T_{1,\text{out}}) \quad \text{or} \quad T_{1,\text{out}} = T_{1,\text{in}} - \frac{\dot{Q} \Delta t}{mc_v}$$

Thus,

$$T_{1,\text{out}} = 101^\circ \text{C} - \frac{(215 \text{ W})(60 \text{ s})}{(0.5 \text{ kg})(450 \text{ J/kg} \cdot \text{K})} = \mathbf{43.7^\circ \text{C}} < 45^\circ \text{C}$$

Finally, the irradiation $G_{2,\text{out}}$ measured by the radiometer at the exit of the cooling chamber is

$$\begin{aligned} G_{2,\text{out}} &= \frac{(\varepsilon \sigma T_{1,\text{out}}^4 + \rho G_1) A_1}{\pi L^2} \\ &= \frac{(0.85)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(43.7 + 273)^4 \text{ K}^4 + (0.15)(300 \text{ W/m}^2)(15 \times 10^{-4} \text{ m}^2)}{\pi (1.5 \text{ m})^2} \\ &= \mathbf{0.112 \text{ W/m}^2} \end{aligned}$$

Discussion The temperature of the parts exiting the cooling chamber is below 45°C , and therefore they are safe for handling in the packing operation.

Having two radiometers monitoring the temperatures of the parts as they enter and exit the cooling chambers allows the engineers to determine the necessary cooling time for the parts. Therefore, the conveyor speed can be optimized while at the same time preventing thermal burn hazards.

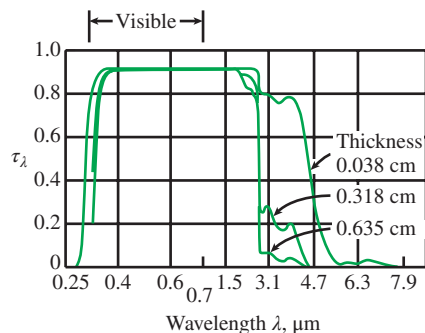


FIGURE 12-37

The spectral transmissivity of low-iron glass at room temperature for different thicknesses.

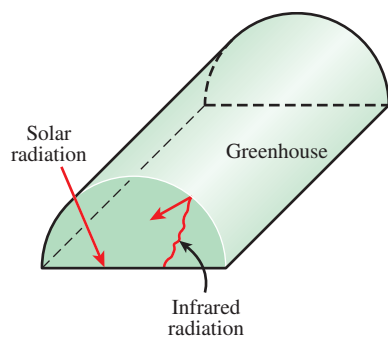


FIGURE 12-38

A greenhouse traps energy by allowing the solar radiation to come in but not allowing the infrared radiation to go out.

The Greenhouse Effect

You have probably noticed that when you leave your car under direct sunlight on a sunny day, the interior of the car gets much warmer than the air outside, and you may have wondered why the car acts like a *heat trap*. The answer lies in the spectral transmissivity curve of the *glass*, which resembles an inverted U, as shown in Fig. 12–37. We observe from this figure that glass at thicknesses encountered in practice transmits over 90 percent of radiation in the visible range and is practically opaque (nontransparent) to radiation in the longer-wavelength infrared regions of the electromagnetic spectrum (roughly $\lambda > 3 \mu\text{m}$). Therefore, glass has a transparent window in the wavelength range $0.3 \mu\text{m} < \lambda < 3 \mu\text{m}$ in which over 90 percent of solar radiation is emitted. On the other hand, the radiation emitted by surfaces at room temperature falls entirely in the infrared region. Consequently, glass allows the solar radiation to enter but does not allow the infrared radiation from the interior surfaces to escape. This causes a rise in the interior temperature as a result of the energy buildup in the car. This heating effect, which is due to the nongray characteristic of glass (or clear plastics), is known as the **greenhouse effect**, since it is used extensively in greenhouses (Fig. 12–38).

The greenhouse effect is also experienced on a larger scale on earth. The surface of the earth, which warms up during the day as a result of the absorption of solar energy, cools down at night by radiating its energy into deep space as infrared radiation. The combustion gases such as CO_2 and water vapor in the atmosphere transmit the bulk of the solar radiation but absorb the infrared radiation emitted by the surface of the earth. Thus, there is concern that the energy trapped on earth will eventually cause global warming and thus drastic changes in weather patterns.

In *humid* places such as coastal areas, there is not a large change between the daytime and nighttime temperatures because the humidity acts as a barrier on the path of the infrared radiation coming from the earth, and thus it slows down the cooling process at night. In areas with clear skies such as deserts, there is a large swing between the daytime and nighttime temperatures because of the absence of such barriers for infrared radiation.

12-6 ■ ATMOSPHERIC AND SOLAR RADIATION

The sun is our primary source of energy. The energy coming off the sun, called *solar energy*, reaches us in the form of electromagnetic waves after experiencing considerable interactions with the atmosphere. The radiation energy emitted or reflected by the constituents of the atmosphere form the *atmospheric radiation*. Here we give an overview of the solar and atmospheric radiation because of their importance and relevance to daily life. Also, our familiarity with solar energy makes it an effective tool in developing a better understanding for some of the new concepts introduced earlier. Detailed treatment of this exciting subject can be found in numerous books devoted to this topic.

The *sun* is a nearly spherical body that has a diameter of $D \approx 1.39 \times 10^9 \text{ m}$ and a mass of $m \approx 2 \times 10^{30} \text{ kg}$ and is located at a mean distance of $L = 1.50 \times 10^{11} \text{ m}$ from the earth. It emits radiation energy continuously at a rate of $E_{\text{sun}} \approx 3.8 \times 10^{26} \text{ W}$. Less than a billionth of this energy (about $1.7 \times 10^{17} \text{ W}$)

strikes the earth, which is sufficient to keep the earth warm and to maintain life through the photosynthesis process. The energy of the sun is due to the continuous *fusion* reaction during which two hydrogen atoms fuse to form one atom of helium. Therefore, the sun is essentially a *nuclear reactor*, with temperatures as high as 40,000,000 K in its core region. The temperature drops to about 5800 K in the outer region of the sun, called the convective zone, as a result of the dissipation of this energy by radiation.

The solar energy reaching the earth's atmosphere is called the **total solar irradiance** G_s , whose value is

$$G_s = 1373 \text{ W/m}^2 \quad (12-49)$$

The total solar irradiance (also called the **solar constant**) represents *the rate at which solar energy is incident on a surface normal to the sun's rays at the outer edge of the atmosphere when the earth is at its mean distance from the sun* (Fig. 12-39).

The value of the total solar irradiance can be used to estimate the effective surface temperature of the sun from the requirement that

$$(4\pi L^2)G_s = (4\pi r^2)\sigma T_{\text{sun}}^4 \quad (12-50)$$

where L is the mean distance between the sun's center and the earth and r is the radius of the sun. The left-hand side of this equation represents the total solar energy passing through a spherical surface whose radius is the mean earth-sun distance, and the right-hand side represents the total energy that leaves the sun's outer surface. The conservation of energy principle requires that these two quantities be equal to each other, since the solar energy experiences no attenuation (or enhancement) on its way through the vacuum (Fig. 12-40). The **effective surface temperature** of the sun is determined from Eq. 12-50 to be $T_{\text{sun}} = 5780 \text{ K}$. That is, the sun can be treated as a blackbody at a temperature of 5780 K. This is also confirmed by the measurements of the spectral distribution of the solar radiation just outside the atmosphere plotted in Fig. 12-41, which shows only small deviations from the idealized blackbody behavior.

The spectral distribution of solar radiation on the ground plotted in Fig. 12-41 shows that the solar radiation undergoes considerable *attenuation* as it passes through the atmosphere as a result of absorption and scattering. About 99 percent of the atmosphere is contained within a distance of 30 km from the earth's surface. The several dips on the spectral distribution of radiation on the earth's surface are due to *absorption* by the gases O_2 , O_3 (ozone), H_2O , and CO_2 . Absorption by *oxygen* occurs in a narrow band about $\lambda = 0.76 \mu\text{m}$. The *ozone* absorbs *ultraviolet* radiation at wavelengths below $0.3 \mu\text{m}$ almost completely, and radiation in the range $0.3\text{--}0.4 \mu\text{m}$ considerably. Thus, the ozone layer in the upper regions of the atmosphere protects biological systems on earth from harmful ultraviolet radiation. In turn, we must protect the ozone layer from the destructive chemicals commonly used as refrigerants, cleaning agents, and propellants in aerosol cans. The use of these chemicals is now banned. The ozone gas also absorbs some radiation in the visible range. Absorption in the infrared region is dominated by *water vapor* and *carbon dioxide*. The dust particles and other pollutants in the atmosphere also absorb radiation at various wavelengths.

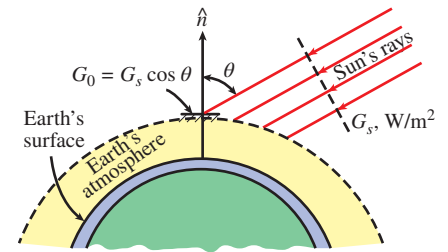


FIGURE 12-39

Solar radiation reaching the earth's atmosphere and the total solar irradiance.

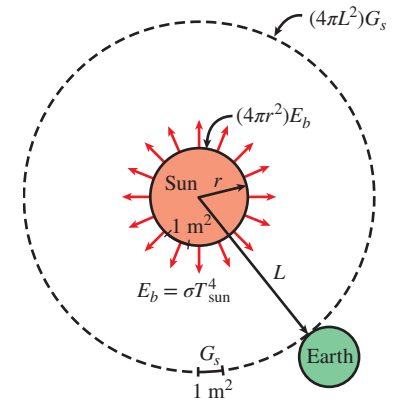


FIGURE 12-40

The total solar energy passing through concentric spheres remains constant, but the energy falling per unit area decreases with increasing radius.

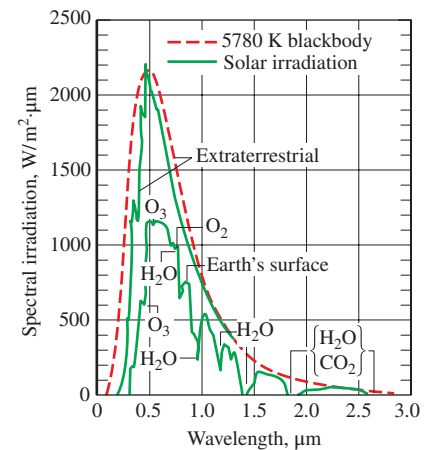


FIGURE 12-41

Spectral distribution of solar radiation just outside the atmosphere, at the surface of the earth on a typical day, and comparison with blackbody radiation at 5780 K.

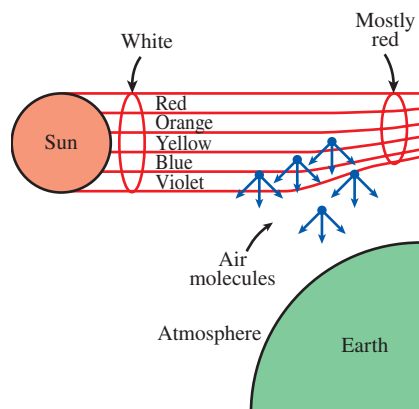


FIGURE 12-42

Air molecules scatter blue light much more than they do red light. At sunset, light travels through a thicker layer of atmosphere, which removes much of the blue from the natural light, allowing the red to dominate.

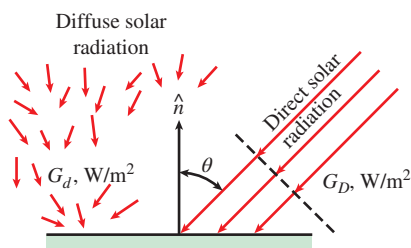


FIGURE 12-43

The direct and diffuse radiation incident on a horizontal surface on earth's surface.

As a result of these absorptions, the solar energy reaching the earth's surface is weakened considerably, to about 950 W/m^2 on a clear day and much less on cloudy or smoggy days. Also, practically all of the solar radiation reaching the earth's surface falls in the wavelength band from 0.3 to $2.5 \mu\text{m}$.

Another mechanism that attenuates solar radiation as it passes through the atmosphere is *scattering* or *reflection* by air molecules and the many other kinds of particles such as dust, smog, and water droplets suspended in the atmosphere. Scattering is mainly governed by the size of the particle relative to the wavelength of radiation. The oxygen and nitrogen molecules primarily scatter radiation at very short wavelengths, comparable to the sizes of the molecules themselves. Therefore, radiation at wavelengths corresponding to violet and blue colors is scattered the most. This molecular scattering in all directions is what gives the sky its bluish color. The same phenomenon is responsible for red sunrises and sunsets. Early in the morning and late in the afternoon, the sun's rays pass through a greater thickness of the atmosphere than they do at midday, when the sun is at the top. Therefore, the violet and blue colors of the light encounter a greater number of molecules by the time they reach the earth's surface, and thus a greater fraction of them are scattered (Fig. 12-42). Consequently, the light that reaches the earth's surface consists primarily of colors corresponding to longer wavelengths such as red, orange, and yellow. The clouds appear in reddish-orange color during sunrise and sunset because the light they reflect is reddish-orange at those times. For the same reason, a red traffic light is visible from a longer distance than is a green light under the same circumstances.

The solar energy incident on a surface on earth is considered to consist of *direct* and *diffuse* parts. The part of solar radiation that reaches the earth's surface without being scattered or absorbed by the atmosphere is called **direct solar radiation** G_D . The scattered radiation is assumed to reach the earth's surface uniformly from all directions and is called **diffuse solar radiation** G_d . Then the *total* solar energy incident on the unit area of a *horizontal surface* on the ground is (Fig. 12-43)

$$G_{\text{solar}} = G_D \cos \theta + G_d \quad (\text{W/m}^2) \quad (12-51)$$

where θ is the angle of incidence of direct solar radiation (the angle that the sun's rays make with the normal of the surface). The diffuse radiation varies from about 10 percent of the total radiation on a clear day to nearly 100 percent on a totally cloudy day.

The gas molecules and the suspended particles in the atmosphere emit radiation as well as absorbing it. The atmospheric emission is primarily due to the CO_2 and H_2O molecules and is concentrated in the regions from 5 to $8 \mu\text{m}$ and above $13 \mu\text{m}$. Although this emission is far from resembling the distribution of radiation from a blackbody, it is convenient in radiation calculations to treat the atmosphere as a blackbody at some lower fictitious temperature that emits an equivalent amount of radiation energy. This fictitious temperature is called the **effective sky temperature** T_{sky} . Then the radiation emission from the atmosphere to the earth's surface is expressed as

$$G_{\text{sky}} = \sigma T_{\text{sky}}^4 \quad (\text{W/m}^2) \quad (12-52)$$

The value of T_{sky} depends on the atmospheric conditions. It ranges from about 230 K for cold, clear-sky conditions to about 285 K for warm, cloudy-sky conditions.

Note that the effective sky temperature does not deviate much from the room temperature. Thus, in the light of Kirchhoff's law, we can take the absorptivity of a surface to be equal to its emissivity at room temperature, $\alpha = \varepsilon$. Then the sky radiation absorbed by a surface can be expressed as

$$E_{\text{sky, absorbed}} = \alpha G_{\text{sky}} = \alpha \sigma T_{\text{sky}}^4 = \varepsilon \sigma T_{\text{sky}}^4 \quad (\text{W/m}^2) \quad (12-53)$$

The net rate of radiation heat transfer to a surface exposed to solar and atmospheric radiation is determined from an energy balance (Fig. 12-44):

$$\begin{aligned} \dot{q}_{\text{net, rad}} &= \sum E_{\text{absorbed}} - \sum E_{\text{emitted}} \\ &= E_{\text{solar, absorbed}} + E_{\text{sky, absorbed}} - E_{\text{emitted}} \\ &= \alpha_s G_{\text{solar}} + \varepsilon \sigma T_{\text{sky}}^4 - \varepsilon \sigma T_s^4 \\ &= \alpha_s G_{\text{solar}} + \varepsilon \sigma (T_{\text{sky}}^4 - T_s^4) \quad (\text{W/m}^2) \end{aligned} \quad (12-54)$$

where T_s is the temperature of the surface in K and ε is its emissivity at room temperature. A positive result for $\dot{q}_{\text{net, rad}}$ indicates a radiation heat gain by the surface and a negative result indicates a heat loss.

The absorption and emission of radiation by the *elementary gases* such as H_2 , O_2 , and N_2 at moderate temperatures are negligible, and a medium filled with these gases can be treated as a *vacuum* in radiation analysis. The absorption and emission of gases with *larger molecules* such as H_2O and CO_2 , however, can be *significant* and may need to be considered when considerable amounts of such gases are present in a medium. For example, a 1-m-thick layer of water vapor at 1 atm pressure and 100°C emits more than 50 percent of the energy that a blackbody would emit at the same temperature.

In solar energy applications, the spectral distribution of incident solar radiation is very different than the spectral distribution of emitted radiation by the surfaces, since the former is concentrated in the short-wavelength region and the latter in the infrared region. Therefore, the radiation properties of surfaces are quite different for the incident and emitted radiation, and the surfaces cannot be assumed to be gray. Instead, the surfaces are assumed to have two sets of properties: one for solar radiation and another for infrared radiation at room temperature. Table 12-3 lists the *emissivity* ε and the *solar absorptivity* α_s of some common materials. Surfaces that are intended to *collect solar energy*, such as the absorber surfaces of solar collectors, are desired to have high α_s but low ε values to maximize the absorption of solar radiation and to minimize the emission of radiation. Surfaces that are intended to *remain cool* under the sun, such as the outer surfaces of fuel tanks and refrigerator trucks, are desired to have just the opposite properties. Surfaces are often given the desired properties by coating them with thin layers of *selective* materials. A surface can be kept cool, for example, by simply painting it white. In practice, engineers pay close attention to the ratio α_s/ε when selecting appropriate materials for the purpose of heat collection or heat rejection. For heat collection,

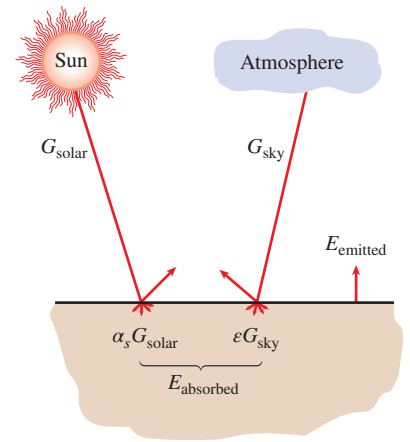


FIGURE 12-44

Radiation interactions of a surface exposed to solar and atmospheric radiation.

TABLE 12-3

Comparison of the solar absorptivity α_s of some surfaces with their emissivity ε at room temperature

Surface	α_s	ε
Aluminum		
Polished	0.09	0.03
Anodized	0.14	0.84
Foil	0.15	0.05
Copper		
Polished	0.18	0.03
Tarnished	0.65	0.75
Stainless steel		
Polished	0.37	0.60
Dull	0.50	0.21
Plated metals		
Black nickel oxide	0.92	0.08
Black chrome	0.87	0.09
Concrete	0.60	0.88
White marble	0.46	0.95
Red brick	0.63	0.93
Asphalt	0.90	0.90
Black paint	0.97	0.97
White paint	0.14	0.93
Snow	0.28	0.97
Human skin (Caucasian)	0.62	0.97

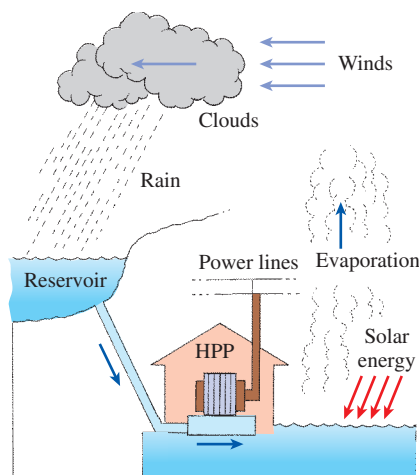


FIGURE 12-45

The cycle that water undergoes in a hydroelectric power plant.

materials with large values of α_s/ε (such as clean galvanized sheet metal with $\alpha_s/\varepsilon = 5.0$) are required. For heat rejection, on the other hand, materials with small values of α_s/ε (such as anodized aluminum with $\alpha_s/\varepsilon = 0.17$) are desirable. Values of α_s/ε together with solar absorptivity for selected materials are listed in Table A-19.

We close this section by pointing out that what we call *renewable energy* is usually nothing more than the manifestation of solar energy in different forms. Such energy sources include wind energy, hydroelectric power, ocean thermal energy, ocean wave energy, and wood. For example, no hydroelectric power plant can generate electricity year after year unless the water evaporates by absorbing solar energy and comes back as rainfall to replenish the water source (Fig. 12-45). Although solar energy is sufficient to meet all of the world's energy needs, currently it is not economical to do so because of the low concentration of solar energy on earth and the high capital cost of harnessing it.

EXAMPLE 12-6 Selective Absorber and Reflective Surfaces

Consider a surface exposed to solar radiation. At a given time, the direct and diffuse components of solar radiation are $G_D = 400$ and $G_d = 300$ W/m², and the direct radiation makes a 20° angle with the normal of the surface. The surface temperature is observed to be 320 K at that time. Assuming an effective sky temperature of 260 K, determine the net rate of radiation heat transfer for these cases (Fig. 12-46):

- $\alpha_s = 0.9$ and $\varepsilon = 0.9$ (gray absorber surface)
- $\alpha_s = 0.1$ and $\varepsilon = 0.1$ (gray reflector surface)
- $\alpha_s = 0.9$ and $\varepsilon = 0.1$ (selective absorber surface)
- $\alpha_s = 0.1$ and $\varepsilon = 0.9$ (selective reflector surface)

SOLUTION A surface is exposed to solar and sky radiation. The net rate of radiation heat transfer is to be determined for four different combinations of emissivities and solar absorptivities.

Analysis The total solar energy incident on the surface is

$$\begin{aligned} G_{\text{solar}} &= G_D \cos \theta + G_d \\ &= (400 \text{ W/m}^2) \cos 20^\circ + (300 \text{ W/m}^2) \\ &= 676 \text{ W/m}^2 \end{aligned}$$

Then the net rate of radiation heat transfer for each of the four cases is determined from:

$$\dot{q}_{\text{net, rad}} = \alpha_s G_{\text{solar}} + \varepsilon \sigma (T_{\text{sky}}^4 - T_s^4)$$

- $\alpha_s = 0.9$ and $\varepsilon = 0.9$ (gray absorber surface):

$$\begin{aligned} \dot{q}_{\text{net, rad}} &= 0.9(676 \text{ W/m}^2) + 0.9(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(260 \text{ K})^4 - (320 \text{ K})^4] \\ &= 307 \text{ W/m}^2 \end{aligned}$$

- $\alpha_s = 0.1$ and $\varepsilon = 0.1$ (gray reflector surface):

$$\begin{aligned} \dot{q}_{\text{net, rad}} &= 0.1(676 \text{ W/m}^2) + 0.1(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(260 \text{ K})^4 - (320 \text{ K})^4] \\ &= 34 \text{ W/m}^2 \end{aligned}$$

(c) $\alpha_s = 0.9$ and $\varepsilon = 0.1$ (selective absorber surface):

$$\begin{aligned}\dot{q}_{\text{net, rad}} &= 0.9(676 \text{ W/m}^2) + 0.1(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(260 \text{ K})^4 - (320 \text{ K})^4] \\ &= \mathbf{575 \text{ W/m}^2}\end{aligned}$$

(d) $\alpha_s = 0.1$ and $\varepsilon = 0.9$ (selective reflector surface):

$$\begin{aligned}\dot{q}_{\text{net, rad}} &= 0.1(676 \text{ W/m}^2) + 0.9(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(260 \text{ K})^4 - (320 \text{ K})^4] \\ &= \mathbf{-234 \text{ W/m}^2}\end{aligned}$$

Discussion Note that the surface of an ordinary gray material of high absorptivity gains heat at a rate of 307 W/m^2 . The amount of heat gain increases to 575 W/m^2 when the surface is coated with a selective material that has the same absorptivity for solar radiation but a low emissivity for infrared radiation. Also note that the surface of an ordinary gray material of high reflectivity still gains heat at a rate of 34 W/m^2 . When the surface is coated with a selective material that has the same reflectivity for solar radiation but a high emissivity for infrared radiation, the surface loses 234 W/m^2 instead. Therefore, the temperature of the surface will decrease when a selective reflector surface is used.

TOPIC OF SPECIAL INTEREST*

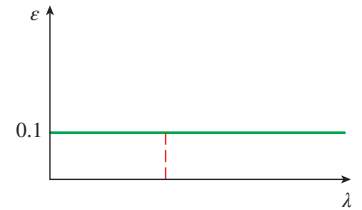
Solar Heat Gain Through Windows

The sun is the primary heat source of the earth, and the solar irradiance on a surface normal to the sun's rays beyond the earth's atmosphere at the mean earth-sun distance of 149.5 million km is called the total solar irradiance or solar constant. The accepted value of the solar constant is 1373 W/m^2 ($435.4 \text{ Btu/h} \cdot \text{ft}^2$), but its value changes by 3.5 percent from a maximum of 1418 W/m^2 on January 3 when the earth is closest to the sun, to a minimum of 1325 W/m^2 on July 4 when the earth is farthest away from the sun. The spectral distribution of solar radiation beyond the earth's atmosphere resembles the energy emitted by a blackbody at 5780°C , with about 9 percent of the energy contained in the ultraviolet region (at wavelengths between 0.29 to $0.4 \mu\text{m}$), 39 percent in the visible region (0.4 to $0.7 \mu\text{m}$), and the remaining 52 percent in the near-infrared region (0.7 to $3.5 \mu\text{m}$). The peak radiation occurs at a wavelength of about $0.48 \mu\text{m}$, which corresponds to the green color portion of the visible spectrum. Obviously a glazing material that transmits the visible part of the spectrum while absorbing the infrared portion is ideally suited for an application that calls for maximum daylight and minimum solar heat gain. Surprisingly, the ordinary window glass approximates this behavior remarkably well (Fig. 12–47).

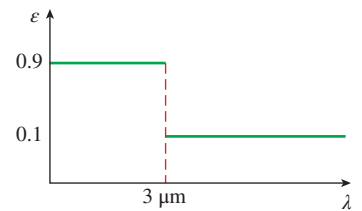
Part of the solar radiation entering the earth's atmosphere is scattered and absorbed by air and water vapor molecules, dust particles, and water droplets in the clouds, and thus the solar radiation incident on earth's surface is less than the solar constant. The extent of the attenuation of solar radiation depends on the length of the path of the rays through the



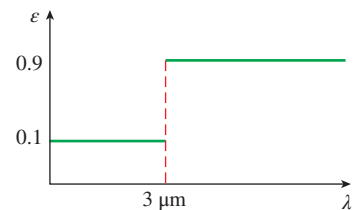
(a)



(b)



(c)

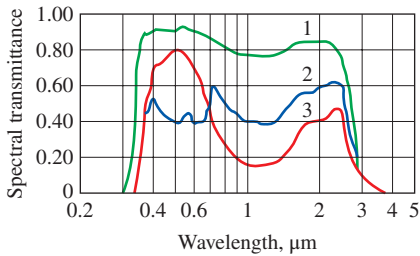


(d)

FIGURE 12–46

Graphical representation of the spectral emissivities of the four surfaces considered in Example 12–6.

* This section can be skipped without a loss of continuity.



1. 3-mm regular sheet
2. 6-mm gray heat-absorbing plate/float
3. 6-mm green heat-absorbing plate/float

FIGURE 12-47

The variation of the transmittance of typical architectural glass with wavelength.

Source: ASHRAE Handbook of Fundamentals, Chap. 27, Fig. 11.

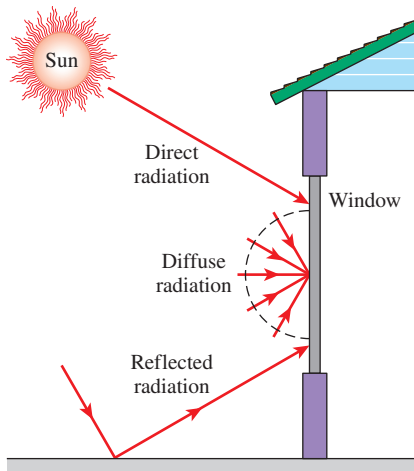


FIGURE 12-48

Direct, diffuse, and reflected components of solar radiation incident on a window.

atmosphere as well as the composition of the atmosphere (the clouds, dust, humidity, and smog) along the path. Most ultraviolet radiation is absorbed by the ozone in the upper atmosphere. At a solar altitude of 41.8° , the total energy of direct solar radiation incident at sea level on a clear day consists of about 3 percent ultraviolet, 38 percent visible, and 59 percent infrared radiation.

The part of solar radiation that reaches the earth's surface without being scattered or absorbed is the *direct radiation*. Solar radiation that is scattered or reemitted by the constituents of the atmosphere is the *diffuse radiation*. Direct radiation comes directly from the sun following a straight path, whereas diffuse radiation comes from all directions in the sky. The entire radiation reaching the ground on an overcast day is diffuse radiation. The radiation reaching a surface, in general, consists of three components: direct radiation, diffuse radiation, and radiation reflected onto the surface from surrounding surfaces (Fig. 12-48). Common surfaces such as grass, trees, rocks, and concrete reflect about 20 percent of the radiation while absorbing the rest. Snow-covered surfaces, however, reflect 70 percent of the incident radiation. Radiation incident on a surface that does not have a direct view of the sun consists of diffuse and reflected radiation. Therefore, at solar noon, solar radiations incident on the east, west, and north surfaces of a south-facing house are identical since they all consist of diffuse and reflected components. The difference between the radiations incident on the south and north walls in this case gives the magnitude of direct radiation incident on the south wall.

When solar radiation strikes a glass surface, part of it (about 8 percent for uncoated clear glass) is reflected back to outdoors, part of it (5 to 50 percent, depending on composition and thickness) is absorbed within the glass, and the remainder is transmitted indoors, as shown in Fig. 12-49. The conservation of energy principle requires that the sum of the transmitted, reflected, and absorbed solar radiations be equal to the incident solar radiation. That is,

$$\tau_s + \rho_s + \alpha_s = 1$$

where τ_s is the transmissivity, ρ_s is the reflectivity, and α_s is the absorptivity of the glass for solar energy, which are the fractions of incident solar radiation transmitted, reflected, and absorbed, respectively. The standard 3-mm (1/8-in)-thick single-pane double-strength clear window glass transmits 86 percent, reflects 8 percent, and absorbs 6 percent of the solar energy incident on it. The radiation properties of materials are usually given for normal incidence, but can also be used for radiation incident at other angles since the transmissivity, reflectivity, and absorptivity of the glazing materials remain essentially constant for incidence angles up to about 60° from the normal.

The hourly variation of solar radiation incident on the walls and windows of a house is given in Table 12-4. Solar radiation that is transmitted indoors is partially absorbed and partially reflected each time it strikes a surface, but all of it is eventually absorbed as sensible heat by the furniture,

walls, people, and so forth. Therefore, the solar energy transmitted inside a building represents a heat gain for the building. Also, the solar radiation absorbed by the glass is subsequently transferred to the indoors and outdoors by convection and radiation. The sum of the *transmitted* solar radiation and the portion of the *absorbed* radiation that flows indoors constitutes the **solar heat gain** of the building.

The fraction of incident solar radiation that enters through the glazing is called the **solar heat gain coefficient** (SHGC) and is expressed as

$$\begin{aligned} \text{SHGC} &= \frac{\text{Solar heat gain through the window}}{\text{Solar radiation incident on the window}} \\ &= \frac{\dot{q}_{\text{solar, gain}}}{\dot{q}_{\text{solar, incident}}} = \tau_s + f_i \alpha_s \end{aligned} \quad (12-55)$$

where α_s is the solar absorptivity of the glass and f_i is the inward flowing fraction of the solar radiation absorbed by the glass. Therefore, the dimensionless quantity SHGC is the sum of the fractions of the directly transmitted (τ_s) and the absorbed and reemitted ($f_i \alpha_s$) portions of solar radiation incident on the window. The value of SHGC ranges from 0 to 1, with 1 corresponding to an opening in the wall (or the ceiling) with no glazing. When the SHGC of a window is known, the total solar heat gain through that window is determined from

$$\dot{Q}_{\text{solar, gain}} = \text{SHGC} \times A_{\text{glazing}} \times \dot{q}_{\text{solar, incident}} \quad (\text{W}) \quad (12-56)$$

where A_{glazing} is the glazing area of the window and $\dot{q}_{\text{solar, incident}}$ is the solar heat flux incident on the outer surface of the window in W/m^2 .

Another way of characterizing the solar transmission characteristics of different kinds of glazing and shading devices is to compare them to a well-known glazing material that can serve as a base case. This is done by taking the standard 3-mm(1/8-in)-thick double-strength clear window glass sheet whose SHGC is 0.87 as the *reference glazing* and defining a **shading coefficient** SC as

$$\begin{aligned} \text{SC} &= \frac{\text{Solar heat gain of product}}{\text{Solar heat gain of reference glazing}} \\ &= \frac{\text{SHGC}}{\text{SHGC}_{\text{ref}}} = \frac{\text{SHGC}}{0.87} = 1.15 \times \text{SHGC} \end{aligned} \quad (12-57)$$

Therefore, the shading coefficient of a single-pane clear glass window is $\text{SC} = 1.0$. The shading coefficients of other commonly used fenestration products are given in Table 12–5 for summer design conditions. The values for winter design conditions may be slightly lower because of the higher heat transfer coefficients on the outer surface due to high winds and thus higher rate of outward flow of solar heat absorbed by the glazing, but the difference is small.

Note that the larger the shading coefficient, the smaller the shading effect, and thus the larger the amount of solar heat gain. A glazing material with a large shading coefficient allows a large fraction of solar radiation to come in.

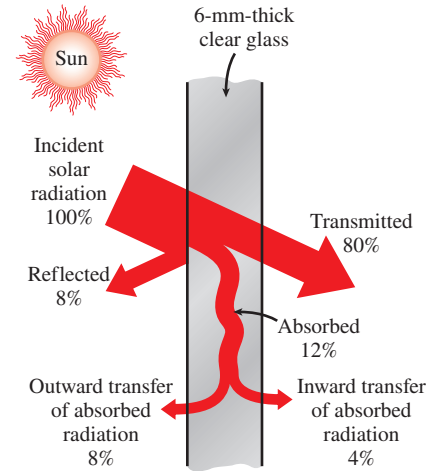


FIGURE 12-49

Distribution of solar radiation incident on a clear glass.

TABLE 12-4

Hourly variation of solar radiation incident on various surfaces and the daily totals throughout the year at 40° latitude

		Solar Radiation Incident on the Surface, * W/m ²															
		Solar Time															
Date	Direction of Surface	5	6	7	8	9	10	11	12 noon	13	14	15	16	17	18	19	Daily Total
Jan.	N	0	0	0	20	43	66	68	71	68	66	43	20	0	0	0	446
	NE	0	0	0	63	47	66	68	71	68	59	43	20	0	0	0	489
	E	0	0	0	402	557	448	222	76	68	59	43	20	0	0	0	1863
	SE	0	0	0	483	811	875	803	647	428	185	48	20	0	0	0	4266
	S	0	0	0	271	579	771	884	922	884	771	579	271	0	0	0	5897
	SW	0	0	0	20	48	185	428	647	803	875	811	483	0	0	0	4266
	W	0	0	0	20	43	59	68	76	222	448	557	402	0	0	0	1863
	NW	0	0	0	20	43	59	68	71	68	66	47	63	0	0	0	489
	Horizontal	0	0	0	51	198	348	448	482	448	348	198	51	0	0	0	2568
	Direct	0	0	0	446	753	865	912	926	912	865	753	446	0	0	0	—
Apr.	N	0	41	57	79	97	110	120	122	120	110	97	79	57	41	0	1117
	NE	0	262	508	462	291	134	123	122	120	110	97	77	52	17	0	2347
	E	0	321	728	810	732	552	293	131	120	110	97	77	52	17	0	4006
	SE	0	189	518	682	736	699	582	392	187	116	97	77	52	17	0	4323
	S	0	18	59	149	333	437	528	559	528	437	333	149	59	18	0	3536
	SW	0	17	52	77	97	116	187	392	582	699	736	682	518	189	0	4323
	W	0	17	52	77	97	110	120	392	293	552	732	810	728	321	0	4006
	NW	0	17	52	77	97	110	120	122	123	134	291	462	508	262	0	2347
	Horizontal	0	39	222	447	640	786	880	911	880	786	640	447	222	39	0	6938
	Direct	0	282	651	794	864	901	919	925	919	901	864	794	651	282	0	—
July	N	3	133	109	103	117	126	134	138	134	126	117	103	109	133	3	1621
	NE	8	454	590	540	383	203	144	138	134	126	114	95	71	39	0	3068
	E	7	498	739	782	701	531	294	149	134	126	114	95	71	39	0	4313
	SE	2	248	460	580	617	576	460	291	155	131	114	95	71	39	0	3849
	S	0	39	76	108	190	292	369	395	369	292	190	108	76	39	0	2552
	SW	0	39	71	95	114	131	155	291	460	576	617	580	460	248	2	3849
	W	0	39	71	95	114	126	134	149	294	531	701	782	739	498	7	4313
	NW	0	39	71	95	114	126	134	138	144	203	383	540	590	454	8	3068
	Horizontal	1	115	320	528	702	838	922	949	922	838	702	528	320	115	1	3902
	Direct	7	434	656	762	818	850	866	871	866	850	818	762	656	434	7	—
Oct.	N	0	0	7	40	62	77	87	90	87	77	62	40	7	0	0	453
	NE	0	0	74	178	84	80	87	90	87	87	62	40	7	0	0	869
	E	0	0	163	626	652	505	256	97	87	87	62	40	7	0	0	2578
	SE	0	0	152	680	853	864	770	599	364	137	66	40	7	0	0	4543
	S	0	0	44	321	547	711	813	847	813	711	547	321	44	0	0	5731
	SW	0	0	7	40	66	137	364	599	770	864	853	680	152	0	0	4543
	W	0	0	7	40	62	87	87	97	256	505	652	626	163	0	0	2578
	NW	0	0	7	40	62	87	87	90	87	80	84	178	74	0	0	869
	Horizontal	0	0	14	156	351	509	608	640	608	509	351	156	14	0	0	3917
	Direct	0	0	152	643	811	884	917	927	917	884	811	643	152	0	0	—

*Multiply by 0.3171 to convert to Btu/h·ft².

Values given are for the 21st of the month for average days with no clouds. The values can be up to 15 percent higher at high elevations under very clear skies and up to 30 percent lower at very humid locations with very dusty industrial atmospheres. Daily totals are obtained using Simpson's rule for integration with 10-min time intervals. Solar reflectance of the ground is assumed to be 0.2, which is valid for old concrete, crushed rock, and bright green grass. For a specified location, use solar radiation data obtained for that location. The direction of a surface indicates the direction a vertical surface is facing. For example, W represents the solar radiation incident on a west-facing wall per unit area of the wall.

Solar time may deviate from the local time. Solar noon at a location is the time when the sun is at the highest location (and thus when the shadows are shortest). Solar radiation data are symmetric about the solar noon: the value on a west wall before the solar noon is equal to the value on an east wall two hours after the solar noon.

Source: ASHRAE Handbook of Fundamentals, Chap. 27, Table 15

Shading devices are classified as *internal shading* and *external shading*, depending on whether the shading device is placed *inside* or *outside*. External shading devices are more effective in reducing the solar heat gain since they intercept the sun's rays before they reach the glazing. The solar heat gain through a window can be reduced by as much as 80 percent by exterior shading. Roof overhangs have long been used for exterior shading of windows. The sun is high in the horizon in summer and low in winter. A properly sized roof overhang or a horizontal projection blocks off the sun's rays completely in summer while letting in most of them in winter, as shown in Fig. 12–50. Such shading structures can reduce the solar heat gain on the south, southeast, and southwest windows in the northern hemisphere considerably. A window can also be shaded from outside by vertical or horizontal architectural projections, insect or shading screens, and sun screens. To be effective, air must be able to move freely around the exterior device to carry away the heat absorbed by the shading and the glazing materials.

Some type of internal shading is used in most windows to provide privacy and aesthetic effects as well as some control over solar heat gain. Internal shading devices reduce solar heat gain by reflecting transmitted solar radiation back through the glazing before it can be absorbed and converted into heat in the building.

Draperies reduce the annual heating and cooling loads of a building by 5 to 20 percent, depending on the type and the user habits. In summer, they reduce heat gain primarily by reflecting back direct solar radiation (Fig. 12–51). The semiclosed airspace formed by the draperies serves as an additional barrier against heat transfer, resulting in a lower *U*-factor for the window and thus a lower rate of heat transfer in summer and winter. The solar optical properties of draperies can be measured accurately, or they can be obtained directly from the manufacturers. The shading coefficient of draperies depends on the openness factor, which is the ratio of the open area between the fibers that permits the sun's rays to pass freely, to the total area of the fabric. Tightly woven fabrics allow little direct radiation to pass through, and thus they have a small openness factor. The *reflectance* of the surface of the drapery facing the glazing has a major effect on the amount of solar heat gain. *Light-colored* draperies made of closed or tightly woven fabrics maximize the back reflection and thus minimize the solar gain. *Dark-colored* draperies made of open or semi-open woven fabrics, on the other hand, minimize the back reflection and thus maximize the solar gain.

The shading coefficients of drapes also depend on the way they are hung. Usually, the width of drapery used is twice the width of the draped area to allow folding of the drapes and to give them their characteristic “full” or “wavy” appearance. A flat drape behaves like an ordinary window shade. A flat drape has a higher reflectance and thus a lower shading coefficient than a full drape.

External shading devices such as overhangs and tinted glazings do not require operation, and they provide reliable service over a long time without significant degradation during their service life. Their operation does not depend on a person or an automated system, and these passive shading devices are considered fully effective when determining the peak cooling load and the

TABLE 12–5

Shading coefficient SC and solar transmissivity τ_{solar} for some common glass types for summer design conditions

	<i>Nominal Thickness</i>			
Type of Glazing	mm	in	τ_{solar}	SC*
<i>(a) Single Glazing</i>				
Clear	3	$\frac{1}{8}$	0.86	1.0
	6	$\frac{1}{4}$	0.78	0.95
	10	$\frac{3}{8}$	0.72	0.92
	13	$\frac{1}{2}$	0.67	0.88
Heat absorbing	3	$\frac{1}{8}$	0.64	0.85
	6	$\frac{1}{4}$	0.46	0.73
	10	$\frac{3}{8}$	0.33	0.64
	13	$\frac{1}{2}$	0.24	0.58
<i>(b) Double Glazing</i>				
Clear in,	3 ^a	$\frac{1}{8}$	0.71 ^b	0.88
clear out	6	$\frac{1}{4}$	0.61	0.82
Clear in, heat absorbing out ^c	6	$\frac{1}{4}$	0.36	0.58

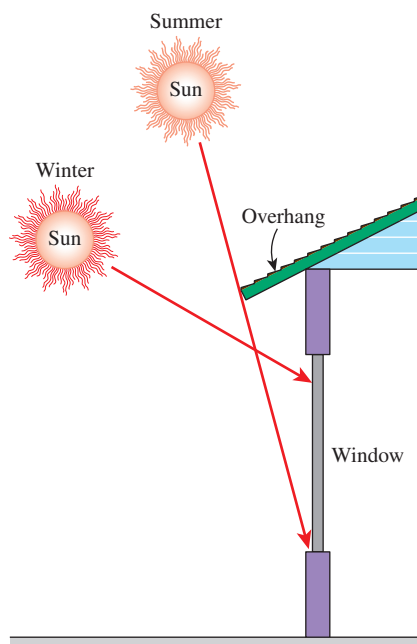
*Multiply by 0.87 to obtain SHGC.

^aThe thickness of each pane of glass.

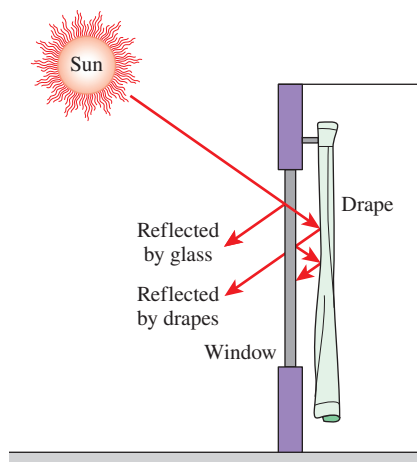
^bCombined transmittance for assembled unit.

^cRefers to gray-, bronze-, and green-tinted heat-absorbing float glass.

Source: ASHRAE Handbook of Fundamentals, Chap. 27, Table 11

**FIGURE 12-50**

A properly sized overhang blocks off the sun's rays completely in summer while letting them in in winter.

**FIGURE 12-51**

Draperies reduce heat gain in summer by reflecting back solar radiation and reduce heat loss in winter by forming an airspace before the window.

annual energy use. The effectiveness of manually operated shading devices, on the other hand, varies greatly depending on the user's habits, and this variation should be considered when evaluating performance.

The primary function of an indoor shading device is to provide *thermal comfort* for the occupants. An unshaded window glass allows most of the incident solar radiation in, and it also dissipates part of the solar energy it absorbs by emitting infrared radiation to the room. The emitted radiation and the transmitted direct sunlight may bother the occupants near the window. In winter, the temperature of the glass is lower than the room air temperature, causing excessive heat loss by radiation from the occupants. A shading device allows the control of direct solar and infrared radiation while providing various degrees of privacy and outward vision. The shading device is also at a higher temperature than the glass in winter and thus reduces radiation loss from occupants. *Glare* from draperies can be minimized by using off-white colors. Indoor shading devices, especially draperies made of a closed-weave fabric, are effective in reducing *sounds* that originate in the room, but they are not as effective against the sounds coming from outside.

The type of climate in an area usually dictates the types of windows to be used in buildings. In *cold climates* where the heating load is much larger than the cooling load, the windows should have the highest transmissivity for the entire solar spectrum, and a high reflectivity (or low emissivity) for the far infrared radiation emitted by the walls and furnishings of the room. Low-e windows are well suited for such heating-dominated buildings. Properly designed and operated windows allow more heat into the building over a heating season than it loses, making them energy contributors rather than energy losers. In *warm climates* where the cooling load is much larger than the heating load, the windows should allow the visible solar radiation (light) in but should block off the infrared solar radiation. Such windows can reduce the solar heat gain by 60 percent with no appreciable loss in daylighting. This behavior is approximated by window glazings that are coated with a heat-absorbing film outside and a low-e film inside (Fig. 12-52). Properly selected windows can reduce the cooling load by 15 to 30 percent compared to windows with clear glass.

Note that radiation heat transfer between a room and its windows is proportional to the emissivity of the glass surface facing the room, ϵ_{glass} , and can be expressed as

$$\dot{Q}_{\text{rad, room-window}} = \epsilon_{\text{glass}} A_{\text{glass}} \sigma (T_{\text{room}}^4 - T_{\text{glass}}^4) \quad (12-58)$$

Therefore, a low-e interior glass will reduce the heat loss by radiation in winter ($T_{\text{glass}} < T_{\text{room}}$) and heat gain by radiation in summer ($T_{\text{glass}} > T_{\text{room}}$).

Tinted glass and glass coated with reflective films reduce solar heat gain in summer and heat loss in winter. The conductive heat gains or losses can be minimized by using multiple-pane windows. Double-pane windows are usually called for in climates where the winter design temperature is less than 7°C (45°F). Double-pane windows with tinted or reflective films are commonly used in buildings with large window areas. Clear glass is preferred for showrooms since it affords maximum visibility from outside, but bronze-, gray-, and green-colored glass are preferred in office buildings since they provide considerable privacy while reducing glare.

EXAMPLE 12-7 Installing Reflective Films on Windows

A manufacturing facility located at 45°N latitude has a glazing area of 40 m² that consists of double-pane windows made of clear glass (SHGC = 0.766). To reduce the solar heat gain in summer, a reflective film that reduces the SHGC to 0.261 is considered. The cooling season consists of June, July, August, and September, and the heating season October through April. The average daily solar heat fluxes incident on the west side at this latitude are 1.86, 2.66, 3.43, 4.00, 4.36, 5.13, 4.31, 3.93, 3.28, 2.80, 1.84, and 1.54 kWh/day·m² for January through December, respectively. Also, the unit cost of electricity and natural gas are \$0.08/kWh and \$0.50/therm, respectively. If the coefficient of performance (COP) of the cooling system is 2.5 and efficiency of the furnace is 0.8, determine the net annual cost savings due to installing reflective coating on the windows. Also, determine the simple payback period if the installation cost of reflective film is \$20/m² (Fig. 12–53).

SOLUTION The net annual cost savings due to installing reflective film on the west windows of a building and the simple payback period are to be determined.

Assumptions 1 The calculations given below are for an average year. 2 The unit costs of electricity and natural gas remain constant.

Analysis Using the daily averages for each month and noting the number of days of each month, the total solar heat flux incident on the glazing during summer and winter months are determined to be

$$\begin{aligned} Q_{\text{solar, summer}} &= 5.13 \times 30 + 4.31 \times 31 + 3.93 \times 31 + 3.28 \times 30 = 508 \text{ kWh/year} \\ Q_{\text{solar, winter}} &= 2.80 \times 31 + 1.84 \times 30 + 1.54 \times 31 + 1.86 \times 31 \\ &\quad + 2.66 \times 28 + 3.43 \times 31 + 4.00 \times 30 \\ &= 548 \text{ kWh/year} \end{aligned}$$

Then the decrease in the annual cooling load and the increase in the annual heating load due to the reflective film become

$$\begin{aligned} \text{Cooling load decrease} &= Q_{\text{solar, summer}} A_{\text{glazing}} (\text{SHGC}_{\text{without film}} - \text{SHGC}_{\text{with film}}) \\ &= (508 \text{ kWh/year})(40 \text{ m}^2)(0.766 - 0.261) \\ &= 10,262 \text{ kWh/year} \\ \text{Heating load increase} &= Q_{\text{solar, winter}} A_{\text{glazing}} (\text{SHGC}_{\text{without film}} - \text{SHGC}_{\text{with film}}) \\ &= (548 \text{ kWh/year})(40 \text{ m}^2)(0.766 - 0.261) \\ &= 11,070 \text{ kWh/year} = 377.7 \text{ therms/year} \end{aligned}$$

since 1 therm = 29.31 kWh. The corresponding decrease in cooling costs and the increase in heating costs are

$$\begin{aligned} \text{Decrease in cooling costs} &= (\text{Cooling load decrease})(\text{Unit cost of electricity})/\text{COP} \\ &= (10,262 \text{ kWh/year})(\$0.08/\text{kWh})/2.5 = \$328/\text{year} \\ \text{Increase in heating costs} &= (\text{Heating load increase})(\text{Unit cost of fuel})/\text{Efficiency} \\ &= (377.7 \text{ therms/year})(\$0.50/\text{therm})/0.80 = \$236/\text{year} \end{aligned}$$

Then the net annual cost savings due to the reflective film become

$$\begin{aligned} \text{Cost savings} &= \text{Decrease in cooling costs} - \text{Increase in heating costs} \\ &= \$328 - \$236 = \mathbf{\$92/\text{year}} \end{aligned}$$

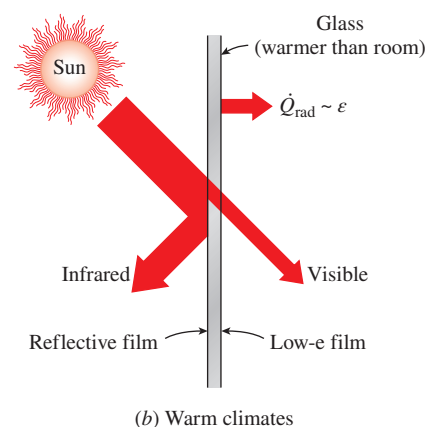
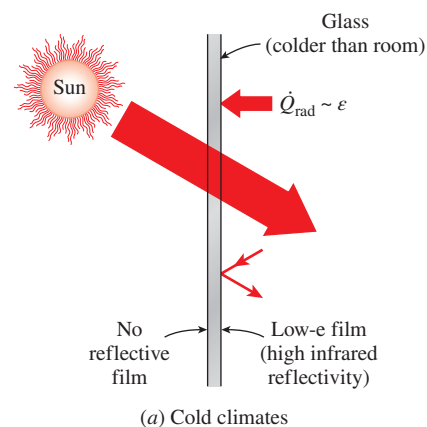


FIGURE 12-52

Radiation heat transfer between a room and its windows is proportional to the emissivity of the glass surface, and low-e coatings on the inner surface of the windows reduce heat loss in winter and heat gain in summer.

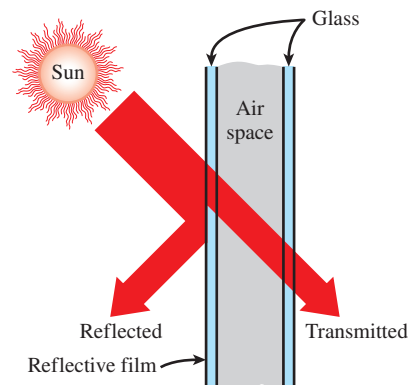


FIGURE 12-53

Schematic for Example 12-7.

The implementation cost of installing films is

$$\text{Implementation cost} = (\$20/\text{m}^2)(40 \text{ m}^2) = \$800$$

This gives a simple payback period of

$$\text{Simple payback period} = \frac{\text{Implementation cost}}{\text{Annual cost savings}} = \frac{\$800}{\$92/\text{year}} = \mathbf{8.7 \text{ years}}$$

Discussion The reflective film will pay for itself in this case in about nine years. This may be unacceptable to most manufacturers since they are not usually interested in any energy conservation measure that does not pay for itself within three years. But the enhancement in thermal comfort and thus the resulting increase in productivity often makes it worthwhile to install reflective film.

SUMMARY

Radiation propagates in the form of electromagnetic waves. The *frequency* ν and *wavelength* λ of electromagnetic waves in a medium are related by $\lambda = c/\nu$, where c is the speed of propagation in that medium. All matter continuously emits *thermal radiation* as a result of vibrational and rotational motions of molecules, atoms, and electrons of a substance.

A *blackbody* is defined as a perfect emitter and absorber of radiation. At a specified temperature and wavelength, no surface can emit more energy than a blackbody. A blackbody absorbs *all* incident radiation, regardless of wavelength and direction. The radiation energy emitted by a blackbody per unit time and per unit surface area is called the *blackbody emissive power* E_b and is expressed by the *Stefan–Boltzmann law* as

$$E_b(T) = \sigma T^4$$

where $\sigma = 5.670 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is the *Stefan–Boltzmann constant* and T is the absolute temperature of the surface in K. At any specified temperature, the spectral blackbody emissive power $E_{b\lambda}$ increases with wavelength, reaches a peak, and then decreases with increasing wavelength. The wavelength at which the peak occurs for a specified temperature is given by *Wien's displacement law* as

$$(\lambda T)_{\text{max power}} = 2897.8 \mu\text{m} \cdot \text{K}$$

The *blackbody radiation function* f_λ represents the fraction of radiation emitted by a blackbody at temperature T in the wavelength band from $\lambda = 0$ to λ . The fraction of radiation energy emitted by a blackbody at temperature T over a finite wavelength band from $\lambda = \lambda_1$ to $\lambda = \lambda_2$ is determined from

$$f_{\lambda_1-\lambda_2}(T) = f_{\lambda_2}(T) - f_{\lambda_1}(T)$$

where $f_{\lambda_1}(T)$ and $f_{\lambda_2}(T)$ are the blackbody radiation functions corresponding to $\lambda_1 T$ and $\lambda_2 T$, respectively.

The magnitude of a viewing angle in space is described by solid angle expressed as $d\omega = dA_n/r^2$. The *radiation intensity* for emitted radiation $I_e(\theta, \phi)$ is defined as the rate at which radiation energy is emitted in the (θ, ϕ) direction per unit area normal to this direction and per unit solid angle about this direction. The *radiation flux* for emitted radiation is the *emissive power* E and is expressed as

$$E = \int_{\text{hemisphere}} dE = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_e(\theta, \phi) \cos \theta \sin \theta d\theta d\phi$$

For a *diffusely emitting* surface, intensity is independent of direction, and thus

$$E = \pi I_e$$

For a blackbody, we have

$$E_b = \pi I_b \quad \text{and} \quad I_b(T) = \frac{E_b(T)}{\pi} = \frac{\sigma T^4}{\pi}$$

The radiation flux incident on a surface from all directions is *irradiation* G , and for diffusely incident radiation of intensity I_i it is expressed as

$$G = \pi I_i$$

The rate at which radiation energy leaves a unit area of a surface in all directions is *radiosity* J , and for a surface that is both a diffuse emitter and a diffuse reflector it is expressed as

$$J = \pi I_{e+r}$$

where I_{e+r} is the sum of the emitted and reflected intensities. The spectral emitted quantities are related to total quantities as

$$I_e = \int_0^\infty I_{\lambda,e} d\lambda \quad \text{and} \quad E = \int_0^\infty E_\lambda d\lambda$$

They reduce for a diffusely emitting surface and for a blackbody to

$$E_\lambda = \pi I_{\lambda, e} \quad \text{and} \quad E_{b\lambda}(\lambda, T) = \pi I_{b\lambda}(\lambda, T)$$

The *emissivity* of a surface represents the ratio of the radiation emitted by the surface at a given temperature to the radiation emitted by a blackbody at the same temperature. Different emissivities are defined as

Spectral directional emissivity:

$$\varepsilon_{\lambda, \theta}(\lambda, \theta, \phi, T) = \frac{I_{\lambda, e}(\lambda, \theta, \phi, T)}{I_{b\lambda}(\lambda, T)}$$

Total directional emissivity:

$$\varepsilon_\theta(\theta, \phi, T) = \frac{I_e(\theta, \phi, T)}{I_b(T)}$$

Spectral hemispherical emissivity:

$$\varepsilon_\lambda(\lambda, T) = \frac{E_\lambda(\lambda, T)}{E_{b\lambda}(\lambda, T)}$$

Total hemispherical emissivity:

$$\varepsilon(T) = \frac{E(T)}{E_b(T)} = \frac{\int_0^\infty \varepsilon_\lambda(\lambda, T) E_{b\lambda}(\lambda, T) d\lambda}{\sigma T^4}$$

Emissivity can also be expressed as a step function by dividing the spectrum into a sufficient number of *wavelength bands* of constant emissivity as, for example,

$$\varepsilon(T) = \varepsilon_1 f_{0-\lambda_1}(T) + \varepsilon_2 f_{\lambda_1-\lambda_2}(T) + \varepsilon_3 f_{\lambda_2-\infty}(T)$$

The *total hemispherical emissivity* ε of a surface is the average emissivity over all directions and wavelengths.

When radiation strikes a surface, part of it is absorbed, part of it is reflected, and the remaining part, if any, is transmitted. The fraction of incident radiation (intensity I_i or irradiation G) absorbed by the surface is called the *absorptivity*, the fraction reflected by the surface is called the *reflectivity*, and the fraction transmitted is called the *transmissivity*. Various absorptivities, reflectivities, and transmissivities for a medium are expressed as

$$\alpha_{\lambda, \theta}(\lambda, \theta, \phi) = \frac{I_{\lambda, \text{abs}}(\lambda, \theta, \phi)}{I_{\lambda, i}(\lambda, \theta, \phi)} \quad \text{and} \quad \rho_{\lambda, \theta}(\lambda, \theta, \phi) = \frac{I_{\lambda, \text{ref}}(\lambda, \theta, \phi)}{I_{\lambda, i}(\lambda, \theta, \phi)}$$

$$\alpha_\lambda(\lambda) = \frac{G_{\lambda, \text{abs}}(\lambda)}{G_\lambda(\lambda)}, \quad \rho_\lambda(\lambda) = \frac{G_{\lambda, \text{ref}}(\lambda)}{G_\lambda(\lambda)}, \quad \text{and} \quad \tau_\lambda(\lambda) = \frac{G_{\lambda, \text{tr}}(\lambda)}{G_\lambda(\lambda)}$$

$$\alpha = \frac{G_{\text{abs}}}{G}, \quad \rho = \frac{G_{\text{ref}}}{G}, \quad \text{and} \quad \tau = \frac{G_{\text{tr}}}{G}$$

The consideration of wavelength and direction dependence of properties makes radiation calculations very complicated. Therefore, the *gray* and *diffuse* approximations are commonly utilized in radiation calculations. A surface is said to be *diffuse* if its properties are independent of direction and *gray* if its properties are independent of wavelength.

The sum of the absorbed, reflected, and transmitted fractions of radiation energy must be equal to unity,

$$\alpha + \rho + \tau = 1$$

For *opaque* surfaces ($\tau = 0$): $\alpha + \rho = 1$

For surfaces with *no reflectance* ($\rho = 0$): $\alpha + \tau = 1$

Surfaces are usually assumed to reflect in a perfectly *specular* or *diffuse* manner for simplicity. In *specular* (or *mirrorlike*) *reflection*, the angle of reflection equals the angle of incidence of the radiation beam. In *diffuse reflection*, radiation is reflected equally in all directions. Reflection from smooth and polished surfaces approximates specular reflection, whereas reflection from rough surfaces approximates diffuse reflection. *Kirchhoff's law* of radiation is expressed as

$$\varepsilon_{\lambda, \theta}(T) = \alpha_{\lambda, \theta}(T), \quad \varepsilon_\lambda(T) = \alpha_\lambda(T), \quad \text{and} \quad \varepsilon(T) = \alpha(T)$$

Gas molecules and the suspended particles in the atmosphere emit radiation as well as absorbing it. The atmosphere can be treated as a blackbody at some lower fictitious temperature, called the *effective sky temperature* T_{sky} that emits an equivalent amount of radiation energy,

$$G_{\text{sky}} = \sigma T_{\text{sky}}^4$$

The net rate of radiation heat transfer to a surface exposed to solar and atmospheric radiation is determined from an energy balance expressed as

$$\dot{q}_{\text{net, rad}} = \alpha_s G_{\text{solar}} + \varepsilon \sigma (T_{\text{sky}}^4 - T_s^4)$$

where T_s is the surface temperature in K, and ε is the surface emissivity at room temperature.

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PROBLEMS*

Electromagnetic and Thermal Radiation

- 12-1C** What is an electromagnetic wave? How does it differ from a sound wave?
- 12-2C** By what properties is an electromagnetic wave characterized? How are these properties related to each other?
- 12-3C** What is thermal radiation? How does it differ from the other forms of electromagnetic radiation?
- 12-4C** How does microwave cooking differ from conventional cooking?
- 12-5C** What is visible light? How does it differ from the other forms of electromagnetic radiation?
- 12-6C** What is the cause of color? Why do some objects appear blue to the eye while others appear red? Is the color of a surface at room temperature related to the radiation it emits?
- 12-7C** Why do skiers get sunburned so easily?
- 12-8C** How do ultraviolet and infrared radiation differ? Do you think your body emits any radiation in the ultraviolet range? Explain.
- 12-9C** Why is radiation usually treated as a surface phenomenon?
- 12-10** A cordless telephone is designed to operate at a frequency of 8.5×10^8 Hz. Determine the wavelength of these telephone waves.
- 12-11** Electricity is generated and transmitted in power lines at a frequency of 50 Hz (1 Hz = 1 cycle per second). Determine the wavelength of the electromagnetic waves generated by the passage of electricity in power lines.
- 12-12** The speed of light in vacuum is given to be 3.0×10^8 m/s. Determine the speed of light in air ($n = 1$), in water ($n = 1.33$), and in glass ($n = 1.5$).
- 12-13** A radio station is broadcasting radio waves at a wavelength of 150 m. Determine the frequency of these waves.

12-14 A microwave oven is designed to operate at a frequency of 2.2×10^9 Hz. Determine the wavelength of these microwaves and the energy of each microwave.

12-15 Consider a radio wave with a wavelength of 10^7 μm and a γ -ray with a wavelength of 10^{-7} μm . Determine the photon energies of the radio wave and the γ -ray, and the photon energy ratio of the γ -ray to the radio wave.

12-16 An electromagnetic wave with a wavelength of 0.5 μm is being propagated in different media: air, water, and glass. The refractive index of air, water, and glass are 1, 1.33, and 1.5 respectively. Determine the photon energy, in eV, of the electromagnetic wave in each medium ($1 \text{ eV} = 1.6022 \times 10^{-19} \text{ J}$).

12-17 The electromagnetic spectrum that lies between 0.40 and 0.76 μm is what we call visible light. Within this spectrum, the color violet has the shortest wavelength while the color red has the longest wavelength. Determine which of these colors, violet ($\lambda = 0.40$ μm) or red ($\lambda = 0.76$ μm), propagates more photon energy.

Blackbody Radiation

- 12-18C** What is a blackbody? Does a blackbody actually exist?
- 12-19C** Define the total and spectral blackbody emissive powers. How are they related to each other? How do they differ?
- 12-20C** Why did we define the blackbody radiation function? What does it represent? For what is it used?
- 12-21C** Consider two identical bodies, one at 1000 K and the other at 1500 K. Which body emits more radiation in the shorter-wavelength region? Which body emits more radiation at a wavelength of 20 μm ?
- 12-22** Consider a surface at a uniform temperature of 1000 K. Determine the maximum rate of thermal radiation that can be emitted by this surface, in W/m^2 .
- 12-23**  A thin-walled tube has its inner surface coated with polypropylene lining. The ASME Code for Process Piping limits the maximum use temperature for polypropylene lining to 107°C (ASME B31.3-2014, Table A323.4.3). The tube has a diameter of 5 cm. Assume the tube behaves as a blackbody, and irradiation on the tube from the surroundings is negligible. Determine the maximum amount of total radiation emission rate per unit length of the tube that can be achieved without exceeding the maximum use temperature for polypropylene lining.

* Problems designated by a “C” are concept questions, and students are encouraged to answer them all. Problems designated by an “E” are in English units, and the SI users can ignore them. Problems with the  icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the  icon are Prevention through Design problems. Problems with the  icon are Codes and Standards problems.

12-24 Consider a $20\text{-cm} \times 20\text{-cm} \times 20\text{-cm}$ cubical body at 900 K suspended in the air. Assuming the body closely approximates a blackbody, determine (a) the rate at which the cube emits radiation energy in W and (b) the spectral blackbody emissive power at a wavelength of $4\text{ }\mu\text{m}$.

12-25 A flame from a match may be approximated as a blackbody at the effective surface temperature of 1700 K , while moonlight may be approximated as a blackbody at the effective surface temperature of 4000 K , respectively. Determine the peak spectral blackbody emissive power for both lighting sources (match flame and moonlight).

12-26  The maximum use temperature for an ASTM A479 904L stainless steel bar is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Assume the bar behaves as a blackbody and determine (a) the spectral blackbody emissive powers at the limits of the electromagnetic spectrum for thermal radiation at the maximum use temperature and (b) the maximum spectral blackbody emissive power that the bar can reach without exceeding its maximum use temperature.

12-27 At a wavelength of $0.7\text{ }\mu\text{m}$, the blackbody emissive power is equal to 10^8 W/m^2 . Determine (a) the temperature of the blackbody and (b) the total emissive power at this temperature.

12-28  The maximum use temperature for an ASTM A240 410S stainless steel plate is 649°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The plate is used in a process where the maximum spectral blackbody emissive power from the plate surface is $9000\text{ W/m}^2 \cdot \mu\text{m}$. Assume the plate behaves as a blackbody, and determine whether the plate is suitable for this process.

12-29  The sun can be treated as a blackbody at 5780 K . Using appropriate software, calculate and plot the spectral blackbody emissive power $E_{b\lambda}$ of the sun versus wavelength in the range of $0.01\text{ }\mu\text{m}$ to $1000\text{ }\mu\text{m}$. Discuss the results.

12-30 A small body is placed inside a spherical stainless steel chamber with a diameter of 3 m . The chamber is evacuated and has a constant surface temperature of 500 K . Determine the radiation incident on the surface of the small body inside the chamber if the interior surface of the chamber is (a) coated black and (b) well polished.

12-31 Consider a black spherical ball, with a diameter of 25 cm , that is suspended in air. Determine the surface temperature of the ball that should be maintained in order to heat 10 kg of air from 20 to 30°C in 5 minutes.

12-32  A tube placed in a vacuum has an outer diameter of 5 cm , and its wall thickness is 1 cm . The tube wall has a thermal conductivity of $5\text{ W/m}\cdot\text{K}$. The tube inner surface is coated with polytetrafluoroethylene (PTFE) lining. The ASME Code for Process Piping limits the maximum use temperature for polypropylene lining to 260°C (ASME

B31.3-2014, Table A323.4.3). The total radiation emission rate per unit length of the tube from the tube outer surface is 680 W/m . Determine whether the use of PTFE lining is suitable for coating the tube's inner surface. Assume the tube behaves as blackbody and irradiation on the tube from the surroundings is negligible.

12-33 A thin vertical copper plate is subjected to a uniform heat flux of 1000 W/m^2 on one side, while the other side is exposed to ambient surroundings at 5°C . The surface of the plate is oxidized black and can be treated as a blackbody. The heat transfer coefficient due to natural convection on the plate surface is $5.3\text{ W/m}^2\cdot\text{K}$. Determine the surface temperature of the plate. Discuss the contribution of the net radiation heat transfer on the total heat loss from the plate.

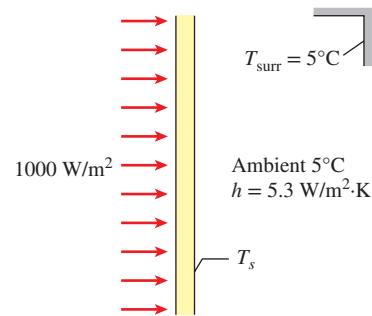


FIGURE P12-33

12-34 A circular ceramic plate that can be modeled as a blackbody is being heated by an electrical heater. The plate is 30 cm in diameter and is situated in a surrounding ambient temperature of 15°C where the natural convection heat transfer coefficient is $12\text{ W/m}^2\cdot\text{K}$. If the efficiency of the electrical heater to transfer heat to the plate is 80 percent, determine the electric power that the heater needs to keep the surface temperature of the plate at 200°C .

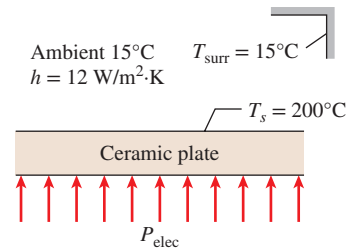


FIGURE P12-34

12-35 We wish an incandescent lightbulb to emit at least 15 percent of its energy at wavelengths shorter than $0.8\text{ }\mu\text{m}$. Determine the minimum temperature to which the filament of the lightbulb must be heated.

12-36 The temperature of the filament of an incandescent lightbulb is 2500 K . Assuming the filament to be a blackbody,

determine the fraction of the radiant energy emitted by the filament that falls in the visible range. Also, determine the wavelength at which the emission of radiation from the filament peaks.

12-37 The temperature of the filament of an incandescent lightbulb is 2800 K. Treating the filament as a blackbody, determine the fraction of the radiant energy emitted by the filament that falls in the visible range. Also, determine the wavelength at which the emission of radiation from the filament peaks.

12-38  Reconsider Prob. 12-37. Using appropriate software, investigate the effect of temperature on the fraction of radiation emitted in the visible range. Let the surface temperature vary from 1000 K to 4000 K, and plot the fraction of radiation emitted in the visible range versus the surface temperature.

12-39E Consider a black surface at 2060°F. Determine the radiation energy emitted by the black surface per unit area (in Btu/h·ft²) for $\lambda \geq 4 \mu\text{m}$.

12-40 It is desired that the radiation energy emitted by a light source reach a maximum in the blue range ($\lambda = 0.47 \mu\text{m}$). Determine the temperature of this light source and the fraction of radiation it emits in the visible range ($\lambda = 0.40 - 0.76 \mu\text{m}$).

12-41E The sun can be treated as a blackbody at an effective surface temperature of 10,400 R. Determine the rate at which infrared radiation energy ($\lambda = 0.76 - 100 \mu\text{m}$) is emitted by the sun, in Btu/h·ft².

12-42 A 3-mm-thick glass window transmits 90 percent of the radiation between $\lambda = 0.3$ and $3.0 \mu\text{m}$ and is essentially opaque for radiation at other wavelengths. Determine the rate of radiation transmitted through a 3-m $\times$ 3-m glass window from blackbody sources at (a) 5800 K and (b) 1000 K.

12-43 Daylight and candlelight may be approximated as a blackbody at the effective surface temperatures of 5800 K and 1800 K, respectively. Determine the radiation energy (in W/m²) emitted by both lighting sources (daylight and candlelight) within the visible light region (0.40 to 0.76 μm).

12-44  The maximum use temperature for an ASTM B335 nickel alloy rod is 427°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Assume the rod behaves as a blackbody. Determine the highest radiation energy per unit area that can be emitted from the rod over a wavelength band from 2 to 10 μm , without exceeding the maximum use temperature.

12-45 Consider the sun, which is considered to be a blackbody with a surface temperature of roughly 5800 K. Determine the percentage of solar energy (a) in the visible range, (b) at wavelengths shorter than the visible range, and (c) at wavelengths longer than the visible range.

Radiation Intensity

12-46C What does a solid angle represent, and how does it differ from a plane angle? What is the value of a solid angle associated with a sphere?

12-47C For a surface, how is irradiation defined? For diffusely incident radiation, how is irradiation on a surface related to the intensity of incident radiation?

12-48C For a surface, how is radiosity defined? For diffusely emitting and reflecting surfaces, how is radiosity related to the intensities of emitted and reflected radiation?

12-49C When the variation of spectral radiation quantity with wavelength is known, how is the corresponding total quantity determined?

12-50C How is the intensity of emitted radiation defined? For a diffusely emitting surface, how is the emissive power related to the intensity of emitted radiation?

12-51 A small surface of area $A_1 = 3 \text{ cm}^2$ emits radiation as a blackbody, and part of the radiation emitted by A_1 strikes another small surface of area $A_2 = 8 \text{ cm}^2$ oriented as shown in Fig. P12-51. If the rate at which radiation emitted by A_1 that strikes A_2 is measured to be $274 \times 10^{-6} \text{ W}$, determine the intensity of the radiation emitted by A_1 , and the temperature of A_1 .

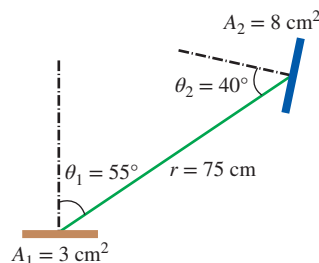


FIGURE P12-51

12-52 A small circular surface of area $A_1 = 2 \text{ cm}^2$ located at the center of a 2-m-diameter sphere emits radiation as a blackbody at $T_1 = 1000 \text{ K}$. Determine the rate at which radiation energy is streaming through a $D_2 = 1\text{-cm}$ -diameter hole located (a) on top of the sphere directly above A_1 and (b) on the side of the sphere such that the line that connects the centers of A_1 and A_2 makes 45° with surface A_1 .

12-53 Repeat Prob. 12-52 for a 4-m-diameter sphere.

12-54 A small surface of area $A_1 = 7 \text{ cm}^2$ emits radiation as a blackbody at $T_1 = 800 \text{ K}$. Part of the radiation emitted by A_1 strikes another small surface of area $A_2 = 7 \text{ cm}^2$ oriented as shown in Fig. P12-54. Determine the solid angle subtended by A_2 when viewed from A_1 , and the rate at which radiation emitted by A_1 strikes A_2 directly. What would your answer be if A_2 were directly above A_1 at a distance of 80 cm?

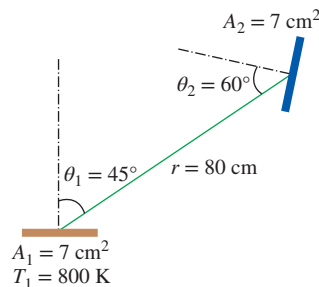


FIGURE P12-54

12-55 A small surface of area $A_1 = 1 \text{ cm}^2$ emits radiation as a blackbody at 2100 K. Determine the rate at which radiation energy is emitted through a band defined by $0 \leq \phi \leq 2\pi$ and $45 \leq \theta \leq 60^\circ$, where θ is the angle a radiation beam makes with the normal of the surface and ϕ is the azimuth angle.

12-56 The intensity of solar radiation incident on earth's surface can be expressed as $I_i = 100 \cos \theta$, where I_i has the units of $\text{W/m}^2 \cdot \text{sr}$. Determine the peak value for the intensity of incident solar radiation and the solar irradiation on earth's surface.

12-57 A small surface of area $A_1 = 3 \text{ cm}^2$ emits radiation as a blackbody with total emissive power of $E_b = 5.67 \times 10^4 \text{ W/m}^2$. Part of the radiation emitted by A_1 strikes another small surface of area $A_2 = 8 \text{ cm}^2$ oriented as shown in Fig. P12-57. Determine the rate at which radiation emitted by A_1 strikes A_2 , and the irradiation on A_2 .

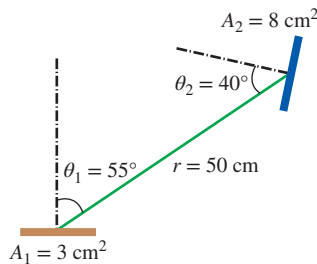


FIGURE P12-57

12-58 A small surface of area $A_1 = 3 \text{ cm}^2$ emits radiation as a blackbody at $T_1 = 1000 \text{ K}$. A radiation detector (A_2) is placed normal to the direction of viewing from surface A_1 at a distance L , as shown in Fig. P12-58. If the radiation detector is measuring an irradiation of 100 W/m^2 , determine the distance between the radiation detector and the surface A_1 .



FIGURE P12-58

12-59 A small surface of area $A_1 = 1 \text{ cm}^2$ is subjected to incident radiation of constant intensity $I_i = 2.2 \times 10^4 \text{ W/m}^2 \cdot \text{sr}$ over the entire hemisphere. Determine the rate at which radiation energy is incident on the surface through (a) $0 \leq \theta \leq 45^\circ$ and (b) $45 \leq \theta \leq 90^\circ$, where θ is the angle a radiation beam makes with the normal of the surface.

12-60 Consider an aperture of 10-mm diameter through which radiation is emitted as a blackbody with an emissive power of $2.87 \times 10^5 \text{ W/m}^2$. A radiation sensor is positioned with a 30° tilt off the normal direction of viewing, as shown in the figure. If the sensor is sensing an irradiation of 50 W/m^2 , determine the distance between the aperture and the sensor.

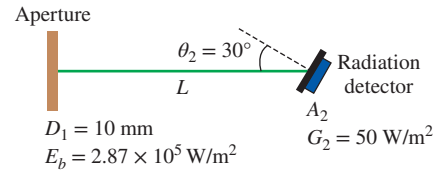


FIGURE P12-60

12-61 A small circular plate has a diameter of 2 cm and can be approximated as a blackbody. To determine the radiation from the plate, a radiometer is placed normal to the direction of viewing from the plate at a distance of 50 cm. If the radiometer measured an irradiation of 85 W/m^2 from the plate, determine the temperature of the plate.

12-62  Heat dissipated from an engine in operation can cause hot spots on its surface. If the outer surface of an engine is situated in a place where oil leakage is possible, then when the leaked oil comes in contact with hot spots above the oil's autoignition temperature, it can ignite spontaneously. Consider an engine outer surface that can be approximated as a blackbody. To prevent fire hazard in the event of an oil leak on the engine's surface, the surface temperature of the engine should be kept below 180°C . A radiometer is placed normal to and at a distance of 1 m from the engine surface to monitor the surface temperature. The radiometer receives radiation from a target area of 1 cm^2 of the engine surface. If the radiometer detects an irradiation of 0.1 W/m^2 , would there be any risk of fire hazard?

12-63 A radiometer can be used to determine the position of an approaching hot object by measuring the amount of irradiation it detects. Consider a radiometer placed at a distance $H = 0.5 \text{ m}$ from the x -axis used to measure the position of an approaching small blackbody object. If the radiometer is measuring 80 percent of the irradiation corresponding to the position of the object directly under the radiometer ($x = 0$), determine the position of the object L .

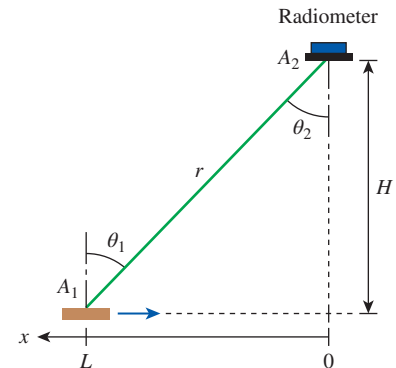


FIGURE P12-63

12-64  Reconsider Prob. 12-63. Using appropriate software, evaluate the effect of the position of the approaching object on the irradiation ratio $G_{2,x}/G_{2,0}$, where $G_{2,x}$ and $G_{2,0}$ are the irradiation on the radiometer when the

object is at position x and at $x = 0$, respectively. By varying x from 0 to 3 m with 0.2-m increments, plot the irradiation ratio $G_{2,x}/G_{2,0}$ as a function of the approaching object position x for the placement of the radiometer at $H = 0.5, 1.0$, and 1.5 m.

12-65 A small circular plate with a surface A_1 can be approximated as a blackbody. A radiometer with a surface A_2 is placed normal to the direction of viewing from the plate at a distance L . Would the irradiation on the radiometer from the plate decrease if the distance L were doubled, and if so, by how much?

12-66 A blackbody plate ($A_1 = 5 \text{ cm}^2$) is subjected to a uniform heat flux of 1000 W/m^2 on the bottom, while the top surface is exposed to an ambient surrounding at 5°C . The heat transfer coefficient due to natural convection on the plate surface is $5 \text{ W/m}^2\cdot\text{K}$. A radiometer is placed 1 m above the plate normal to the direction of viewing from the plate. Determine the irradiation that the radiometer would detect.

12-67 **PtD** A long metal sheet that can be approximated as a blackbody is being conveyed through a water bath to be cooled. In order to prevent thermal burns to people handling the sheet, it must exit the water bath at a temperature below 45°C . A radiometer is placed normal to and at a distance of 0.5 m from the sheet's surface to monitor the exit temperature. The radiometer receives radiation from a target area of 1 cm^2 of the metal sheet surface. When the radiometer detects that the metal sheet temperature is not below 45°C , an alarm will go off to warn that the sheet is not safe to touch. Determine the irradiation on the radiometer that should trigger the warning alarm.

12-68 **PtD** A radiometer is used to monitor the temperature of manufactured parts ($A_1 = 10 \text{ cm}^2$) on a conveyor. The radiometer is placed at a distance of 1 m from and normal to the manufactured parts. When a part moves to the position normal to the radiometer, the sensor measures the radiation emitted from the part. In order to prevent thermal burns to people handling the manufactured parts at the end of the conveyor, the temperature of the parts should be below 45°C . An array of spray heads is programmed to discharge mist to cool the parts when the radiometer detects a temperature of 45°C or higher on a part. If the manufactured parts can be approximated as blackbodies, determine the irradiation on the radiometer that should trigger the spray heads to release cooling mist when the temperature is not below 45°C .

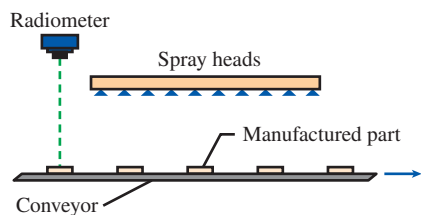


FIGURE P12-68

12-69 **PtD** A long metal bar ($c_p = 450 \text{ J/kg}\cdot\text{K}$, $\rho = 7900 \text{ kg/m}^3$) is being conveyed through a water bath to be quenched. The metal bar has a cross section of $30 \text{ mm} \times 15 \text{ mm}$,

and it enters the water bath at 700°C . During the quenching process, 500 kW of heat is released from the bar in the water bath. In order to prevent thermal burns to people handling the metal bar, it must exit the water bath at a temperature below 45°C . A radiometer is placed normal to and at a distance of 1 m from the bar to monitor the exit temperature. The radiometer receives radiation from a target area of 1 cm^2 of the bar surface. An irradiation signal detected by the radiometer is used to control the speed of the bar being conveyed through the water bath so that the exit temperature is safe for handling. If the radiometer detects an irradiation of 0.015 W/m^2 , determine the speed of the bar being conveyed through the water bath. Assume that the metal bar can be approximated as a blackbody.

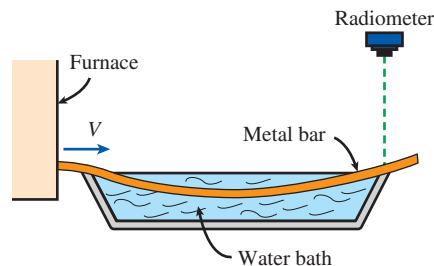


FIGURE P12-69

12-70 **C&S** A radiometer is placed normal to a CPVC pipe at a distance of 1 m to monitor the pipe surface temperature. The radiometer measures an irradiation of $500 \mu\text{W/m}^2$ from a target area of 1 mm^2 on the CPVC pipe surface. The ASME Code for Process Piping specifies a maximum use temperature of 93.3°C for CPVC pipe (ASME B31.3-2014, Table A-1M). Determine whether the surface temperature of the CPVC pipe complies with the ASME code.

12-71 **C&S** A radiometer is placed normal to an ASTM A240 410S stainless steel sheet at a distance of 0.7 m to monitor the surface temperature. The maximum use temperature for an ASTM A240 410S stainless steel sheet is 649°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The stainless steel sheet has an emissivity of 0.4, and an irradiation of 0.2 W/m^2 from a target area of 5 mm^2 is sensed by the radiometer. Determine whether the surface temperature of the stainless steel sheet complies with the ASME code.

Radiation Properties

12-72C What is a graybody? How does it differ from a blackbody? What is a diffuse gray surface?

12-73C Define the properties emissivity and absorptivity. When are these two properties equal to each other?

12-74C Define the properties reflectivity and transmissivity and discuss the different forms of reflection.

12-75C What is the greenhouse effect? Why is it a matter of great concern among atmospheric scientists?

12-76C We can see the inside of a microwave oven during operation through its glass door, which indicates that visible

radiation is escaping the oven. Do you think that the harmful microwave radiation might also be escaping?

12-77E A 5-in-diameter spherical ball is known to emit radiation at a rate of 480 Btu/h when its surface temperature is 950 R. Determine the average emissivity of the ball at this temperature.

12-78 A small surface of area $A_1 = 5 \text{ cm}^2$ emits radiation as a blackbody at $T_1 = 1000 \text{ K}$. A radiation sensor of area $A_2 = 3 \text{ cm}^2$ is placed normal to the direction of viewing from surface A_1 at a distance L . An optical filter with the following spectral transmissivity is placed in front of the sensor:

$$\tau_\lambda = \begin{cases} \tau_1 = 0, & 0 \leq \lambda < 2 \text{ } \mu\text{m} \\ \tau_2 = 0.5, & 2 \text{ } \mu\text{m} \leq \lambda < \infty \end{cases}$$

If the distance between the radiation sensor and surface A_1 is $L = 0.5 \text{ m}$, determine the irradiation measured by the sensor.

12-79 **P1D** A tank is filled with chemicals undergoing an exothermic reaction. The reaction heats up the tank wall, and to prevent thermal burn on individuals, the tank's outer surface temperature should be below 45°C . The emissivity of the tank surface is 0.85, and irradiation on the tank surface from the surroundings is estimated to be 390 W/m^2 . A radiometer is placed normal to and at a distance of 0.5 m from the tank surface to monitor the surface temperature. The radiometer receives radiation from a target area of 1 cm^2 of the tank surface. If the irradiation measured by the radiometer is 0.085 W/m^2 , determine whether or not the tank surface is safe to touch.

12-80 A furnace that has a $40\text{-cm} \times 40\text{-cm}$ glass window can be considered to be a blackbody at 1200 K . If the transmissivity of the glass is 0.7 for radiation at wavelengths less than $3 \text{ } \mu\text{m}$ and zero for radiation at wavelengths greater than $3 \text{ } \mu\text{m}$, determine the fraction and the rate of radiation coming from the furnace and transmitted through the window.

12-81 The spectral emissivity function of an opaque surface at 1000 K is approximated as

$$\epsilon_\lambda = \begin{cases} \epsilon_1 = 0.4, & 0 \leq \lambda < 3 \text{ } \mu\text{m} \\ \epsilon_2 = 0.7, & 3 \text{ } \mu\text{m} \leq \lambda < 6 \text{ } \mu\text{m} \\ \epsilon_3 = 0.3, & 6 \text{ } \mu\text{m} \leq \lambda < \infty \end{cases}$$

Determine the average emissivity of the surface and the rate of radiation emission from the surface, in W/m^2 .

12-82 **C&S** The maximum use temperature for an ASTM A240 410S stainless steel plate is 649°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The plate is treated such that it exhibits the following spectral emissivity:

$$\epsilon_\lambda = \begin{cases} \epsilon_1 = 0.8, & 0 \leq \lambda < 3.25 \text{ } \mu\text{m} \\ \epsilon_2 = 0.6, & 3.25 \text{ } \mu\text{m} \leq \lambda < 10.3 \text{ } \mu\text{m} \\ \epsilon_3 = 0.3, & 10.3 \text{ } \mu\text{m} \leq \lambda < \infty \end{cases}$$

Determine the highest rate of radiation emission, in W/m^2 , that the plate can achieve without violating the ASME code.

12-83 **C&S** A carbon steel (ASTM A992) plate has a maximum use temperature of 427°C specified by the

ASME Code for Process Piping (ASME B31.3-2014, Table A-1M). The plate was surface treated such that it exhibits the following spectral reflectivity:

$$\rho_\lambda = \begin{cases} \rho_1 = 0.2, & 0 \leq \lambda < 4 \text{ } \mu\text{m} \\ \rho_2 = 0.5, & 4 \text{ } \mu\text{m} \leq \lambda < 15 \text{ } \mu\text{m} \\ \rho_3 = 0.8, & 15 \text{ } \mu\text{m} \leq \lambda < \infty \end{cases}$$

Assume the spectral form of Kirchhoff's law applies to the plate. Determine the highest rate of radiation emission, in W/m^2 , that the plate can achieve without violating the ASME code.

12-84 The emissivity of a tungsten filament can be approximated to be 0.5 for radiation at wavelengths less than $1 \text{ } \mu\text{m}$ and 0.15 for radiation at greater than $1 \text{ } \mu\text{m}$. Determine the average emissivity of the filament at (a) 1500 K and (b) 2500 K . Also, determine the absorptivity and reflectivity of the filament at both temperatures.

12-85 The variations of the spectral emissivity of two surfaces are as given in Fig. P12-85. Determine the average emissivity of each surface at $T = 3000 \text{ K}$. Also, determine the average absorptivity and reflectivity of each surface for radiation coming from a source at 3000 K . Which surface is more suitable to serve as a solar absorber?

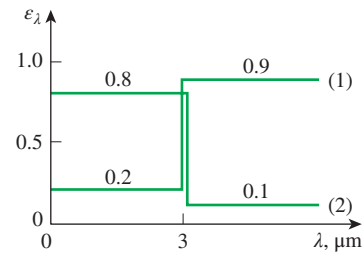


FIGURE P12-85

12-86 The emissivity of a surface coated with aluminum oxide can be approximated to be 0.15 for radiation at wavelengths less than $5 \text{ } \mu\text{m}$ and 0.9 for radiation at wavelengths greater than $5 \text{ } \mu\text{m}$. Determine the average emissivity of this surface at (a) 5800 K and (b) 300 K . What can you say about the absorptivity of this surface for radiation coming from sources at 5800 K and 300 K ? *Answers: (a) 0.154, (b) 0.89*

12-87 The variation of the spectral absorptivity of a surface is as given in Fig. P12-87. Determine the average absorptivity and reflectivity of the surface for radiation that originates from a source at $T = 2500 \text{ K}$. Also, determine the average emissivity of this surface at 3000 K .

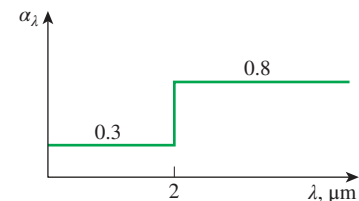


FIGURE P12-87

12–88 The reflectivity of aluminum coated with lead sulfate is 0.35 for radiation at wavelengths less than $3\text{ }\mu\text{m}$ and 0.95 for radiation greater than $3\text{ }\mu\text{m}$. Determine the average reflectivity of this surface for solar radiation ($T \approx 5800\text{ K}$) and radiation coming from surfaces at room temperature ($T \approx 300\text{ K}$). Also, determine the emissivity and absorptivity of this surface at both temperatures. Do you think this material is suitable for use in solar collectors?

12–89  The maximum use temperature for an ASTM A992 carbon steel plate is 427°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The plate was surface treated such that it exhibits the following spectral reflectivity:

$$\rho_\lambda = \begin{cases} \rho_1 = 0.45, & 0 \leq \lambda < 1\text{ }\mu\text{m} \\ \rho_2 = 0.65, & 1\text{ }\mu\text{m} \leq \lambda < 6\text{ }\mu\text{m} \\ \rho_3 = 0.80, & 6\text{ }\mu\text{m} \leq \lambda < \infty \end{cases}$$

Assume the spectral form of Kirchhoff's law applies to the plate. Solar irradiation ($T \approx 5800\text{ K}$) on the plate surface is 1000 W/m^2 , and the net radiation heat flux leaving the plate surface is 6000 W/m^2 . Determine whether the carbon steel plate is suitable for the given conditions so that it complies with the ASME code.

12–90 The variation of the spectral transmissivity of a 0.6-cm-thick glass window is as given in Fig. P12–90. Determine the average transmissivity of this window for solar radiation ($T \approx 5800\text{ K}$) and radiation coming from surfaces at room temperature ($T \approx 300\text{ K}$). Also, determine the amount of solar radiation transmitted through the window for incident solar radiation of 650 W/m^2 . *Answers: 0.870, 0.00016, 566 W/m^2*

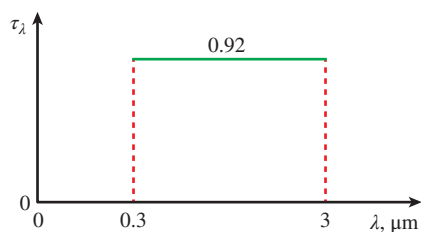


FIGURE P12–90

12–91 Consider an opaque horizontal plate that is well insulated on the edges and the lower surface. The plate is maintained at 500 K , has a total hemispherical absorptivity of 0.51, and has the following spectral emissivity function:

$$\varepsilon_\lambda = \begin{cases} \varepsilon_1 = 0.4, & 0 \leq \lambda < 4\text{ }\mu\text{m} \\ \varepsilon_2 = 0.8, & 4\text{ }\mu\text{m} \leq \lambda < \infty \end{cases}$$

If the plate is subjected to an irradiation of 5600 W/m^2 , determine (a) the total hemispherical emissivity and (b) the radiosity of the plate surface.

12–92 An opaque horizontal plate is well insulated on the edges and the lower surface. The irradiation on the plate is 3000 W/m^2 , of which 500 W/m^2 is reflected. The plate has a uniform temperature of 700 K and has an emissive power of 5000 W/m^2 . Determine the total emissivity and absorptivity of the plate.

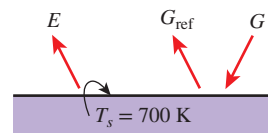


FIGURE P12–92

12–93 Irradiation on a semitransparent medium is at a rate of 520 W/m^2 . If 160 W/m^2 of the irradiation is reflected from the medium and 130 W/m^2 is transmitted through the medium, determine the medium's absorptivity, reflectivity, transmissivity, and emissivity.

12–94 Consider an opaque horizontal plate that is well insulated on the edges and the lower surface. The plate is uniformly irradiated from above while air at $T_\infty = 300\text{ K}$ flows over the surface, providing a uniform convection heat transfer coefficient of $40\text{ W/m}^2\cdot\text{K}$. Under steady-state conditions, the surface has a radiosity of 4000 W/m^2 , and the plate temperature is maintained uniformly at 350 K . If the total absorptivity of the plate is 0.40, determine (a) the irradiation on the plate, (b) the total reflectivity of the plate, (c) the emissive power of the plate, and (d) the total emissivity of the plate.

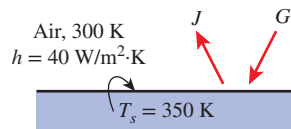


FIGURE P12–94

12–95 Consider an opaque plate that is well insulated on the edges and heated at the bottom with an electric heater. The plate has an emissivity of 0.67 and is situated in an ambient surrounding temperature of 7°C where the natural convection heat transfer coefficient is $7\text{ W/m}^2\cdot\text{K}$. To maintain a surface temperature of 80°C , the electric heater supplies 1000 W/m^2 of uniform heat flux to the plate. Determine the radiosity of the plate under these conditions.

12–96 A horizontal plate is experiencing uniform irradiation on both upper and lower surfaces. The ambient air temperature surrounding the plate is 290 K with a convection heat transfer coefficient of $30\text{ W/m}^2\cdot\text{K}$. Both upper and lower surfaces of the plate have a radiosity of 4000 W/m^2 , and the plate temperature is maintained uniformly at 390 K . If the plate is not opaque and has an absorptivity of 0.527, determine the irradiation and emissivity of the plate.

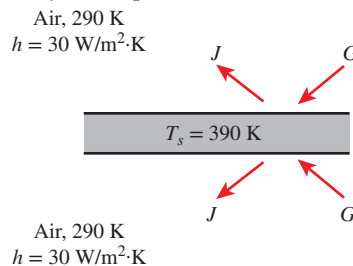


FIGURE P12–96

12–97 A semitransparent plate ($A_1 = 2 \text{ cm}^2$) has an irradiation of 500 W/m^2 , where 30 percent of the irradiation is reflected away from the plate and 50 percent of the irradiation is transmitted through the plate. A radiometer is placed 0.5 m above the plate normal to the direction of viewing from the plate. If the temperature of the plate is uniform at 350 K , determine the irradiation that the radiometer would detect.

Atmospheric and Solar Radiation

12–98C When the earth is closest to the sun, we have winter in the northern hemisphere. Explain why. Also explain why we have summer in the northern hemisphere when the earth is farthest away from the sun.

12–99C Explain why surfaces usually have quite different absorptivities for solar radiation and for radiation originating from the surrounding bodies.

12–100C You have probably noticed warning signs on the highways stating that bridges may be icy even when the roads are not. Explain how this can happen.

12–101C What changes would you notice if the sun emitted radiation at an effective temperature of 2000 K instead of 5762 K ?

12–102C What is the solar constant? How is it used to determine the effective surface temperature of the sun? How would the value of the solar constant change if the distance between the earth and the sun doubled?

12–103C Explain why the sky is blue and the sunset is yellow-orange.

12–104C What is the effective sky temperature?

12–105E Solar radiation is incident on the outer surface of a spaceship at a rate of $400 \text{ Btu/h}\cdot\text{ft}^2$. The surface has an absorptivity of $\alpha_s = 0.10$ for solar radiation and an emissivity of $\varepsilon = 0.6$ at room temperature. The outer surface radiates heat into space at 0 R . If there is no net heat transfer into the spaceship, determine the equilibrium temperature of the surface. *Answer: 444 R*

12–106 The air temperature on a clear night is observed to remain at about 4°C . Yet water is reported to have frozen that night due to radiation effect. Taking the convection heat transfer coefficient to be $18 \text{ W/m}^2\cdot\text{K}$, determine the value of the maximum effective sky temperature that night.

12–107 A surface has an absorptivity of $\alpha_s = 0.72$ for solar radiation and an emissivity of $\varepsilon = 0.6$ at room temperature. The surface temperature is observed to be 350 K when the direct and the diffuse components of solar radiation are $G_D = 350$ and $G_d = 400 \text{ W/m}^2$, respectively, and the direct radiation makes a 30° angle with the normal of the surface. Taking the effective sky temperature to be 280 K , determine the net rate of radiation heat transfer to the surface at that time.

12–108 The absorber surface of a solar collector is made of aluminum coated with black chrome ($\alpha_s = 0.87$ and $\varepsilon = 0.09$). Solar radiation is incident on the surface at a rate of 720 W/m^2 .

The air and the effective sky temperatures are 25°C and 15°C , respectively, and the convection heat transfer coefficient is $10 \text{ W/m}^2\cdot\text{K}$. For an absorber surface temperature of 70°C , determine the net rate of solar energy delivered by the absorber plate to the water circulating behind it.

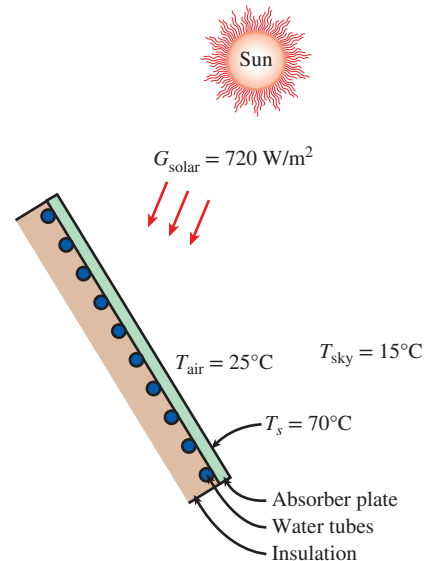


FIGURE P12–108

12–109 Determine the equilibrium temperature of the absorber surface in Prob. 12–108 if the back side of the absorber is insulated.

12–110  Reconsider Prob. 12–108. Using appropriate software, plot the net rate of solar energy transferred to water as a function of the absorptivity of the absorber plate. Let the absorptivity vary from 0.5 to 1.0 , and discuss the results.

Special Topic: Solar Heat Gain Through Windows

12–111C What fraction of the solar energy is in the visible range (*a*) outside the earth's atmosphere and (*b*) at sea level on earth? Answer the same question for infrared radiation.

12–112C Describe the solar radiation properties of a window that is ideally suited for minimizing the air-conditioning load.

12–113C What is the effect of a low-e coating on the inner surface of a window glass on the (*a*) heat loss in winter and (*b*) heat gain in summer through the window?

12–114C What is a shading device? Is an internal or external shading device more effective in reducing the solar heat gain through a window? How does the color of the surface of a shading device facing outside affect the solar heat gain?

12–115C What does the SC (shading coefficient) of a device represent? How do the SCs of clear glass and heat-absorbing glass compare?

12–116C Define the SHGC (solar heat gain coefficient), and explain how it differs from the SC (shading coefficient). What are the values of the SHGC and SC of a single-pane clear-glass window?

12–117 Consider a building in New York (40°N latitude) that has 76 m^2 of window area on its south wall. The windows are double-pane heat-absorbing type and are equipped with light-colored venetian blinds with a shading coefficient of $\text{SC} = 0.30$. Determine the total solar heat gain of the building through the south windows at solar noon in April. What would your answer be if there were no blinds at the windows?

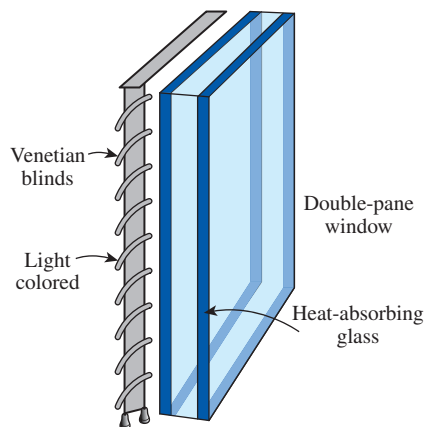


FIGURE P12–117

12–118 A typical winter day in Reno, Nevada (39°N latitude), is cold but sunny, and thus the solar heat gain through the windows can be more than the heat loss through them during daytime. Consider a house with double-door-type windows that are double paned with 3-mm-thick glasses and 6.4 mm of airspace and have aluminum frames and spacers. The house is maintained at 22°C at all times. Determine if the house is losing more or less heat than it is gaining from the sun through an east window on a typical day in January for a 24-h period if the average outdoor temperature is 10°C . *Answer: less*

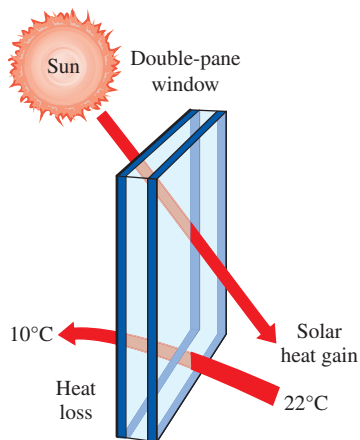


FIGURE P12–118

12–119 Repeat Prob. 12–118 for a south window.

12–120 Consider a building located near 40°N latitude that has equal window areas on all four sides. The building owner is considering coating the south-facing windows with reflective film to reduce the solar heat gain and thus the cooling load. But someone suggests that the owner will reduce the cooling load even more if she coats the west-facing windows instead. What do you think?

12–121E Determine the rate of net heat gain (or loss) through a 9-ft-high, 15-ft-wide, fixed $\frac{1}{8}$ -in single-glass window with aluminum frames on the west wall at 3 PM solar time during a typical day in January at a location near 40°N latitude when the indoor and outdoor temperatures are 78°F and 20°F , respectively.

12–122 A manufacturing facility located at 32°N latitude has a glazing area of 60 m^2 facing west that consists of double-pane windows made of clear glass ($\text{SHGC} = 0.766$). To reduce the solar heat gain in summer, a reflective film that will reduce the SHGC to 0.35 is considered. The cooling season consists of June, July, August, and September, and the heating season is October through April. The average daily solar heat fluxes incident on the west side at this latitude are 2.35, 3.03, 3.62, 4.00, 4.20, 4.24, 4.16, 3.93, 3.48, 2.94, 2.33, and $2.07\text{ kWh/day}\cdot\text{m}^2$ for January through December, respectively. Also, the unit costs of electricity and natural gas are $\$0.15/\text{kWh}$ and $\$0.90/\text{therm}$, respectively. If the coefficient of performance of the cooling system is 3.2 and the efficiency of the furnace is 0.90, determine the net annual cost savings due to installing reflective coating on the windows. Also, determine the simple payback period if the installation cost of reflective film is $\$15/\text{m}^2$.

12–123 A house located in Boulder, Colorado (40°N latitude), has ordinary double-pane windows with 6-mm-thick glasses, and the total window areas are 8, 6, 6, and 4 m^2 on the south, west, east, and north walls, respectively. Determine the total solar heat gain of the house at 9:00, 12:00, and 15:00 solar time in July. Also, determine the total amount of solar heat gain per day for an average day in January.

12–124 Repeat Prob. 12–123 for double-pane windows that are gray-tinted.

Review Problems

12–125 A 1-m-diameter spherical cavity is maintained at a uniform temperature of 600 K. Now a 5-mm-diameter hole is drilled. Determine the maximum rate of radiation energy streaming through the hole. What would your answer be if the diameter of the cavity were 3 m?

12–126 Daylight and incandescent light may be approximated as blackbodies at the effective surface temperatures of 5800 K and 2800 K, respectively. Determine the wavelength at maximum emission of radiation for each of the lighting sources.

12-127 The human skin is “selective” when it comes to the absorption of the solar radiation that strikes it perpendicularly. The skin absorbs only 50 percent of the incident radiation with wavelengths between $\lambda_1 = 0.517 \mu\text{m}$ and $\lambda_2 = 1.552 \mu\text{m}$. The radiation with wavelengths shorter than λ_1 and longer than λ_2 is fully absorbed. The solar surface may be modeled as a blackbody with effective surface temperature of 5800 K. Calculate the fraction of the incident solar radiation that is absorbed by the human skin.

12-128 Consider a small black surface of area $A = 3.5 \text{ cm}^2$ maintained at 750 K. Determine the rate at which radiation energy is emitted by the surface through a ring-shaped opening defined by $0 \leq \phi \leq 2\pi$ and $40 \leq \theta \leq 50^\circ$, where ϕ is the azimuth angle and θ is the angle a radiation beam makes with the normal of the surface.

12-129 A small surface of area $A = 3 \text{ cm}^2$ emits radiation with an intensity that can be expressed as $I_e(\theta, \phi) = 100 \phi \cos \theta$ where I_e has the units of $\text{W/m}^2 \cdot \text{sr}$. Determine the emissive power from the surface into the hemisphere surrounding it and the rate of radiation emission from the surface.

12-130 A radiation sensor and a small surface of area A_1 that emits radiation as a blackbody are oriented as shown in Fig. P12-130. Determine the distance L at which the sensor is measuring two-thirds of the radiation rate emitted from surface A_1 corresponding to the position directly under the sensor at $L = 0$.

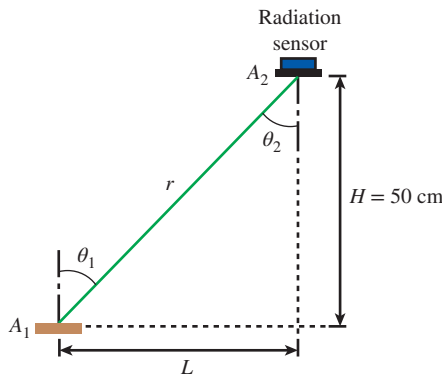


FIGURE P12-130

12-131 The spectral emissivity of an opaque surface at 1200 K is approximated as

$$\begin{aligned} \epsilon_1 &= 0 & \text{for } \lambda < 2 \mu\text{m} \\ \epsilon_2 &= 0.85 & \text{for } 2 \leq \lambda \leq 6 \mu\text{m} \\ \epsilon_3 &= 0 & \text{for } \lambda > 6 \mu\text{m} \end{aligned}$$

Determine the total emissivity and the emissive flux of the surface.

12-132 The spectral absorptivity of an opaque surface is as shown in Fig. P12-132. Determine the absorptivity of the surface for radiation emitted by a source at (a) 1000 K and (b) 3000 K.

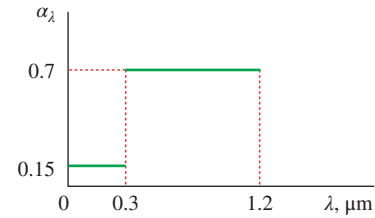


FIGURE P12-132

12-133 The surface in Prob. 12-132 receives solar radiation at a rate of 470 W/m^2 . Determine the solar absorptivity of the surface and the rate of absorption of solar radiation.

12-134 The spectral transmissivity of a 3-mm-thick regular glass can be expressed as

$$\begin{aligned} \tau_1 &= 0 & \text{for } \lambda < 0.35 \mu\text{m} \\ \tau_2 &= 0.85 & \text{for } 0.35 < \lambda < 2.5 \mu\text{m} \\ \tau_3 &= 0 & \text{for } \lambda > 2.5 \mu\text{m} \end{aligned}$$

Determine the transmissivity of this glass for solar radiation. What is the transmissivity of this glass for light?

12-135 The spectral transmissivity of a glass cover used in a solar collector is given as

$$\begin{aligned} \tau_1 &= 0 & \text{for } \lambda < 0.3 \mu\text{m} \\ \tau_2 &= 0.9 & \text{for } 0.3 < \lambda < 3 \mu\text{m} \\ \tau_3 &= 0 & \text{for } \lambda > 3 \mu\text{m} \end{aligned}$$

Solar radiation is incident at a rate of 950 W/m^2 , and the absorber plate, which can be considered to be black, is maintained at 340 K by the cooling water. Determine (a) the solar flux incident on the absorber plate, (b) the transmissivity of the glass cover for radiation emitted by the absorber plate, and (c) the rate of heat transfer to the cooling water if the glass cover temperature is also 340 K.

12-136 A small surface of area $A_1 = 3 \text{ cm}^2$ and a radiation sensor of area $A_2 = 1 \text{ cm}^2$ are oriented as shown in Fig. P12-136. Surface A_1 is opaque (diffuse emitter and reflector) and maintains a constant temperature of $T_1 = 400 \text{ K}$. The absorptivity and emissivity of surface A_1 are 0.5 and 0.7, respectively. If surface A_1 is experiencing 2000 W/m^2 of irradiation, determine the rate at which radiation emitted from A_1 is intercepted by the sensor.

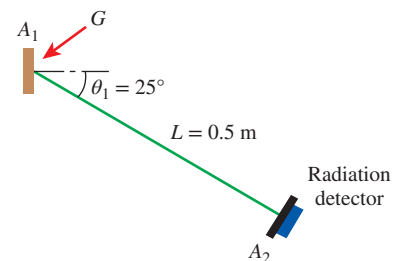


FIGURE P12-136

12–137 Solar radiation is incident on the front surface of a thin plate with direct and diffuse components of 300 and 250 W/m², respectively. The direct radiation makes a 30° angle with the normal of the surface. The plate surfaces have a solar absorptivity of 0.63 and an emissivity of 0.93. The air temperature is 5°C and the convection heat transfer coefficient is 20 W/m²·K. The effective sky temperature for the front surface is –33°C while the surrounding surfaces are at 5°C for the back surface. Determine the equilibrium temperature of the plate.

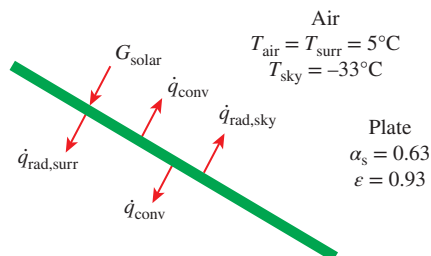


FIGURE P12–137

12–138 A horizontal opaque flat plate is well insulated on the edges and the lower surface. The top surface has an area of 5 m², and it experiences uniform irradiation at a rate of 5000 W. The plate absorbs 4000 W of the irradiation, and the surface is losing heat at a rate of 500 W by convection. If the plate maintains a uniform temperature of 350 K, determine the absorptivity, reflectivity, and emissivity of the plate.

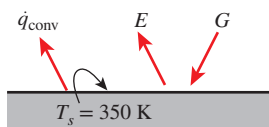


FIGURE P12–138

Fundamentals of Engineering (FE) Exam Problems

12–139 Consider a surface at –10°C in an environment at 25°C. The maximum rate of heat that can be emitted from this surface by radiation is

- (a) 152 W/m² (b) 176 W/m² (c) 211 W/m²
(d) 271 W/m² (e) 324 W/m²

12–140 Consider a surface at 900 K. The spectral blackbody emissive power at a wavelength of 50 μm is

- (a) 3.2 W/m²·μm (b) 9.6 W/m²·μm (c) 24 W/m²·μm
(d) 76 W/m²·μm (e) 108 W/m²·μm

12–141 The wavelength at which the blackbody emissive power reaches its maximum value at 300 K is

- (a) 5.1 μm (b) 9.7 μm (c) 15.5 μm
(d) 38.0 μm (e) 73.1 μm

12–142 A surface absorbs 10 percent of radiation at wavelengths less than 3 μm and 50 percent of radiation at

wavelengths greater than 3 μm. The average absorptivity of this surface for radiation emitted by a source at 3000 K is

- (a) 0.14 (b) 0.22 (c) 0.30 (d) 0.38 (e) 0.42

12–143 A surface at 300°C has an emissivity of 0.7 in the wavelength range of 0 – 4.4 μm and 0.3 over the rest of the wavelength range. At a temperature of 300°C, 19 percent of the blackbody emissive power is in the wavelength range up to 4.4 μm. The total emissivity of this surface is

- (a) 0.300 (b) 0.376 (c) 0.624 (d) 0.700 (e) 0.500

12–144 Consider a 4-cm-diameter and 6-cm-long cylindrical rod at 1200 K. If the emissivity of the rod surface is 0.75, the total amount of radiation emitted by all surfaces of the rod in 20 min is

- (a) 88 kJ (b) 118 kJ (c) 6661 kJ
(d) 1064 kJ (e) 1418 kJ

12–145 Solar radiation is incident on a semitransparent body at a rate of 500 W/m². If 150 W/m² of this incident radiation is reflected back and 225 W/m² is transmitted across the body, the absorptivity of the body is

- (a) 0 (b) 0.25 (c) 0.30 (d) 0.45 (e) 1

12–146 Solar radiation is incident on an opaque surface at a rate of 400 W/m². The emissivity of the surface is 0.65 and the absorptivity to solar radiation is 0.85. The convection coefficient between the surface and the environment at 25°C is 6 W/m²·K. If the surface is exposed to atmosphere with an effective sky temperature of 250 K, the equilibrium temperature of the surface is

- (a) 281 K (b) 298 K (c) 303 K (d) 317 K (e) 339 K

12–147 A surface is exposed to solar radiation. The direct and diffuse components of solar radiation are 480 and 250 W/m², and the direct radiation makes a 35° angle with the normal of the surface. The solar absorptivity and the emissivity of the surface are 0.24 and 0.41, respectively. If the surface is observed to be at 315 K and the effective sky temperature is 256 K, the net rate of radiation heat transfer to the surface is

- (a) –79 W/m² (b) –22 W/m² (c) 25 W/m²
(d) 154 W/m² (e) 643 W/m²

Design and Essay Problems

12–148 Write an essay on the radiation properties of selective surfaces used on the absorber plates of solar collectors. Find out about the various kinds of such surfaces, and discuss the performance and cost of each type. Recommend a selective surface that optimizes cost and performance.

12–149 According to an Atomic Energy Commission report, a hydrogen bomb can be approximated as a large fireball at a temperature of 7200 K. You are to assess the impact if such a bomb exploded 5 km above a city. Assume the diameter of the fireball to be 1 km and the blast to last 15 s. Investigate the level of radiation energy people, plants, and houses will be exposed to, and how adversely they will be affected by the blast.